

THE *CONICS* OF APOLLONIUS.



THE CONE.

IF a straight line indefinite in length, and passing always through a fixed point, be made to move round the circumference of a circle which is not in the same plane with the point, so as to pass successively through every point of that circumference, the moving straight line will trace out the surface of a **double cone**, or two similar cones lying in opposite directions and meeting in the fixed point, which is the **apex** of each cone.

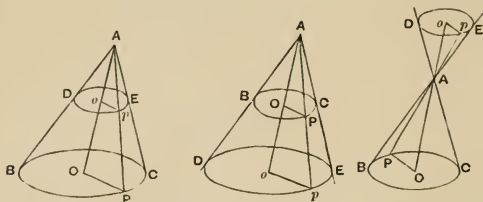
The circle about which the straight line moves is called the **base** of the cone lying between the said circle and the fixed point, and the **axis** is defined as the straight line drawn from the fixed point or the apex to the centre of the circle forming the base.

The cone so described is a **scalene** or **oblique** cone except in the particular case where the axis is perpendicular to the base. In this latter case the cone is a **right** cone.

If a cone be cut by a plane passing through the apex, the resulting section is a triangle, two sides being straight lines lying on the surface of the cone and the third side being the straight line which is the intersection of the cutting plane and the plane of the base.

Let there be a cone whose apex is A and whose base is the circle BC , and let O be the centre of the circle, so that AO is the axis of the cone. Suppose now that the cone is cut by any plane parallel to the plane of the base BC , as DE , and let

the axis AO meet the plane DE in o . Let p be any point on the intersection of the plane DE and the surface of the cone. Join Ap and produce it to meet the circumference of the circle BC in P . Join OP , op .



Then, since the plane passing through the straight lines AO , AP cuts the two parallel planes BC , DE in the straight lines OP , op respectively, OP , op are parallel.

$$\therefore op : OP = Ao : AO.$$

And, BPC being a circle, OP remains constant for all positions of p on the curve DpE , and the ratio $Ao : AO$ is also constant.

Therefore op is constant for all points on the section of the surface by the plane DE . In other words, that section is a circle.

Hence *all sections of the cone which are parallel to the circular base are circles.* [I. 4.]*

Next, let the cone be cut by a plane passing through the axis and perpendicular to the plane of the base BC , and let the section be the triangle ABC . Conceive another plane HK drawn at right angles to the plane of the triangle ABC and cutting off from it the triangle AHK such that AHK is similar to the triangle ABC but lies in the contrary sense, i.e. such that the angle AKH is equal to the angle ABC . Then the section of the cone by the plane HK is called a **subcontrary** section (*ὑπεναντία τομή*).

* The references in this form, here and throughout the book, are to the original propositions of Apollonius.

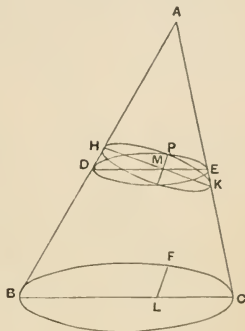
Let P be any point on the intersection of the plane HK with the surface, and F any point on the circumference of the circle BC . Draw PM , FL each perpendicular to the plane of the triangle ABC , meeting the straight lines HK , BC respectively in M , L . Then PM , FL are parallel.

Draw through M the straight line DE parallel to BC , and it follows that the plane through DME , PM is parallel to the base BC of the cone.

Thus the section DPE is a circle, and $DM \cdot ME = PM^2$.

But, since DE is parallel to BC , the angle ADE is equal to the angle ABC which is by hypothesis equal to the angle AKH .

Therefore in the triangles HDM , EKM the angles HDM , EKM are equal, as also are the vertical angles at M .



Therefore the triangles HDM , EKM are similar.

Hence $HM : MD = EM : MK$.

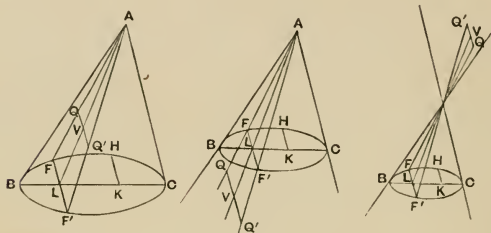
$$\therefore HM \cdot MK = DM \cdot ME = PM^2.$$

And P is any point on the intersection of the plane HK with the surface. Therefore the section made by the plane HK is a circle.

Thus there are two series of circular sections of an oblique cone, one series being parallel to the base, and the other consisting of the sections subcontrary to the first series. [I. 5.]

Suppose a cone to be cut by any plane through the axis making the triangular section ABC , so that BC is a diameter of the circular base. Let H be any point on the circumference of the base, let HK be perpendicular to the diameter BC , and let a parallel to HK be drawn from any point Q on the surface of the cone but not lying in the plane of the axial triangle. Further, let AQ be joined and produced, if necessary, to meet

the circumference of the base in F , and let FLF' be the chord perpendicular to BC . Join AL , AF' . Then the straight line through Q parallel to HK is also parallel to FLF' ; it follows therefore that the parallel through Q will meet both AL and AF' . And AL is in the plane of the axial triangle ABC . Therefore the parallel through Q will meet both the plane of the axial triangle and the other side of the surface of the cone, since AF' lies on the cone.



Let the points of intersection be V , Q' respectively.

Then $QV : VQ' = FL : LF'$, and $FL = LF'$.

$$\therefore QV = VQ',$$

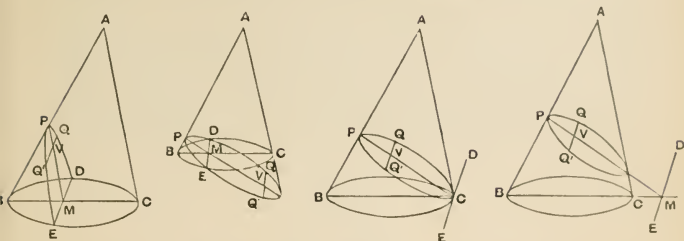
or QQ' is bisected by the plane of the axial triangle. [I. 6.]

Again, let the cone be cut by another plane not passing through the apex but intersecting the plane of the base in a straight line DME perpendicular to BC , the base of any axial triangle, and let the resulting section of the surface of the cone be DPE , the point P lying on either of the sides AB , AC of the axial triangle. The plane of the section will then cut the plane of the axial triangle in the straight line PM joining P to the middle point of DE .

Now let Q be any point on the curve of section, and through Q draw a straight line parallel to DE .

Then this parallel will, if produced to meet the other side of the surface in Q' , meet, and be bisected by, the axial

triangle. But it lies also in the plane of the section DPE ; it will therefore meet, and be bisected by, PM .



Therefore PM bisects any chord of the section which is parallel to DE .

Now a straight line bisecting each of a series of parallel chords of a section of a cone is called a **diameter**.

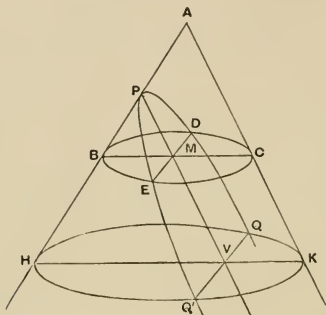
Hence, *if a cone be cut by a plane which intersects the circular base in a straight line perpendicular to the base of any axial triangle, the intersection of the cutting plane and the plane of the axial triangle will be a diameter of the resulting section of the cone.* [I. 7.]

If the cone be a *right* cone it is clear that the diameter so found will, for all sections, be at right angles to the chords which it bisects.

If the cone be *oblique*, the angle between the diameter so found and the parallel chords which it bisects will in general not be a right angle, but will be a right angle in the particular case only where the plane of the axial triangle ABC is at right angles to the plane of the base.

Again, if PM be the diameter of a section made by a plane cutting the circular base in the straight line DME perpendicular to BC , and if PM be in such a direction that it does not meet AC though produced to infinity, i.e. if PM be either parallel to AC , or makes with PB an angle less than the angle BAC and therefore meets CA produced beyond the apex of the cone, the section made by the said plane extends to infinity.

For, if we take any point V on PM produced and draw through it HK parallel to BC , and QQ' parallel to DE , the plane through HK, QQ' is parallel to that through DE, BC , i.e. to the base. Therefore the section $HQQK'$ is a circle. And D, E, Q, Q' are all on the surface of the cone and are also on the cutting plane. Therefore the section DPE extends to the circle HQK , and in like manner to the circular section through any point on PM produced, and therefore to any distance from P . [I. 8.]



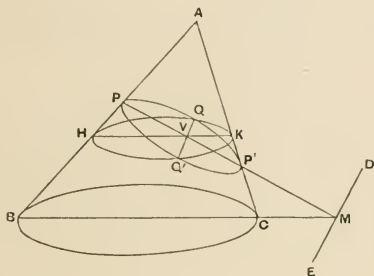
[It is also clear that $DM^2 = BM \cdot MC$, and $QV^2 = HV \cdot VK$; and $HV \cdot VK$ becomes greater as V is taken more distant from P . For, in the case where PM is parallel to AC , VK remains constant while HV increases; and in the case where the diameter PM meets CA produced beyond the apex of the cone, both HV, VK increase together as V moves away from P . Thus QV increases indefinitely as the section extends to infinity.]

If on the other hand PM meets AC , the section does not extend to infinity. In that case the section will be a circle if its plane is parallel to the base or subcontrary. But, if the section is neither parallel to the base nor subcontrary, it will not be a circle. [I. 9.]

For let the plane of the section meet the plane of the base in DME , a straight line perpendicular to BC , a diameter of the

circular base. Take the axial triangle through BC meeting the plane of section in the straight line PP' . Then P, P', M are all points in the plane of the axial triangle and in the plane of section. Therefore $PP'M$ is a straight line.

If possible, let the section PP' be a circle. Take any point Q on it and draw QQ' parallel to DME . Then if QQ' meets the axial triangle in V , $QV = VQ'$. Therefore PP' is the diameter of the supposed circle.



Let $HQKQ'$ be the circular section through QQ' parallel to the base.

Then, from the circles, $QV^2 = HV \cdot VK$,

$$QV^2 = PV \cdot VP'.$$

$$\therefore HV \cdot VK = PV \cdot VP',$$

so that

$$HV : VP = P'V : VK.$$

$\therefore$ the triangles VPH, VKP' are similar, and

$$\angle PHV = \angle KP'V;$$

$\therefore \angle KP'V = \angle ABC$, and the section PP' is subcontrary: which contradicts the hypothesis.

$\therefore PQP'$ is not a circle.

It remains to investigate the character of the sections mentioned on the preceding page, viz. (a) those which extend to infinity, (b) those which are finite but are not circles.

Suppose, as usual, that the plane of section cuts the circular base in a straight line DME and that ABC is the axial triangle

whose base BC is that diameter of the base of the cone which bisects DME at right angles at the point M . Then, if the plane of the section and the plane of the axial triangle intersect in the straight line PM , PM is a diameter of the section bisecting all chords of the section, as QQ' , which are drawn parallel to DE .

If QQ' is so bisected in V , QV is said to be an **ordinate**, or a straight line **drawn ordinate-wise** (τεταγμένως κατηγμένη), to the diameter PM ; and the length PV cut off from the diameter by any ordinate QV will be called the **abscissa** of QV .

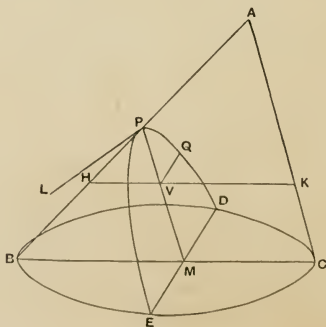
Proposition 1.

[I. 11.]

First let the diameter PM of the section be parallel to one of the sides of the axial triangle as AC , and let QV be any ordinate to the diameter PM . Then, if a straight line PL (supposed to be drawn perpendicular to PM in the plane of the section) be taken of such a length that $PL : PA = BC^2 : BA \cdot AC$, it is to be proved that

$$QV^2 = PL \cdot PV.$$

Let HK be drawn through V parallel to BC . Then, since QV is also parallel to DE , it follows that the plane through H, Q, K is parallel to the base of the cone and therefore



produces a circular section whose diameter is HK . Also QV is at right angles to HK .

$$\therefore HV \cdot VK = QV^2.$$

Now, by similar triangles and by parallels,

$$HV : PV = BC : AC$$

and

$$VK : PA = BC : BA.$$

$$\therefore HV \cdot VK : PV \cdot PA = BC^2 : BA \cdot AC.$$

Hence

$$QV^2 : PV \cdot PA = PL : PA$$

$$= PL \cdot PV : PV \cdot PA.$$

$$\therefore QV^2 = PL \cdot PV.$$

It follows that the square on any ordinate to the fixed diameter PM is equal to a rectangle applied (*παραβάλλειν*) to the fixed straight line PL drawn at right angles to PM with altitude equal to the corresponding abscissa PV . Hence the section is called a **PARABOLA**.

The fixed straight line PL is called the **latus rectum** (*ὀρθία*) or the **parameter of the ordinates** (*παρ' ἣν δύνανται αἱ καταγόμεναι τεταγμένως*).

This parameter, corresponding to the diameter PM , will for the future be denoted by the symbol p .

Thus

$$QV^2 = p \cdot PV,$$

or

$$QV^2 \propto PV.$$

Proposition 2.

[I. 12.]

Next let PM not be parallel to AC but let it meet CA produced beyond the apex of the cone in P' . Draw PL at right angles to PM in the plane of the section and of such a length that $PL : PP' = BF \cdot FC : AF^2$, where AF is a straight line through A parallel to PM and meeting BC in F . Then, if VR be drawn parallel to PL and $P'L$ be joined and produced to meet VR in R , it is to be proved that

$$QV^2 = PV \cdot VR.$$

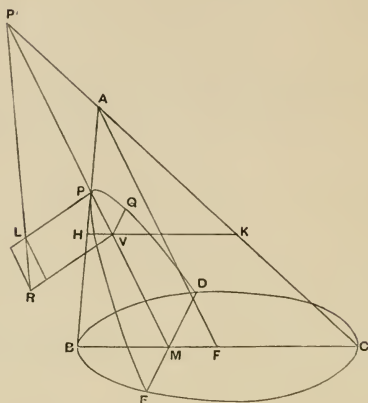
As before, let HK be drawn through V parallel to BC , so that

$$QV^2 = HV \cdot VK.$$

Then, by similar triangles,

$$HV : PV = BF : AF,$$

$$VK : P'V = FC : AF.$$



$$\therefore HV.VK : PV.P'V = BF.FC : AF^2.$$

Hence

$$\begin{aligned} QV^2 : PV.P'V &= PL : PP' \\ &= VR : P'V \\ &= PV.VR : PV.P'V. \\ \therefore QV^2 &= PV.VR. \end{aligned}$$

It follows that the square on the ordinate is equal to a rectangle whose height is equal to the abscissa and whose base lies along the fixed straight line PL but overlaps ($\acute{\upsilon}\pi\epsilon\rho\beta\acute{\alpha}\lambda\lambda\epsilon\iota$) it by a length equal to the difference between VR and PL *. Hence the section is called a **HYPERBOLA**.

* Apollonius describes the rectangle PR as *applied to the latus rectum but exceeding by a figure similar and similarly situated to that contained by PP' and PL* , i.e. exceeding the rectangle VL by the rectangle LR . Thus, if $QV=y$, $PV=x$, $PL=p$, and $PP'=d$,

$$y^2 = px + \frac{p}{d} \cdot x^2,$$

which is simply the Cartesian equation of the hyperbola referred to oblique axes consisting of a diameter and the tangent at its extremity.

PL is called the **latus rectum** or the **parameter of the ordinates** as before, and PP' is called the **transverse** (ἡ πλαγία). The fuller expression **transverse diameter** (ἡ πλαγία διάμετρος) is also used; and, even more commonly, Apollonius speaks of the diameter and the corresponding parameter together, calling the latter the **latus rectum** (i.e. the *erect side*, ἡ ὀρθία πλευρά), and the former the **transverse side** (ἡ πλαγία πλευρά), of the **figure** (εἶδος) **on**, or **applied to, the diameter** (πρὸς τῇ διαμέτρῳ), i.e. of the rectangle contained by PL , PP' as drawn.

The parameter PL will in future be denoted by p .

[COR. It follows from the proportion

$$QV^2 : PV \cdot P'V = PL : PP'$$

that, for any fixed diameter PP' ,

$$QV^2 : PV \cdot P'V \text{ is a constant ratio,}$$

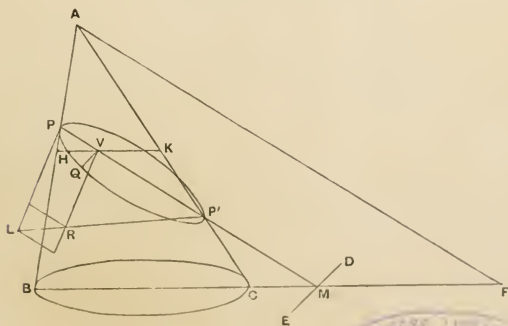
or QV^2 varies as $PV \cdot P'V$.]

Proposition 3.

[I. 13.]

If PM meets AC in P' and BC in M , draw AF parallel to PM meeting BC produced in F , and draw PL at right angles to PM in the plane of the section and of such a length that $PL : PP' = BF \cdot FC : AF^2$. Join $P'L$ and draw VR parallel to PL meeting $P'L$ in R . It will be proved that

$$QV^2 = PV \cdot VR.$$



Draw HK through V parallel to BC . Then, as before,

$$QV^2 = HV \cdot VK.$$

Now, by similar triangles,

$$HV : PV = BF : AF,$$

$$VK : P'V = FC : AF.$$

$$\therefore HV \cdot VK : PV \cdot P'V = BF \cdot FC : AF^2.$$

Hence

$$QV^2 : PV \cdot P'V = PL : PP'$$

$$= VR : P'V$$

$$= PV \cdot VR : PV \cdot P'V.$$

$$\therefore QV^2 = PV \cdot VR.$$

Thus the square on the ordinate is equal to a rectangle whose height is equal to the abscissa and whose base lies along the fixed straight line PL but falls short of it ($\epsilon\lambda\lambda\epsilon\acute{\iota}\pi\epsilon\iota$) by a length equal to the difference between VR and PL *. The section is therefore called an ELLIPSE.

As before, PL is called the **latus rectum**, or the **parameter** of the ordinates to the diameter PP' , and PP' itself is called the **transverse** (with or without the addition of **diameter** or **side** of the **figure**, as explained in the last proposition).

PL will henceforth be denoted by p .

[COR. It follows from the proportion

$$QV^2 : PV \cdot P'V = PL : PP'$$

that, for any fixed diameter PP' ,

$$QV^2 : PV \cdot P'V \text{ is a constant ratio,}$$

or QV^2 varies as $PV \cdot P'V$.]

* Apollonius describes the rectangle PR as *applied to the latus rectum but falling short by a figure similar and similarly situated to that contained by PP' and PL* , i.e. falling short of the rectangle VL by the rectangle LR .

If $QV=y$, $PV=x$, $PL=p$, and $PP'=d$,

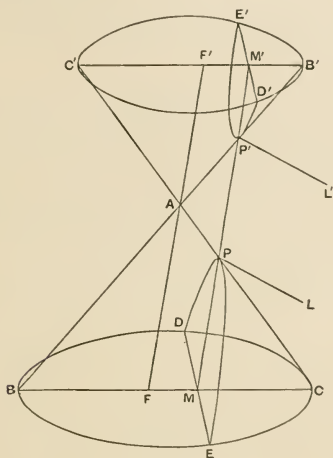
$$y^2 = px - \frac{p}{d} \cdot x^2.$$

Thus Apollonius' enunciation simply expresses the Cartesian equation referred to a diameter and the tangent at its extremity as (oblique) axes.

Proposition 4.

[I. 14.]

If a plane cuts both parts of a double cone and does not pass through the apex, the sections of the two parts of the cone will both be hyperbolas which will have the same diameter and equal latera recta corresponding thereto. And such sections are called OPPOSITE BRANCHES.



Let BC be the circle about which the straight line generating the cone revolves, and let $B'C'$ be any parallel section cutting the opposite half of the cone. Let a plane cut both halves of the cone, intersecting the base BC in the straight line DE and the plane $B'C'$ in $D'E'$. Then $D'E'$ must be parallel to DE .

Let BC be that diameter of the base which bisects DE at right angles, and let a plane pass through BC and the apex A cutting the circle $B'C'$ in $B'C'$, which will therefore be a diameter of that circle and will cut $D'E'$ at right angles, since $B'C'$ is parallel to BC , and $D'E'$ to DE .

Let FAF' be drawn through A parallel to MM' , the straight line joining the middle points of DE , $D'E'$ and meeting CA , $B'A$ respectively in P , P' .

Draw perpendiculars PL , $P'L'$ to MM' in the plane of the section and of such length that

$$PL : PP' = BF \cdot FC : AF^2,$$

$$P'L' : P'P = B'F' \cdot F'C' : AF'^2.$$

Since now MP , the diameter of the section DPE , when produced, meets BA produced beyond the apex, the section DPE is a hyperbola.

Also, since $D'E'$ is bisected at right angles by the base of the axial triangle $AB'C'$, and $M'P$ in the plane of the axial triangle meets $C'A$ produced beyond the apex A , the section $D'P'E'$ is also a hyperbola.

And the two hyperbolas have the same diameter $MPP'M'$.

It remains to prove that $PL = P'L'$.

We have, by similar triangles,

$$BF : AF = B'F' : AF',$$

$$FC : AF = F'C' : AF'.$$

$$\therefore BF \cdot FC : AF^2 = B'F' \cdot F'C' : AF'^2.$$

Hence

$$PL : PP' = P'L' : P'P.$$

$$\therefore PL = P'L'.$$

THE DIAMETER AND ITS CONJUGATE.

Proposition 5.

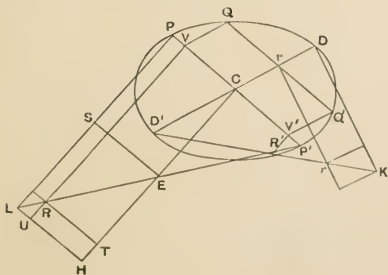
[I. 15.]

If through C , the middle point of the diameter PP' of an ellipse, a double ordinate DCD' be drawn to PP' , DCD' will bisect all chords parallel to PP' , and will therefore be a diameter the ordinates to which are parallel to PP' .

In other words, if the diameter bisect all chords parallel to a second diameter, the second diameter will bisect all chords parallel to the first.

Also the parameter of the ordinates to DCD' will be a third proportional to DD' , PP' .

(1) Let QV be any ordinate to PP' , and through Q draw QQ' parallel to PP' meeting DD' in v and the ellipse in Q' ; and let $Q'V'$ be the ordinate drawn from Q' to PP' .



Then, if PL is the parameter of the ordinates, and if $P'L$ is joined and $VR, CE, V'R'$ drawn parallel to PL to meet $P'L$, we have [Prop. 3]

$$QV^2 = PV \cdot VR,$$

$$QV'^2 = PV' \cdot V'R';$$

and $QV = Q'V'$, because QV is parallel to $Q'V'$ and QQ' to PP' .

$$\therefore PV \cdot VR = PV' \cdot V'R'.$$

$$\text{Hence } PV : PV' = V'R' : VR = P'V' : P'V.$$

$$\therefore PV : PV' \sim PV = P'V' : P'V \sim P'V',$$

or

$$PV : VV' = P'V' : VV'.$$

$$\therefore PV = P'V'.$$

$$\text{Also } CP = CP'.$$

$$\text{By subtraction, } CV = CV',$$

and $\therefore Qv = vQ'$, so that QQ' is bisected by DD' .

(2) Draw DK at right angles to DD' and of such a length that $DD' : PP' = PP' : DK$. Join $D'K$ and draw vr parallel to DK to meet $D'K$ in r .

Also draw TR, LUH and ES parallel to PP' .

Then, since $PC = CP', PS = SL$ and $CE = EH$;

$\therefore$ the parallelogram $(PE) = (SH)$.

$$\text{Also } (PR) = (VS) + (SR) = (SU) + (RH).$$

$$\text{By subtraction, } (PE) - (PR) = (RE);$$

$$\therefore CD^2 - QV^2 = RT \cdot TE.$$

$$\text{But } CD^2 - QV^2 = CD^2 - Cv^2 = D'v \cdot vD.$$

$$\therefore D'v \cdot vD = RT \cdot TE \dots\dots\dots(A).$$

$$\text{Now } DD' : PP' = PP' : DK, \text{ by hypothesis.}$$

$$\begin{aligned} \therefore DD' : DK &= DD'^2 : PP'^2 \\ &= CD^2 : CP^2 \\ &= PC \cdot CE : CP^2 \\ &= RT \cdot TE : RT^2, \end{aligned}$$

and

$$\begin{aligned} DD' : DK &= D'v : vr \\ &= D'v \cdot vD : vD \cdot vr; \end{aligned}$$

$$\therefore D'v \cdot vD : Dv \cdot vr = RT \cdot TE : RT^2.$$

But

$$D'v \cdot vD = RT \cdot TE, \text{ from (A) above;}$$

$$\therefore Dv \cdot vr = RT^2 = CV^2 = Qv^2.$$

Thus DK is the parameter of the ordinates to DD' , such as Qv .

Therefore the parameter of the ordinates to DD' is a third proportional to DD' , PP' .

$$\begin{aligned}\text{COR. We have} \quad CD^2 &= PC \cdot CE \\ &= \frac{1}{2}PP' \cdot \frac{1}{2}PL;\end{aligned}$$

$$\therefore DD'^2 = PP' \cdot PL,$$

or

$$PP' : DD' = DD' : PL,$$

and PL is a third proportional to PP' , DD' .

Thus the relations of PP' , DD' and the corresponding parameters are reciprocal.

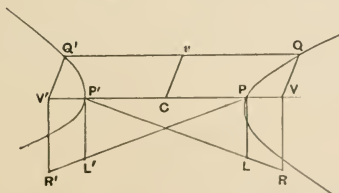
DEF. Diameters such as PP' , DD' , each of which bisects all chords parallel to the other, are called **conjugate diameters**.

Proposition 6.

[I. 16.]

If from the middle point of the diameter of a hyperbola with two branches a line be drawn parallel to the ordinates to that diameter, the line so drawn will be a diameter conjugate to the former one.

If any straight line be drawn parallel to PP' , the given diameter, and meeting the two branches of the hyperbola in Q , Q' respectively, and if from C , the middle point of PP' , a straight line be drawn parallel to the ordinates to PP' meeting QQ in v , we have to prove that QQ is bisected in v .



Let QV , $Q'V'$ be ordinates to PP' , and let PL , $P'L'$ be the parameters of the ordinates in each branch so that [Prop. 4]

$PL = P'L$. Draw VR , $V'R'$ parallel to PL , $P'L'$, and let PL , $P'L$ be joined and produced to meet $V'R'$, VR respectively in R' , R .

$$\begin{aligned}\text{Then we have} \quad QV^2 &= PV \cdot VR, \\ Q'V'^2 &= P'V' \cdot V'R' .\end{aligned}$$

$$\therefore PV \cdot VR = P'V' \cdot V'R', \text{ and } V'R' : VR = PV : P'V'.$$

$$\text{Also } PV' : V'R' = PP' : P'L' = P'P : PL = P'V : VR.$$

$$\begin{aligned}\therefore PV' : P'V &= V'R' : VR \\ &= PV : P'V', \text{ from above;}\end{aligned}$$

$$\therefore PV' : PV = P'V : P'V',$$

$$\text{and} \quad PV' + PV : PV = P'V + P'V' : P'V',$$

$$\text{or} \quad VV' : PV = VV' : P'V';$$

$$\therefore PV = P'V'.$$

$$\text{But} \quad CP = CP';$$

$$\therefore \text{ by addition, } CV = CV',$$

$$\text{or} \quad Qv = Q'v.$$

Hence Cv is a diameter conjugate to PP' .

[More shortly, we have, from the proof of Prop. 2,

$$QV^2 : PV \cdot P'V = PL : PP',$$

$$Q'V'^2 : P'V' \cdot PV' = P'L' : PP',$$

$$\text{and} \quad QV = Q'V', \quad PL = P'L';$$

$$\therefore PV \cdot P'V = P'V' \cdot P'V', \text{ or } PV : PV' = P'V' : P'V,$$

$$\text{whence, as above,} \quad PV = P'V'.]$$

DEF. The middle point of the diameter of an ellipse or hyperbola is called the **centre**; and the straight line drawn parallel to the ordinates of the diameter, of a length equal to the mean proportional between the diameter and the parameter, and bisected at the centre, is called the **secondary diameter** (*δευτέρα διάμετρος*).

Proposition 7.

[I. 20.]

In a parabola the square on an ordinate to the diameter varies as the abscissa.

This is at once evident from Prop. 1.

Proposition 8.

[I. 21.]

In a hyperbola, an ellipse, or a circle, if QV be any ordinate to the diameter PP' ,

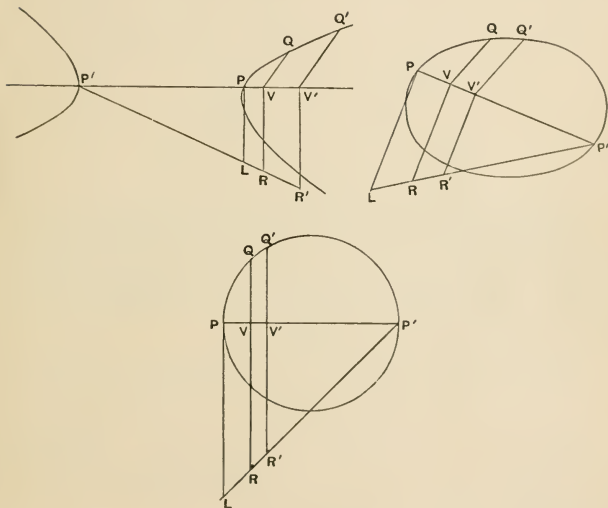
$$QV^2 \propto PV \cdot P'V.$$

[This property is at once evident from the proportion

$$QV^2 : PV \cdot P'V = PL : PP'$$

obtained in the course of Props. 2 and 3; but Apollonius gives a separate proof, starting from the property $QV^2 = PV \cdot VR$ which forms the basis of the definition of the conic, as follows.]

Let QV , $Q'V'$ be two ordinates to the diameter PP' .



Then

$$QV^2 = PV \cdot VR,$$

$$Q'V'^2 = P'V' \cdot V'R';$$

$$\begin{aligned} \therefore QV^2 : PV \cdot P'V &= PV \cdot VR : PV \cdot P'V \\ &= VR : P'V = PL : PP'. \end{aligned}$$

Similarly $Q'V'^2 : PV'.P'V' = PL : PP'$.

$$\therefore QV^2 : Q'V'^2 = PV.P'V : PV'.P'V';$$

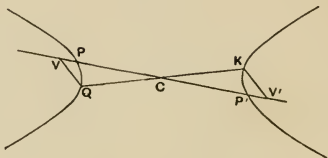
and $QV^2 : PV.P'V$ is a constant ratio,

or $QV^2 \propto PV.P'V$.

Proposition 9.

[I. 29.]

If a straight line through the centre of a hyperbola with two branches meet one branch, it will, if produced, meet the other also.



Let PP' be the given diameter and C the centre. Let CQ meet one branch in Q . Draw the ordinate QV to PP' , and set off CV' along PP' on the other side of the centre equal to CV . Let $V'K$ be the ordinate to PP' through V' . We shall prove that QCK is a straight line.

Since $CV = CV'$, and $CP = CP'$, it follows that $PV = P'V'$;

$$\therefore PV.P'V = P'V'.PV'.$$

But $QV^2 : KV'^2 = PV.P'V : P'V'.PV'$. [Prop. 8]

$\therefore QV = KV'$; and QV, KV' are parallel, while $CV = CV'$.

Therefore QCK is a straight line.

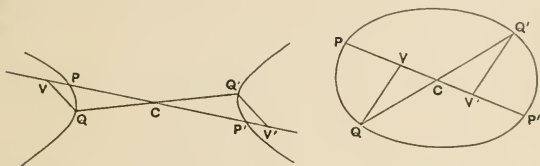
Hence QC , if produced, will cut the opposite branch.

Proposition 10.

[I. 30.]

In a hyperbola or an ellipse any chord through the centre is bisected at the centre.

Let PP' be the diameter and C the centre; and let QQ' be any chord through the centre. Draw the ordinates QV , $Q'V'$ to the diameter PP' .



Then

$$PV \cdot P'V : P'V' \cdot PV' = QV^2 : Q'V'^2 \\ = CV^2 : CV'^2, \text{ by similar triangles.}$$

$$\therefore CV^2 \pm PV \cdot P'V : CV^2 = CV'^2 \pm P'V' \cdot PV' : CV'^2$$

(where the upper sign applies to the ellipse and the lower to the hyperbola).

$$\therefore CP^2 : CV^2 = CP'^2 : CV'^2.$$

But

$$CP^2 = CP'^2;$$

$$\therefore CV^2 = CV'^2, \text{ and } CV = CV'.$$

And QV , $Q'V'$ are parallel;

$$\therefore CQ = CQ'.$$

TANGENTS.

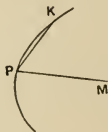
Proposition 11.

[I. 17, 32.]

If a straight line be drawn through the extremity of the diameter of any conic parallel to the ordinates to that diameter, the straight line will touch the conic, and no other straight line can fall between it and the conic.

It is first proved that the straight line drawn in the manner described will fall without the conic.

For, if not, let it fall within it, as PK , where PM is the given diameter. Then KP , being drawn from a point K on the conic parallel to the ordinates to PM , will meet PM and will be bisected by it. But KP produced falls without the conic; therefore it will not be bisected at P .



Therefore the straight line PK must fall without the conic and will therefore touch it.

It remains to be proved that no straight line can fall between the straight line drawn as described and the conic.

(1) Let the conic be a *parabola*, and let PF be parallel to the ordinates to the diameter PV . If possible, let PK fall between PF and the parabola, and draw KV parallel to the ordinates, meeting the curve in Q .

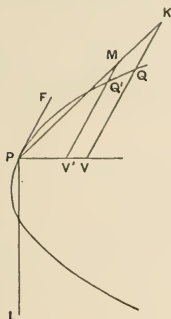
$$\begin{aligned} \text{Then} \quad KV^2 : PV^2 &> QV^2 : PV^2 \\ &> PL \cdot PV : PV^2 \\ &> PL : PV. \end{aligned}$$

Let V' be taken on PV such that

$$KV^2 : PV^2 = PL : PV',$$

and let $V'Q'M$ be drawn parallel to QV , meeting the curve in Q' and PK in M .

$$\begin{aligned}
 \text{Then } KV^2 : PV^2 &= PL : PV' \\
 &= PL . PV' : PV'^2 \\
 &= Q'V'^2 : PV'^2,
 \end{aligned}$$



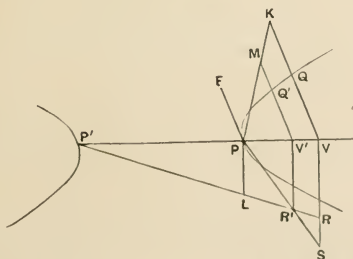
and $KV^2 : PV^2 = MV'^2 : PV'^2$, by parallels.

Therefore $MV'^2 = Q'V'^2$, and $MV' = Q'V'$.

Thus PK cuts the curve in Q' , and therefore does not fall outside it: which is contrary to the hypothesis.

Therefore no straight line can fall between PF' and the curve.

(2) Let the curve be a *hyperbola* or an *ellipse* or a *circle*.

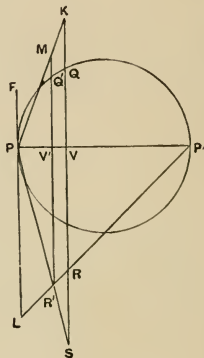
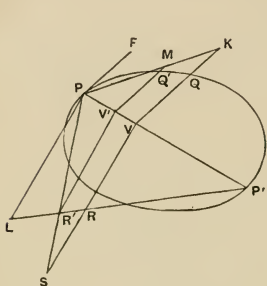


Let PF be parallel to the ordinates to PP' , and, if possible, let PK fall between PF and the curve. Draw KV parallel to the ordinates, meeting the curve in Q , and draw VR per-

pendicular to PV . Join $P'L$ and let it (produced if necessary) meet VR in R .

Then $QV'^2 = PV \cdot VR$, so that $KV^2 > PV \cdot VR$.

Take a point S on VR produced such that $KV^2 = PV \cdot VS$. Join PS and let it meet $P'R$ in R' . Draw $R'V'$ parallel to PL meeting PV in V' , and through V' draw $V'Q'M$ parallel to QV , meeting the curve in Q' and PK in M .



Now $KV^2 = PV \cdot VS$,

$$\therefore VS : KV = KV : PV,$$

so that

$$VS : PV = KV^2 : PV^2.$$

Hence, by parallels,

$$V'R' : PV' = MV'^2 : PV'^2,$$

or MV' is a mean proportional between PV' , $V'R'$,

i.e.

$$MV'^2 = PV' \cdot V'R'$$

$$= Q'V'^2, \text{ by the property of the conic.}$$

$$\therefore MV' = Q'V'.$$

Thus PK cuts the curve in Q' , and therefore does not fall outside it: which is contrary to the hypothesis.

Hence no straight line can fall between PF' and the curve.

Proposition 12.

[I. 33, 35.]

If a point T be taken on the diameter of a parabola outside the curve and such that $TP = PV$, where V is the foot of the ordinate from Q to the diameter PV , the line TQ will touch the parabola.

We have to prove that the straight line TQ or TQ produced does not fall within the curve on either side of Q .

For, if possible, let K , a point on TQ or TQ produced, fall within the curve*, and through K draw $Q'KV'$ parallel to an ordinate and meeting the diameter in V' and the curve in Q' .

$$\begin{aligned} \text{Then } Q'V'^2 : QV^2 \\ &> KV'^2 : QV^2, \text{ by hypothesis,} \\ &> TV'^2 : TV^2. \\ \therefore PV' : PV &> TV'^2 : TV^2. \end{aligned}$$

Hence

$$4TP \cdot PV' : 4TP \cdot PV > TV'^2 : TV^2,$$

and, since $TP = PV$,

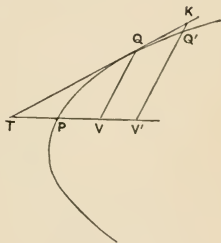
$$\begin{aligned} 4TP \cdot PV &= TV^2, \\ \therefore 4TP \cdot PV' &> TV'^2. \end{aligned}$$

But, since by hypothesis TV' is not bisected in P ,

$$4TP \cdot PV' < TV'^2,$$

(which is absurd.

Therefore TQ does not at any point fall within the curve, and is therefore a tangent.



* Though the proofs of this proposition and the next follow in form the method of *reductio ad absurdum*, it is easily seen that they give in fact the direct demonstration that, if K is any point on the tangent other than Q , the point of contact, K lies outside the curve because, if $KQ'V'$ be parallel to QV , it is proved that $KV' > Q'V'$. The figures in both propositions have accordingly been drawn in accordance with the facts instead of representing the incorrect assumption which leads to the absurdity in each case.

Conversely, if the tangent at Q meet the diameter produced outside the curve in the point T , $TP = PV$. Also no straight line can fall between TQ and the curve.

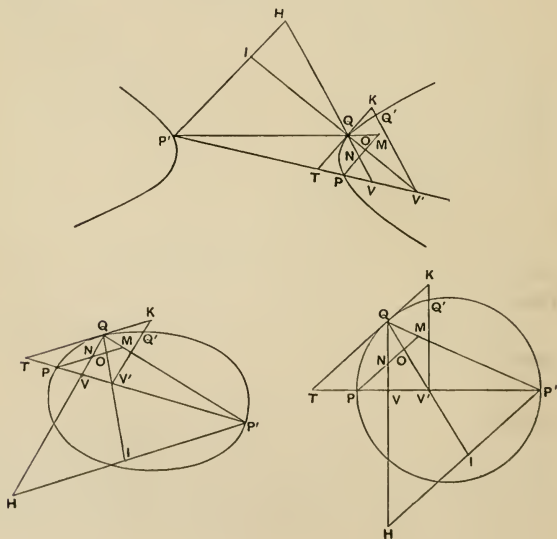
[Apollonius gives a separate proof of this, using the method of *reductio ad absurdum*.]

Proposition 13.

[I. 34, 36.]

In a hyperbola, an ellipse, or a circle, if PP' be the diameter and QV an ordinate to it from a point Q , and if a point T be taken on the diameter but outside the curve such that $TP : TP' = PV : VP'$, then the straight line TQ will touch the curve.

We have to prove that no point on TQ or TQ produced falls within the curve.



If possible, let a point K on TQ or $T'Q$ produced fall within the curve*; draw $Q'KV'$ parallel to an ordinate meeting the curve in Q' . Join $P'Q$, $V'Q$, producing them if necessary, and draw through P' , P parallels to TQ meeting $V'Q$, VQ in I , O and H , N respectively. Also let the parallel through P meet $P'Q$ in M .

Now, by hypothesis, $P'V : PV = TP' : TP$;

$$\therefore \text{by parallels, } P'H : PN = P'Q : QM \\ = P'H : NM.$$

Therefore $PN = NM$.

Hence $PN \cdot NM > PO \cdot OM$,

or $NM : MO > OP : PN$;

$$\therefore P'H : P'I > OP : PN,$$

or $P'H \cdot PN > P'I \cdot OP$.

It follows that $P'H \cdot PN : TQ^2 > P'I \cdot OP : TQ^2$;

$\therefore$ by similar triangles

$$P'V \cdot PV : TV^2 > P'V' \cdot PV' : TV'^2,$$

or $P'V \cdot PV : P'V' \cdot PV' > TV^2 : TV'^2$;

$$\therefore QV^2 : Q'V'^2 > TV^2 : TV'^2$$

$$> QV^2 : KV'^2.$$

$\therefore Q'V' < KV'$, which is contrary to the hypothesis.

Thus TQ does not cut the curve, and therefore it touches it.

Conversely, if the tangent at a point Q meet the diameter PP' outside the section in the point T , and QV is the ordinate from Q ,

$$TP : TP' = PV : VP'.$$

Also no other straight line can fall between TQ and the curve.

[This again is separately proved by Apollonius by a simple *reductio ad absurdum*.]

* See the note on the previous proposition.

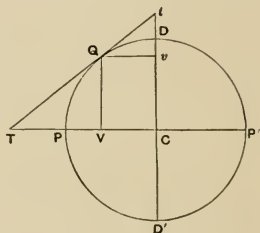
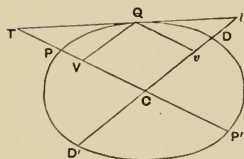
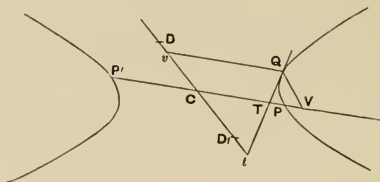
Proposition 14.

[I. 37, 39.]

In a hyperbola, an ellipse, or a circle, if QV be an ordinate to the diameter PP' , and the tangent at Q meet PP' in T , then

$$(1) \quad CV \cdot CT = CP^2,$$

$$(2) \quad QV^2 : CV \cdot VT = p : PP' \text{ [or } CD^2 : CP^2].$$



(1) Since QT is the tangent at Q ,

$$TP : TP' = PV : P'V, \quad [\text{Prop. 13}]$$

$$\therefore TP + TP' : TP \sim TP' = PV + P'V : PV \sim P'V;$$

thus, for the *hyperbola*,

$$2CP : 2CT = 2CV : 2CP;$$

and for the *ellipse* or *circle*,

$$2CT : 2CP = 2CP : 2CV;$$

therefore for all three curves

$$CV \cdot CT = CP^2.$$

(2) Since $CV : CP = CP : CT$,

$$CV \sim CP : CV = CP \sim CT : CP,$$

whence $PV : CV = PT : CP$,

or $PV : PT = CV : CP$.

$$\therefore PV : PV + PT = CV : CV + CP,$$

or $PV : VT = CV : P'V$,

and $CV \cdot VT = PV \cdot P'V$.

But $QV^2 : PV \cdot P'V = p : PP'$ (or $CD^2 : CP^2$). [Prop. 8]

$$\therefore QV^2 : CV \cdot VT = p : PP' \text{ (or } CD^2 : CP^2).$$

COR. It follows at once that $QV : VT$ is equal to the ratio compounded of the ratios $p : PP'$ (or $CD^2 : CP^2$) and $CV : QV$.

Proposition 15.

[I. 38, 40.]

If Qv be the ordinate to the diameter conjugate to PP' , and QT , the tangent at Q , meet that conjugate diameter in t , then

$$(1) \quad Cv \cdot Ct = CD^2,$$

$$(2) \quad Qv^2 : Cv \cdot vt = PP' : p \text{ [or } CP^2 : CD^2],$$

$$(3) \quad tD : tD' = vD' : vD \text{ for the hyperbola,}$$

and $tD : tD' = vD : vD'$ for the ellipse and circle.

Using the figures drawn for the preceding proposition, we have (1)

$$QV^2 : CV \cdot VT = CD^2 : CP^2. \quad [\text{Prop. 14}]$$

But $QV : CV = Cv : CV$,

and $QV : VT = Ct : CT$;

$$\therefore QV^2 : CV \cdot VT = Cv \cdot Ct : CV \cdot CT.$$

Hence $Cv \cdot Ct : CV \cdot CT = CD^2 : CP^2$.

And $CV \cdot CT = CP^2$; [Prop. 14]

$$\therefore Cv \cdot Ct = CD^2.$$

(2) As before,

$$QV^2 : CV \cdot VT = CD^2 : CP^2 \text{ (or } p : PP').$$

But $QV : CV = Cv : Qv$,

and

$$QV : VT = vt : Qv;$$

$$\therefore QV^2 : CV \cdot VT = Cv \cdot vt : Qv^2.$$

Hence

$$\begin{aligned} Qv^2 : Cv \cdot vt &= CP^2 : CD^2 \\ &= PP' : p. \end{aligned}$$

(3) Again,

$$Ct \cdot Cv = CD^2 = CD \cdot CD';$$

$$\therefore Ct : CD = CD' : Cv,$$

and $\therefore Ct + CD : Ct \sim CD = CD' + Cv : CD' \sim Cv$.

Thus

$$tD : tD' = vD' : vD \text{ for the hyperbola,}$$

and

$$tD' : tD = vD' : vD \text{ for the ellipse and circle.}$$

COR. It follows from (2) that $Qv : Cv$ is equal to the ratio compounded of the ratios $PP' : p$ (or $CP^2 : CD^2$) and $vt : Qv$.

PROPOSITIONS LEADING TO THE REFERENCE OF
A CONIC TO ANY NEW DIAMETER AND THE
TANGENT AT ITS EXTREMITY.

Proposition 16.

[I. 41.]

In a hyperbola, an ellipse, or a circle, if equiangular parallelograms (VK), (PM) be described on QV, CP respectively, and their sides are such that $\frac{QV}{QK} = \frac{p}{PP'} \cdot \frac{CP}{CM}$ [i.e. $\frac{CD^2}{CP^2} \cdot \frac{CP}{CM}$], and if (VN) be the parallelogram on CV similar and similarly situated to (PM), then

$$(VN) \pm (VK) = (PM),$$

the lower sign applying to the hyperbola.

Suppose O to be so taken on KQ produced that

$$QV : QO = p : PP',$$

so that

$$QV^2 : QV \cdot QO = QV^2 : PV \cdot PV'.$$

Thus

$$QV \cdot QO = PV \cdot P'V \dots\dots\dots (1).$$

Also $QV : QK = (CP : CM) \cdot (p : PP') = (CP : CM) \cdot (QV : QO)$,

or $(QV : QO) \cdot (QO : QK) = (CP : CM) \cdot (QV : QO)$;

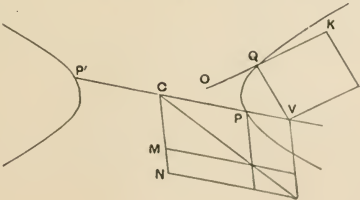
$$\therefore QO : QK = CP : CM \dots\dots\dots (2).$$

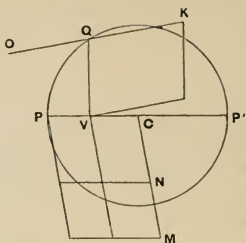
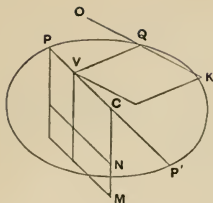
But

$$QO : QK = QV \cdot QO : QV \cdot QK$$

and

$$CP : CM = CP^2 : CP \cdot CM;$$





$$\begin{aligned}\therefore CP^2 : CP \cdot CM &= QV \cdot QO : QV \cdot QK \\ &= PV \cdot P'V : QV \cdot QK, \text{ from (1).}\end{aligned}$$

Therefore, since PM, VK are equiangular,

$$CP^2 : PV \cdot P'V = (PM) : (VK) \dots\dots\dots (3).$$

Hence $CP^2 \mp PV \cdot P'V : CP^2 = (PM) \mp (VK) : (PM)$,
where the upper sign applies to the *ellipse* and *circle* and the lower to the *hyperbola*.

$$\therefore CV^2 : CP^2 = (PM) \mp (VK) : (PM),$$

$$\text{and hence } (VN) : (PM) = (PM) \mp (VK) : (PM),$$

$$\text{so that } (VN) = (PM) \mp (VK),$$

$$\text{or } (VN) \pm (VK) = (PM).$$

[The above proof is reproduced as given by Apollonius in order to show his method of dealing with a somewhat complicated problem by purely geometrical means. The proposition is more shortly proved by a method more akin to algebra as follows.

$$\text{We have } QV^2 : CV^2 \sim CP^2 = CD^2 : CP^2,$$

$$\text{and } \frac{QV}{QK} = \frac{CD^2 \cdot CP}{CP^2 \cdot CM}, \quad \text{or } QV = QK \cdot \frac{CD^2}{CP \cdot CM};$$

$$\therefore QV \cdot QK \cdot \frac{CD^2}{CP \cdot CM} : CV^2 \sim CP^2 = CD^2 : CP^2,$$

$$\text{or } QV \cdot QK = CP \cdot CM \left(\frac{CV^2}{CP^2} \sim 1 \right).$$

$$\therefore (VK) = (VN) \sim (PM),$$

$$\text{or } (VN) \pm (VK) = (PM).]$$

Proposition 17.

[I. 42.]

In a parabola, if QV , RW be ordinates to the diameter through P , and QT , the tangent at Q , and RU parallel to it meet the diameter in T , U respectively; and if through Q a parallel to the diameter be drawn meeting RW produced in F and the tangent at P in E , then

$\triangle RUW = \text{the parallelogram } (EW)$.

Since QT is a tangent,

$$TV = 2PV; [\text{Prop. 12}]$$

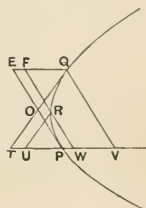
$$\therefore \triangle QTV = (EV) \dots \dots (1).$$

Also $QV^2 : RW^2 = PV : PW;$

$$\therefore \triangle QTV : \triangle RUW = (EV) : (EW),$$

and $\triangle QTV = (EV)$, from (1);

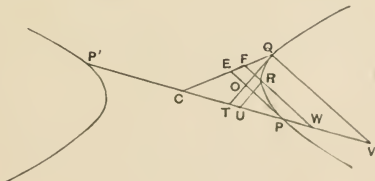
$$\therefore \triangle RUW = (EW).$$

**Proposition 18.**

[I. 43, 44.]

In a hyperbola, an ellipse, or a circle, if the tangent at Q and the ordinate from Q meet the diameter in T , V , and if RW be the ordinate from any point R and RU be parallel to QT ; if also RW and the parallel to it through P meet CQ in F , E respectively, then

$$\triangle CFW \sim \triangle CPE = \triangle RUW.$$



Hence, if x, y are the coordinates of R ,

$$\alpha x^2 + \beta xy + \gamma y^2 = A,$$

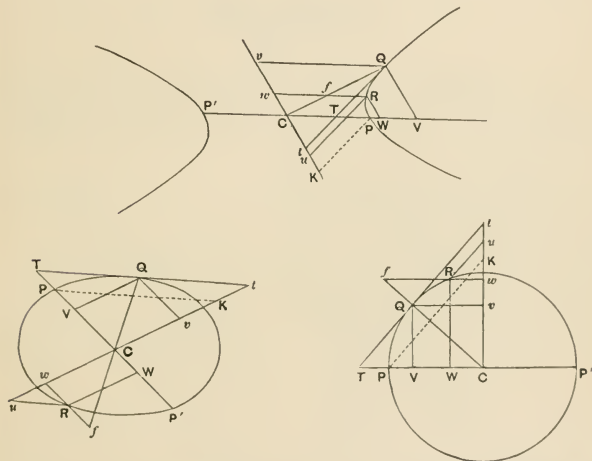
which is the Cartesian equation referred to the centre as origin and any two diameters as axes.]

Proposition 19.

[I. 45.]

If the tangent at Q and the straight line through R parallel to it meet the secondary diameter in t, u respectively, and Qv, Rw be parallel to the diameter PP' , meeting the secondary diameter in v, w ; if also Rw meet CQ in f , then

$$\triangle Cfw = \triangle Ruw \sim \triangle CQt.$$



[Let PK be drawn parallel to Qt meeting the secondary diameter in K , so that the triangle CPK is similar to the triangle vQt .]

We have [Prop. 14, Cor.]

$$\begin{aligned} QV : CV &= (p : PP') \cdot (VT : QV) \\ &= (p : PP') \cdot (Qv : vt), \end{aligned}$$

and the triangles QvC , Qvt are the halves of equiangular parallelograms on Cv (or QV) and Qv (or CV) respectively: also CPK is the triangle on CP similar to Qvt .

Therefore [by Prop. 16], $\triangle CQv = \triangle Qvt \sim \triangle CPK$,

and clearly $\triangle CQv = \triangle Qvt \sim \triangle CQt$;

$$\therefore \triangle CPK = \triangle CQt.$$

Again, the triangle Cfw is similar to the triangle CQv , and the triangle Rwu to the triangle Qvt . Therefore, for the ordinate RW ,

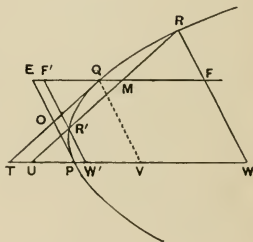
$$\triangle Cfw = \triangle Ruw \sim \triangle CPK = \triangle Ruw \sim \triangle CQt.$$

Proposition 20.

[I. 46.]

In a parabola the straight line drawn through any point parallel to the diameter bisects all chords parallel to the tangent at the point.

Let RR' be any chord parallel to the tangent at Q and let it meet the diameter PV in U . Let QM drawn parallel to PV meet RR' in M , and the straight lines drawn ordinate-wise through R , R' , P in F , F' , E respectively.



We have then [Prop. 17]

$$\triangle RUW = \square EW,$$

and $\triangle R'UW' = \square EW'.$

Therefore, by subtraction, the figure $RR'W'W = \square F'W$. Take away the common part $R'W'WFM$, and we have

$$\triangle RMF = \triangle R'MF'.$$

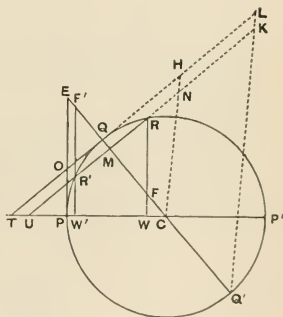
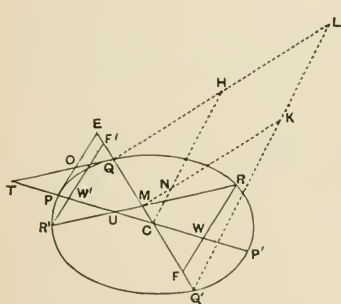
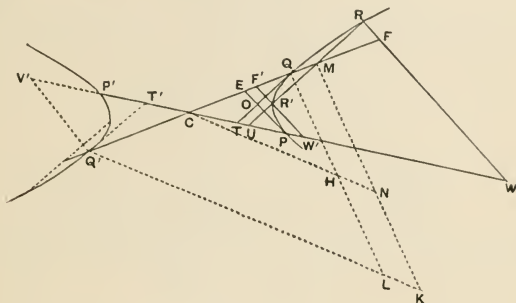
And $R'F'$ is parallel to RF ;

$$\therefore RM = MR'.$$

Proposition 21.

[I. 47, 48.]

In a hyperbola, an ellipse, or a circle, the line joining any point to the centre bisects the chords parallel to the tangent at the point.



If QT be the given tangent and RR' any parallel chord, let RW , $R'W'$, PE be drawn ordinate-wise to PP' , and let CQ meet them in F , F' , E respectively. Further let CQ meet RR' in M .

Then we have [by Prop. 18]

$$\triangle CFW \sim \triangle CPE = \triangle RUW,$$

and

$$\triangle CF'W' \sim \triangle CPE = \triangle R'UW'.$$

Proposition 22.

[I. 49.]

Let the tangent to a parabola at P , the extremity of the original diameter, meet the tangent at any point Q in O , and the parallel through Q to the diameter in E ; and let RR' be any chord parallel to the tangent at Q meeting PT in U and EQ produced in M ; then, if p' be taken such that

$$OQ : QE = p' : 2QT,$$

it is to be proved that

$$RM^2 = p' \cdot QM.$$

In the figure of Prop. 20 draw the ordinate QV .

Then we have, by hypothesis,

$$OQ : QE = p' : 2TQ.$$

Also

$$QE = PV = TP.$$

Therefore the triangles EOQ , POT are equal.

Add to each the figure $QOPWF$;

$\therefore$ the quadrilateral $QTFW = \square(EW) = \triangle RUW$. [Prop. 17]

Subtract the quadrilateral $MUWF$;

$$\therefore \square QU = \triangle RMF,$$

and hence

$$RM \cdot MF = 2QM \cdot QT \dots \dots \dots (1).$$

But

$$RM : MF = OQ : QE = p' : 2QT,$$

or

$$RM^2 : RM \cdot MF = p' \cdot QM : 2QM \cdot QT.$$

Therefore, from (1), $RM^2 = p' \cdot QM$.

Proposition 23.

[I. 50.]

If in a hyperbola, an ellipse, or a circle, the tangents at P , Q meet in O , and the tangent at P meet the line joining Q to the centre in E ; if also a length $QL (= p')$ be taken such that

$$OQ : QE = QL : 2TQ$$

and erected perpendicular to QC ; if further $Q'L$ be joined (where Q' is on QC produced and $CQ = CQ'$), and MK be drawn parallel to QL to meet $Q'L$ in K (where M is the point of concurrence of CQ and RR' , a chord parallel to the tangent at Q): then it is to be proved that

$$RM^2 = QM \cdot MK.$$

In the figures of Prop. 21 draw CHN parallel to $Q'L$, meeting QL in H and MK in N , and let RW be an ordinate to PP' , meeting CQ in F .

Then, since $CQ = CQ'$, $QH = HL$.

$$\begin{aligned} \text{Also} \quad OQ : QE &= QL : 2QT \\ &= QH : QT; \end{aligned}$$

$$\therefore RM : MF = QH : QT \dots\dots\dots(A).$$

Now

$$\triangle RUW = \triangle CFW \sim \triangle CPE = \triangle CFW \sim \triangle CQT^* ;$$

$\therefore$ in the figures as drawn

(1) for the *hyperbola*,

$$\triangle RUW = QTWF,$$

$\therefore$, subtracting $MUWF$,

we have

$$\triangle RMF = QTUM.$$

(2) for the *ellipse* and *circle*,

$$\triangle RUW = \triangle CQT - \triangle CFW ;$$

$$\therefore \triangle CQT = \text{quadrilateral } RUCF ;$$

and, subtracting $\triangle MUC$, we have

$$\triangle RMF = QTUM.$$

$$\therefore RM \cdot MF = QM(QT + MU) \dots\dots\dots(B).$$

* It will be observed that Apollonius here assumes the equality of the two triangles CPE , CQT , though it is not until Prop. 53 [III. 1] that this equality is actually proved. But Eutocius gives another proof of Prop. 18 which, he says, appears in some copies, and which begins by proving these two triangles to be equal by exactly the same method as is used in our text of the later proof. If then the alternative proof is genuine, we have an explanation of the assumption here. If not, we should be tempted to suppose that Apollonius quoted the property as an obvious *limiting case* of Prop. 18 [I. 43, 44] where R coincides with Q ; but this would be contrary to the usual practice of Greek geometers who, no doubt for the purpose of securing greater stringency, preferred to give separate proofs of the limiting cases, though the parallelism of the respective proofs suggests that they were not unaware of the connexion between the general theorem and its limiting cases. Compare Prop. 81 [V. 2], where Apollonius proves separately the case where P coincides with B , though we have for the sake of brevity only mentioned it as a limiting case.

Now $QT : MU = CQ : CM = QH : MN,$

$$\therefore QH + MN : QT + MU = QH : QT \\ = RM : MF \text{ [from (A)]};$$

$$\therefore QM(QH + MN) : QM(QT + MU) = RM^2 : RM \cdot MF;$$

$$\therefore \text{[by (B)]} \quad RM^2 = QM(QH + MN) \\ = QM \cdot MK.$$

The same is true for the opposite branch of the hyperbola. The tangent at Q' is parallel to QT , and $P'E'$ to PE .

[Prop. 21, *Note*.]

$$\therefore O'Q' : Q'E' = OQ : QE = p' : 2QT = p' : 2Q'T',$$

whence the proposition follows.

It results from the propositions just proved that in a *parabola* all straight lines drawn parallel to the original diameter are diameters, and in the *hyperbola* and *ellipse* all straight lines drawn through the centre are diameters; also that the conics can each be referred indifferently to any diameter and the tangent at its extremity as axes.

CONSTRUCTION OF CONICS FROM CERTAIN DATA.

Proposition 24. (Problem.)

[I. 52, 53.]

Given a straight line in a fixed plane and terminating in a fixed point, and another straight line of a certain length, to find a parabola in the plane such that the first straight line is a diameter, the second straight line is the corresponding parameter, and the ordinates are inclined to the diameter at a given angle.

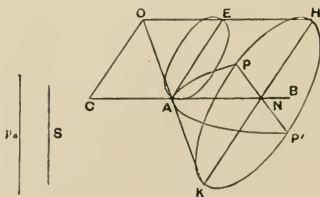
First, let the given angle be a right angle, so that the given straight line is to be the *axis*.

Let AB be the given straight line terminating at A , p_a the given length.

Produce BA to C so that $AC > \frac{p_a}{4}$, and let S be a mean proportional between AC and p_a . (Thus $p_a : AC = S^2 : AC^2$, and $AC > \frac{1}{4}p_a$, whence $AC^2 > \frac{S^2}{4}$, or $2AC > S$, so that it is possible to describe an isosceles triangle having two sides equal to AC and the third equal to S .)

Let AOC be an isosceles triangle in a plane perpendicular to the given plane and such that $AO = AC$, $OC = S$.

Complete the parallelogram $ACOE$, and about AE as diameter, in a plane perpendicular to that of the triangle AOC , describe a circle, and let a cone be drawn with O as



apex and the said circle as base. Then the cone is a right cone because $OE = AC = OA$.

Produce OE , OA to H , K , and draw HK parallel to AE , and let the cone be cut by a plane through HK parallel to the base of the cone. This plane will produce a circular section, and will intersect the original plane in a line PP' , cutting AB at right angles in N .

$$\begin{aligned}\text{Now } p_a : AE &= AE : AO, \text{ since } AE = OC = S, AO = AC; \\ \therefore p_a : AO &= AE^2 : AO^2 \\ &= AE^2 : AO \cdot OE.\end{aligned}$$

Hence PAP' is a parabola in which p_a is the parameter of the ordinates to AB . [Prop. 1]

Secondly, let the given angle not be right. Let the line which is to be the diameter be PM , let p be the length of the parameter, and let MP be produced to F so that $PF = \frac{1}{2}p$. Make the angle FPT equal to the given angle and draw FT perpendicular to TP . Draw TN parallel to PM , and PN perpendicular to TN ; bisect TN in A and draw LAE through A perpendicular to FP meeting PT in O ; and let

$$NA \cdot AL = PN^2.$$

Now with axis AN and parameter AL describe a parabola, as in the first case.

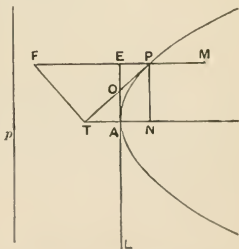
This will pass through P since $PN^2 = LA \cdot AN$. Also PT will be a tangent to it since $AT = AN$. And PM is parallel to AN . Therefore PM is a diameter of the parabola bisecting chords parallel to the tangent PT , which are therefore inclined to the diameter at the given angle.

Again the triangles FTP , OEP are similar;

$$\begin{aligned}\therefore OP : PE &= FP : PT, \\ &= p : 2PT,\end{aligned}$$

by hypothesis.

Therefore p is the parameter of the parabola corresponding to the diameter PM . [Prop. 22]



Proposition 25. (Problem.)

[I. 54, 55, 59.]

Given a straight line AA' in a plane, and also another straight line of a certain length; to find a hyperbola in the plane such that the first straight line is a diameter of it and the second equal to the corresponding parameter, while the ordinates to the diameter make with it a given angle.

First, let the given angle be a right angle.

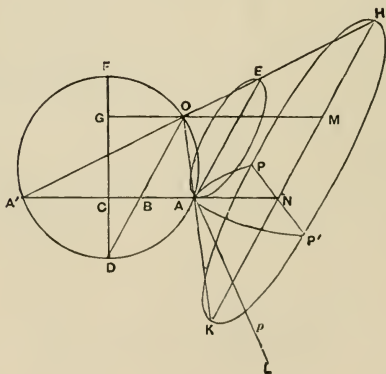
Let AA' , p_a be the given straight lines, and let a circle be drawn through A , A' in a plane perpendicular to the given plane and such that, if C be the middle point of AA' and DF the diameter perpendicular to AA' ,

$$DC : CF \nless AA' : p_a.$$

Then, if $DC : CF = AA' : p_a$, we should use the point F for our construction, but, if not, suppose

$$DC : CG = AA' : p_a \text{ (CG being less than CF).}$$

Draw GO parallel to AA' , meeting the circle in O . Join AO ,



$A'O$, DO . Draw AE parallel to DO meeting $A'O$ produced in E . Let DO meet AA' in B .

Then $\angle OEA = \angle A'OD = \angle AOD = \angle OAE$;

$$\therefore OA = OE.$$

Let a cone be described with O for apex and for base the circle whose diameter is AE and whose plane is perpendicular to that of the circle AOD . The cone will therefore be right, since $OA = OE$.

Produce OE, OA to H, K and draw HK parallel to AE . Draw a plane through HK perpendicular to the plane of the circle AOD . This plane will be parallel to the base of the cone, and the resulting section will be a circle cutting the original plane in PP' at right angles to $A'A$ produced. Let GO meet HK in M .

Then, because NA meets HO produced beyond O , the curve PAP' is a hyperbola.

$$\begin{aligned} \text{And} \quad AA' : p_a &= DC : CG \\ &= DB : BO \\ &= DB \cdot BO : BO^2 \\ &= A'B \cdot BA : BO^2. \end{aligned}$$

$$\text{But} \quad \left. \begin{aligned} A'B : BO &= OM : MH \\ BA : BO &= OM : MK \end{aligned} \right\} \text{by similar triangles,}$$

$$\therefore A'B \cdot BA : BO^2 = OM^2 : HM \cdot MK.$$

$$\text{Hence} \quad AA' : p_a = OM^2 : HM \cdot MK.$$

Therefore p_a is the parameter of the hyperbola PAP' corresponding to the diameter AA' . [Prop. 2]

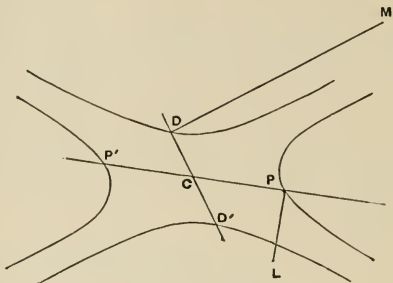
Secondly, let the given angle not be a right angle. Let PP' , p be the given straight lines, CPT the given angle, and C the middle point of PP' . On CP describe a semicircle, and let N be such a point on it that, if NH is drawn parallel to PT to meet CP produced in H ,

$$NH^2 : CH \cdot HP = p : PP'^*.$$

* This construction is assumed by Apollonius without any explanation; but we may infer that it was arrived at by a method similar to that adopted for

From P draw PL perpendicular to PP' and of such a length that $PP' \cdot PL = DD'^2$; then, as in Prop. 25, describe a double hyperbola with diameter PP' and parameter PL and such that the ordinates in it to PP' are parallel to DD' .

Then PP' , DD' are conjugate diameters of the hyperbola so constructed.



Again, draw DM perpendicular to DD' of such a length that $DM \cdot DD' = PP'^2$; and, with DD' as diameter, and DM as the corresponding parameter, describe a double hyperbola such that the ordinates in it to DD' are parallel to PP' .

Then DD' , PP' are conjugate diameters to this hyperbola, and DD' is the transverse, while PP' is the secondary diameter.

The two hyperbolas so constructed are called **conjugate hyperbolas**, and that last drawn is the hyperbola **conjugate** to the first.

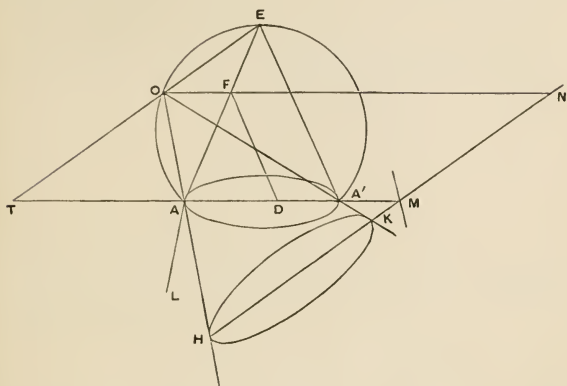
Proposition 27. (Problem.)

[I. 56, 57, 58.]

Given a diameter of an ellipse, the corresponding parameter, and the angle of inclination between the diameter and its ordinates: to find the ellipse.

First, let the angle of inclination be a right angle, and let the diameter be greater than its parameter.

Let AA' be the diameter and AL , a straight line of length p_a perpendicular to it, the parameter.



In a plane at right angles to the plane containing the diameter and parameter describe a segment of a circle on AA' as base,

Take AD on AA' equal to AL . Draw $AE, A'E$ to meet at E , the middle point of the segment. Draw DF parallel to $A'E$ meeting AE in F , and OFN parallel to AA' meeting the circumference in O . Join EO and produce it to meet $A'A$ produced in T . Through any point H on OA produced draw $HKMN$ parallel to OE meeting OA', AA', OF in K, M, N respectively.

Now

$$\begin{aligned}\angle TOA &= \angle OEA + \angle OAE = \angle AA'O + \angle OA'E = \angle AA'E \\ &= \angle EAA' = \angle EOA',\end{aligned}$$

and HK is parallel to OE ,

whence

$$\angle OHK = \angle OKH,$$

and

$$OH = OK.$$

With O as vertex, and as base the circle drawn with diameter HK and in a plane perpendicular to that of the triangle OHK , let a cone be described. This cone will be a right cone because $OH = OK$.

Consider the section of this cone by the plane containing AA' , AL . This will be an ellipse.

$$\begin{aligned} \text{And} \quad p_a : AA' &= AD : AA' \\ &= AF : AE \\ &= TO : TE \\ &= TO^2 : TO \cdot TE \\ &= TO^2 : TA \cdot TA'. \end{aligned}$$

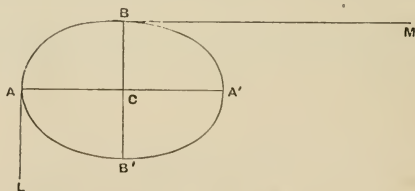
$$\begin{aligned} \text{Now} \quad TO : TA &= HN : NO, \\ \text{and} \quad TO : TA' &= NK : NO, \text{ by similar triangles,} \\ \therefore TO^2 : TA \cdot TA' &= HN \cdot NK : NO^2, \end{aligned}$$

$$\text{so that} \quad p_a : AA' = HN \cdot NK : NO^2,$$

or p_a is the parameter of the ordinates to AA' . [Prop. 3]

Secondly, if the angle of inclination of the ordinates be still a right angle, but the given diameter less than the parameter, let them be BB' , BM respectively.

Let C be the middle point of BB' , and through it draw AA' , perpendicular to BB' and bisected at C , such that



$$AA'^2 = BB' \cdot BM;$$

and draw AL , parallel to BM , such that

$$BM : BB' = AA' : AL;$$

thus $AA' > AL$.

Now with AA' as diameter and AL as the corresponding parameter describe an ellipse in which the ordinates to AA' are perpendicular to it, as above.

This will be the ellipse required, for

(1) it passes through B, B' because

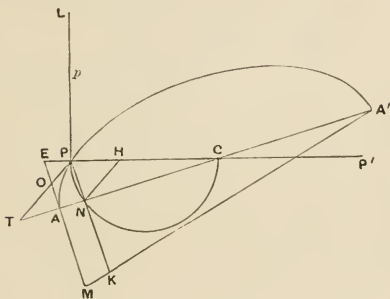
$$\begin{aligned} AL : AA' &= BB' : BM \\ &= BB'^2 : AA'^2 \\ &= BC^2 : AC \cdot CA', \end{aligned}$$

(2)

$$\begin{aligned} BM : BB' &= AC^2 : BC^2 \\ &= AC^2 : BC \cdot CB', \end{aligned}$$

so that BM is the parameter corresponding to BB' .

Thirdly, let the given angle not be a right angle but



equal to the angle CPT , where C is the middle point of the given diameter PP' ; and let PL be the parameter corresponding to PP' .

Take a point N , on the semicircle which has CP for its diameter, such that NH drawn parallel to PT satisfies the relation

$$NH^2 : CH, HP = PL : PP'^*.$$

* This construction like that in Prop. 25 is assumed without explanation. If NH be supposed to meet the other semicircle on CP as diameter in N' , the

Join CN and produce it to meet PT in T . Take A , on CT , such that $CT.CN = CA^2$, and produce AC to A' so that $AC = CA'$. Join PN and produce it to K so that $AN.NK = PN^2$. Join $A'K$. Draw EAM through A perpendicular to CA (and therefore parallel to NK) meeting CP produced in E , PT in O , and $A'K$ produced in M .

Then with axis AA' and parameter AM describe an ellipse as in the first part of this proposition. This will be the ellipse required.

For (1) it will pass through $P \because PN^2 = AN.NK$. For a similar reason, it will pass through $P' \because CP' = CP$ and $CA' = CA$.

(2) PT will be the tangent at $P \because CT.CN = CA^2$.

(3) We have $p : 2CP = NH^2 : CH.HP$,

and

$$2CP : 2PT = CH : HN$$

$$= CH.HP : NH.HP;$$

$$\therefore \text{ex aequali} \quad p : 2PT = NH^2 : NH.HP$$

$$= NH : HP$$

$$= OP : PE.$$

Therefore p is the parameter corresponding to PP' .

[Prop. 23]

problem here reduces to drawing NHN' in a given direction (parallel to PT) so that

$$N'H : NH = PP' : p,$$

and the construction can be effected by the method shown in the note to Prop. 25 *mutatis mutandis*.

ASYMPTOTES.

Proposition 28.

[II. 1, 15, 17, 21.]

(1) *If PP' be a diameter of a hyperbola and p the corresponding parameter, and if on the tangent at P there be set off on each side equal lengths PL, PL' , such that*

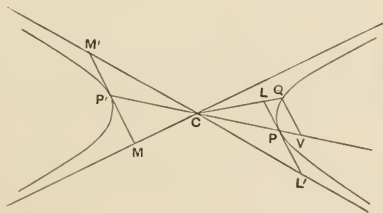
$$PL^2 = PL'^2 = \frac{1}{4}p \cdot PP' [= CD^2],$$

*then CL, CL' produced will not meet the curve in any finite point and are accordingly defined as **asymptotes**.*

(2) *The opposite branches have the same asymptotes.*

(3) *Conjugate hyperbolas have their asymptotes common.*

(1) If possible, let CL meet the hyperbola in Q . Draw the



ordinate QV , which will accordingly be parallel to LL' .

Now

$$\begin{aligned} p : PP' &= p \cdot PP' : PP'^2 \\ &= PL^2 : CP^2 \\ &= QV^2 : CV^2. \end{aligned}$$

But $p : PP' = QV^2 : PV \cdot P'V$.

$$\therefore PV \cdot P'V = CV^2,$$

i.e. $CV^2 - CP^2 = CV^2$, which is absurd.

Therefore CL does not meet the hyperbola in any finite point, and the same is true for CL' .

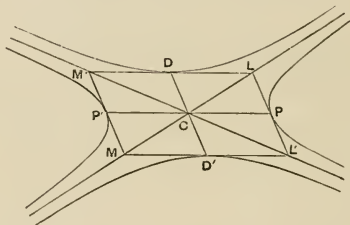
In other words, CL , CL' are **asymptotes**.

(2) If the tangent at P' (on the opposite branch) be taken, and $P'M$, $P'M'$ measured on it such that $P'M^2 = P'M'^2 = CD^2$, it follows in like manner that CM , CM' are asymptotes.

Now MM' , LL' are parallel, $PL = P'M$, and PCP' is a straight line. Therefore LCM is a straight line.

So also is $L'CM'$, and therefore the opposite branches have the same asymptotes.

(3) Let PP' , DD' be conjugate diameters of two conjugate



hyperbolas. Draw the tangents at P , P' , D , D' . Then [Prop. 11 and Prop. 26] the tangents form a parallelogram, and the diagonals of it, LM , $L'M'$, pass through the centre.

Also $PL = PL' = P'M = P'M' = CD$.

Therefore LM , $L'M'$ are the asymptotes of the hyperbola in which PP' is a transverse diameter and DD' its conjugate.

Similarly $DL = DM' = D'L' = D'M = CP$, and LM , $L'M'$ are the asymptotes of the hyperbola in which DD' is a transverse diameter and PP' its conjugate, i.e. the conjugate hyperbola.

Therefore conjugate hyperbolas have their asymptotes common.

Proposition 30.

[II. 3.]

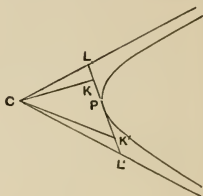
If a straight line touch a hyperbola at P , it will meet the asymptotes in two points L, L' ; LL' will be bisected at P , and $PL^2 = \frac{1}{4}p \cdot PP'$ [= CD^2].

[This proposition is the converse of Prop. 28 (1) above.]

For, if the tangent at P does not meet the asymptotes in the points L, L' described, take on the tangent lengths PK, PK' each equal to CD .

Then CK, CK' are asymptotes; which is impossible.

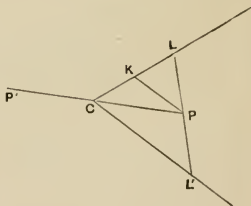
Therefore the points K, K' must be identical with the points L, L' on the asymptotes.

**Proposition 31. (Problem.)**

[II. 4.]

Given the asymptotes and a point P on a hyperbola, to find the curve.

Let CL, CL' be the asymptotes, and P the point. Produce PC to P' so that $CP = CP'$. Draw PK parallel to CL' meeting CL in K , and let CL be made equal to twice CK . Join LP and produce it to L' .



Take a length p such that $LL'^2 = p \cdot PP'$, and with diameter PP' and parameter p describe a hyperbola such that the ordinates to PP' are parallel to LL' . [Prop. 25]

Proposition 32.

[II. 8, 10.]

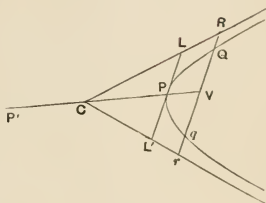
If Qq be any chord, it will, if produced both ways, meet the asymptotes in two points as R, r , and

(1) QR, qr will be equal,

(2) $RQ \cdot Qr = \frac{1}{4}p \cdot PP' [= CD^2]$.

Take V the middle point of Qq , and join CV meeting the curve in P . Then CV is a diameter and the tangent at P is parallel to Qq . [Prop. 11]

Also the tangent at P meets the asymptotes (in L, L'). Therefore Qq parallel to it also meets the asymptotes.



Then (1), since Qq is parallel to LL' , and $LP = PL'$, it follows that $RV = Vr$.

But $QV = Vq$;
therefore, subtracting, $QR = qr$.

(2) We have

$$\begin{aligned} p : PP' &= PL^2 : CP^2 \\ &= RV^2 : CV^2, \end{aligned}$$

and $p : PP' = QV^2 : CV^2 - CP^2$; [Prop. 8]

$$\begin{aligned} \therefore PL^2 : CP^2 &= p : PP' = RV^2 - QV^2 : CP^2 \\ &= RQ \cdot Qr : CP^2; \end{aligned}$$

thus
$$\begin{aligned} RQ \cdot Qr &= PL^2 \\ &= \frac{1}{4}p \cdot PP' = CD^2. \end{aligned}$$

Similarly $rq \cdot qR = CD^2$.

Proposition 33.

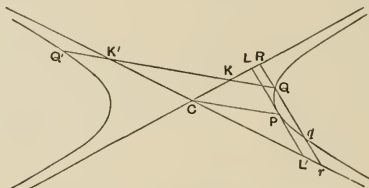
[II. 11, 16.]

If Q, Q' are on opposite branches, and QQ' meet the asymptotes in K, K' , and if CP be the semidiameter parallel to QQ' , then

$$(1) \quad KQ \cdot QK' = CP^2,$$

$$(2) \quad QK = Q'K'.$$

Draw the tangent at P meeting the asymptotes in L, L' , and



let the chord Qq parallel to LL' meet the asymptotes in R, r . Qq is therefore a double ordinate to CP .

Then we have

$$\begin{aligned} PL^2 : CP^2 &= (PL : CP) \cdot (PL' : CP) \\ &= (RQ : KQ) \cdot (Qr : QK') \\ &= RQ \cdot Qr : KQ \cdot QK'. \end{aligned}$$

$$\text{But} \quad PL^2 = RQ \cdot Qr; \quad [\text{Prop. 32}]$$

$$\therefore KQ \cdot QK' = CP^2.$$

$$\text{Similarly} \quad K'Q' \cdot Q'K = CP^2.$$

$$(2) \quad KQ \cdot QK' = CP^2 = K'Q' \cdot Q'K;$$

$$\therefore KQ \cdot (KQ + KK') = K'Q' \cdot (K'Q' + KK'),$$

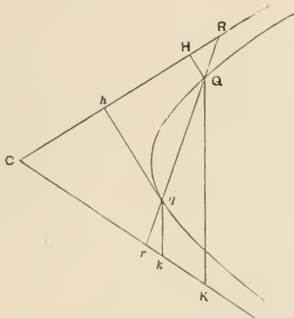
whence it follows that $KQ = K'Q'$.

Proposition 34.

[II. 12.]

If Q, q be any two points on a hyperbola, and parallel straight lines QH, qh be drawn to meet one asymptote at any angle, and QK, qk (also parallel to one another) meet the other asymptote at any angle, then

$$HQ \cdot QK = hq \cdot qk.$$



Let Qq meet the asymptotes in R, r .

We have $RQ \cdot Qr = Rq \cdot qr$; [Prop. 32]

$$\therefore RQ : Rq = qr : Qr.$$

But $RQ : Rq = HQ : hq$,

and $qr : Qr = qk : QK$;

$$\therefore HQ : hq = qk : QK,$$

or $HQ \cdot QK = hq \cdot qk.$

Proposition 35.

[II. 13.]

If in the space between the asymptotes and the hyperbola a straight line be drawn parallel to one of the asymptotes, it will meet the hyperbola in one point only.

Let E be a point on one asymptote, and let EF be drawn parallel to the other.

Then EF produced shall meet the curve in one point only.

For, if possible, let it not meet the curve.

Take Q , any point on the curve, and draw QH , QK each parallel to one asymptote and meeting the other; let a point F be taken on EF such that

$$HQ \cdot QK = CE \cdot EF.$$

Join CF and produce it to meet the curve in q ; and draw qh , qk respectively parallel to QH , QK .

Then $hq \cdot qk = HQ \cdot QK$, [Prop. 34]

and $HQ \cdot QK = CE \cdot EF$, by hypothesis,

$$\therefore hq \cdot qk = CE \cdot EF;$$

which is impossible, $\because hq > EF$, and $qk > CE$.

Therefore EF will meet the hyperbola in one point, as R .

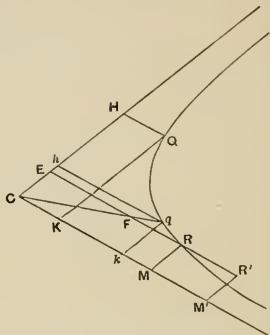
Again, EF will not meet the hyperbola in any other point.

For, if possible, let EF meet it in R' as well as R , and let RM , $R'M'$ be drawn parallel to QK .

Then $ER \cdot RM = ER' \cdot R'M'$: [Prop. 34]

which is impossible, $\because ER' > ER$.

Therefore EF does not meet the hyperbola in a second point R' .



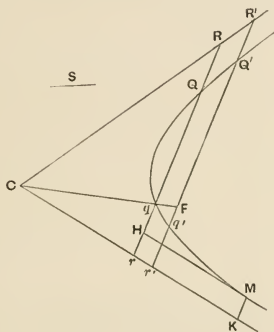
Proposition 36.

[II. 14.]

The asymptotes and the hyperbola, as they pass on to infinity, approach continually nearer, and will come within a distance less than any assignable length.

Let S be the given length.

Draw two parallel chords Qq , $Q'q'$ meeting the asymptotes in R, r and R', r' . Join Cq and produce it to meet $Q'q'$ in F .



Then

$$r'q' \cdot q'R' = rq \cdot qR,$$

and

$$q'R' > qR;$$

$$\therefore q'r' < qr,$$

and hence, as successive chords are taken more and more distant from the centre, qr becomes smaller and smaller.

Take now on rq a length rH less than S , and draw HM parallel to the asymptote Cr .

HM will then meet the curve [Prop. 35] in a point M . And, if MK be drawn parallel to Qq to meet Cr in K ,

$$MK = rH,$$

whence

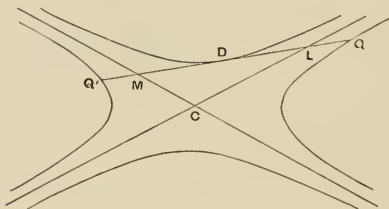
$$MK < S.$$

Proposition 37.

[II. 19.]

Any tangent to the conjugate hyperbola will meet both branches of the original hyperbola and be bisected at the point of contact.

(1) Let a tangent be drawn to either branch of the conjugate hyperbola at a point D .



This tangent will then meet the asymptotes [Prop. 30], and will therefore meet both branches of the original hyperbola.

(2) Let the tangent meet the asymptotes in L , M and the original hyperbola in Q , Q' .

Then [Prop. 30] $DL = DM$.

Also [Prop. 33] $LQ = MQ'$;

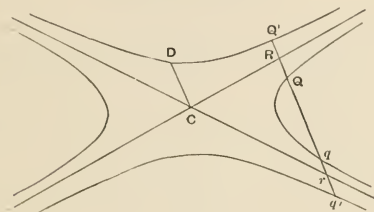
whence, by addition, $DQ = DQ'$.

Proposition 38.

[II. 23.]

If a chord Qq in one branch of a hyperbola meet the asymptotes in R, r and the conjugate hyperbola in Q', q' , then

$$Q'Q \cdot Qq' = 2CD^2.$$



Let CD be the parallel semi-diameter. Then we have
[Props. 32, 33]

$$RQ \cdot Qr = CD^2,$$

$$RQ' \cdot Q'r = CD^2;$$

$$\begin{aligned} \therefore 2CD^2 &= RQ \cdot Qr + RQ' \cdot Q'r \\ &= (RQ + RQ') Qr + RQ' \cdot QQ' \\ &= QQ' \cdot (Qr + RQ') \\ &= QQ' (Qr + rq') \\ &= QQ' \cdot Qq'. \end{aligned}$$

TANGENTS, CONJUGATE DIAMETERS AND AXES.

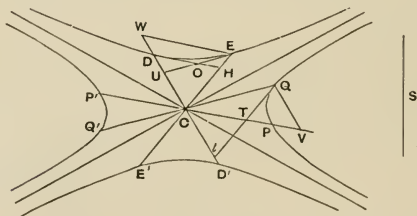
Proposition 39.

[II. 20.]

If Q be any point on a hyperbola, and CE be drawn from the centre parallel to the tangent at Q to meet the conjugate hyperbola in E , then

- (1) *the tangent at E will be parallel to CQ , and*
- (2) *CQ, CE will be conjugate diameters.*

Let PP', DD' be the conjugate diameters of reference, and let QV be the ordinate from Q to PP' , and EW the ordinate



from E to DD' . Let the tangent at Q meet PP', DD' in T, t respectively, let the tangent at E meet DD' in U , and let the tangent at D meet EU, CE in O, H respectively.

Let p, p' be the parameters corresponding to PP', DD' in the two hyperbolas, and we have

$$(1) \quad PP' : p = p' : DD',$$

$$[\because p \cdot PP' = DD'^2, \quad p' \cdot DD' = PP'^2]$$

and
$$PP' : p = CV . VT : QV^2,$$

$$p' : DD' = EW^2 : CW . WU. \quad [\text{Prop. 14}]$$

$$\therefore CV . VT : QV^2 = EW^2 : CW . WU.$$

But, by similar triangles,

$$VT : QV = EW : CW.$$

Therefore, by division,

$$CV : QV = EW : WU.$$

And in the triangles CVQ , EWU the angles at V , W are equal.

Therefore the triangles are similar, and

$$\angle QCV = \angle UEW.$$

But $\angle VCE = \angle CEW$, since EW , CV are parallel.

Therefore, by subtraction, $\angle QCE = \angle CEU$.

Hence EU is parallel to CQ .

(2) Take a straight line S of such length that

$$HE : EO = EU : S,$$

so that S is equal to half the parameter of the ordinates to the diameter EE' of the conjugate hyperbola. [Prop. 23]

Also $Ct . QV = CD^2$, (since $QV = Cv$),

or $Ct : QV = Ct^2 : CD^2$.

Now $Ct : QV = tT : TQ = \triangle tCT : \triangle CQT$,

and $Ct^2 : CD^2 = \triangle tCT : \triangle CDH = \triangle tCT : \triangle CEU$

[as in Prop. 23].

It follows that $\triangle CQT = \triangle CEU$.

And $\angle CQT = \angle CEU$.

$$\therefore CQ . QT = CE . EU \dots\dots\dots (A).$$

But $S : EU = OE : EH$

$$= CQ : QT.$$

$$\therefore S . CE : CE . EU = CQ^2 : CQ . QT.$$

Hence, by (A), $S . CE = CQ^2$.

$$\therefore 2S . EE' = QQ'^2,$$

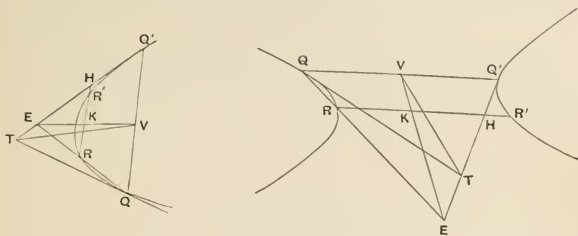
where $2S$ is the parameter corresponding to EE' .

And similarly it may be proved that EE'^2 is equal to the rectangle contained by QQ' and the corresponding parameter.

Therefore QQ' , EE' are conjugate diameters. [Prop. 26]

Also, since RR' is a chord parallel to QQ' bisected by the diameter EV , $RK = KR'$.

Therefore $KR' = KH$: which is impossible.



Therefore EV is not a diameter, and it may be proved in like manner that no other straight line through V is a diameter except TV .

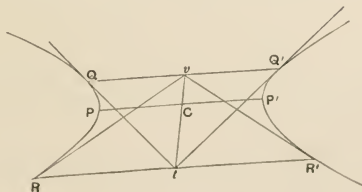
Conversely, the diameter of the conic drawn through T , the point of intersection of the tangents, will bisect the chord of contact QQ' .

[This is separately proved by Apollonius by means of an easy *reductio ad absurdum*.]

Proposition 42.

[II. 40.]

If tQ , tQ' be tangents to opposite branches of a hyperbola, and a chord RR' be drawn through t parallel to QQ' , then the lines joining R , R' to v , the middle point of QQ' , will be tangents at R , R' .



Join vt . vt is then the diameter conjugate to the transverse diameter drawn parallel to QQ' , i.e. to PP' .

But, since the tangent Qt meets the secondary diameter in t ,

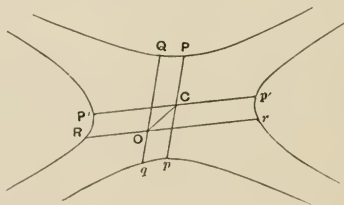
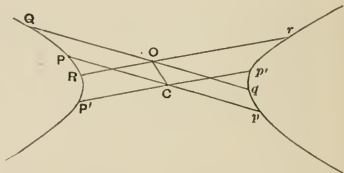
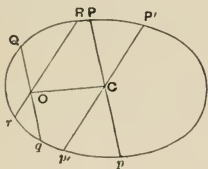
$$Cv \cdot Ct = \frac{1}{4}p \cdot PP' [= CD^2]. \quad [\text{Prop. 15}]$$

Therefore the relation between v and t is reciprocal, and the tangents at R , R' intersect in v .

Proposition 43.

[II. 26, 41, 42.]

In a conic, or a circle, or in conjugate hyperbolas, if two chords not passing through the centre intersect, they do not bisect each other.



Let Qq , Rr , two chords not passing through the centre, meet in O . Join CO , and draw the diameters Pp , $P'p'$ respectively parallel to Qq , Rr .

Then Qq , Rr shall not bisect one another. For, if possible, let each be bisected in O .

Then, since Qq is bisected in O and Pp is a diameter parallel to it, CO , Pp are conjugate diameters.

Therefore the tangent at P is parallel to CO .

Similarly it can be proved that the tangent at P' is parallel to CO .

Therefore the tangents at P , P' are parallel: which is impossible, since PP' is not a diameter.

Therefore Qq , Rr do not bisect one another.

Proposition 44. (Problem.)

[II. 44, 45.]

To find a diameter of a conic, and the centre of a central conic.

(1) Draw two parallel chords and join their middle points. The joining line will then be a diameter.

(2) Draw any two diameters; and these will meet in, and so determine, the centre.

Proposition 45. (Problem.)

[II. 46, 47.]

To find the axis of a parabola, and the axes of a central conic.

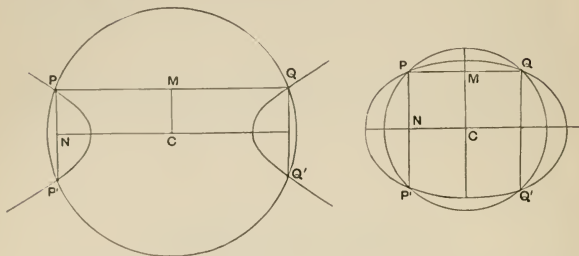
(1) In the case of the *parabola*, let PD be any diameter. Draw any chord QQ' perpendicular to PD , and let N be its middle point. Then AN drawn through N parallel to PD will be the axis.

For, being parallel to PD , AN is a diameter, and, inasmuch as it bisects QQ' at right angles, it is the axis.

And there is only *one* axis because there is only one diameter which bisects QQ' .



(2) In the case of a *central conic*, take any point P on the conic, and with centre C and radius CP describe a circle cutting the conic in P, P', Q, Q' .



Let PP', PQ be two common chords not passing through the centre, and let N, M be their middle points respectively. Join CN, CM .

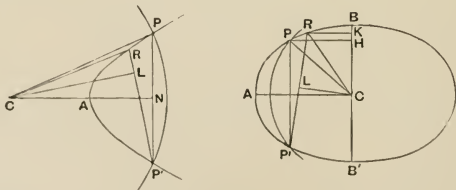
Then CN, CM will both be axes because they are both diameters bisecting chords at right angles. They are also conjugate because each bisects chords parallel to the other.

Proposition 46.

[II. 48.]

No central conic has more than two axes.

If possible, let there be another axis CL . Through P' draw $P'L$ perpendicular to CL , and produce $P'L$ to meet the



curve again in R . Join CP, CR .

Then, since CL is an axis, $P'L = LR$; therefore also

$$CP = CP' = CR.$$

Now in the case of the *hyperbola* it is clear that the circle PP' cannot meet the same branch of the hyperbola in any other points than P, P' . Therefore the assumption is absurd.

In the *ellipse* draw RK, PH perpendicular to the (minor) axis which is parallel to PP' .

Then, since it was proved that $CP = CR$,

$$CP^2 = CR^2,$$

or
$$CH^2 + HP^2 = CK^2 + KR^2.$$

$$\therefore CK^2 - CH^2 = HP^2 - KR^2 \dots\dots\dots (1).$$

Now $BK \cdot KB' + CK^2 = CB^2,$

and $BH \cdot HB' + CH^2 = CB^2,$

$$\therefore CK^2 - CH^2 = BH \cdot HB' - BK \cdot KB'.$$

Hence $HP^2 - KR^2 = BH \cdot HB' - BK \cdot KB',$ from (1).

But, since PH, RK are ordinates to BB' ,

$$PH^2 : BH \cdot HB' = RK^2 : BK \cdot KB',$$

and the difference between the antecedents has been proved equal to the difference between the consequents.

$$\therefore PH^2 = BH \cdot HB',$$

and $RK^2 = BK \cdot KB'.$

$\therefore P, R$ are points on a circle with diameter BB' : which is absurd.

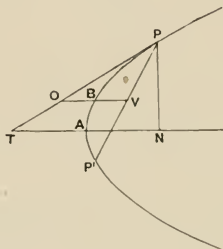
Hence CL is not an axis.

Proposition 47. (Problem.)

[II. 49.]

To draw a tangent to a parabola through any point on or outside the curve.

(1) Let the point be P on the curve. Draw PN perpendicular to the axis, and produce NA to T so that $AT = AN$. Join PT .



Then, since $AT = AN$, PT is the tangent at P . [Prop. 12]

In the particular case where P coincides with A , the vertex, the perpendicular to the axis through A is the tangent.

(2) Let the given point be any external point O . Draw the diameter OBV meeting the curve at B , and make BV equal to OB . Then draw through V the straight line VP parallel to the tangent at B [drawn as in (1)] meeting the curve in P . Join OP .

OP is the tangent required, because PV , being parallel to the tangent at B , is an ordinate to BV , and $OB = BV$.

[Prop. 12]

[This construction obviously gives the *two* tangents through O .]

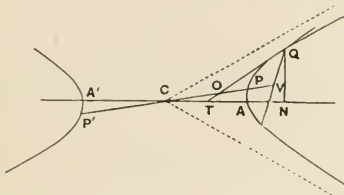
Proposition 48. (Problem.)

[II. 49.]

To draw a tangent to a hyperbola through any point on or outside the curve.

There are here four cases.

Case I. Let the point be Q on the curve.



Draw QN perpendicular to the axis AA' produced, and take on AA' a point T such that $A'T : AT = A'N : AN$. Join TQ .

Then TQ is the tangent at Q . [Prop. 13]

In the particular case where Q coincides with A or A' the perpendicular to the axis at that point is the tangent.

Case II. Let the point be any point O within the angle contained by the asymptotes.

Join CO and produce it both ways to meet the hyperbola in P, P' . Take a point V on CP produced such that

$$P'V : PV = OP' : OP,$$

and through V draw VQ parallel to the tangent at P [drawn as in Case I.] meeting the curve in Q . Join OQ .

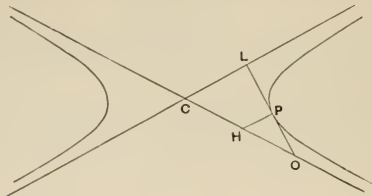
Then, since QV is parallel to the tangent at P , QV is an ordinate to the diameter $P'P$, and moreover

$$P'V : PV = OP' : OP.$$

Therefore OQ is the tangent at Q . [Prop. 13]

[This construction obviously gives the *two* tangents through O .]

Case III. Let the point O be on one of the asymptotes. Bisect CO at H , and through H draw HP parallel to the other



asymptote meeting the curve in P . Join OP and produce it to meet the other asymptote in L .

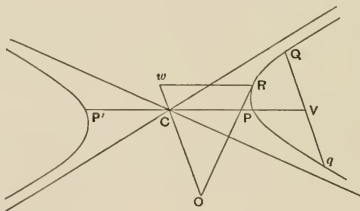
Then, by parallels,

$$OP : PL = OH : HC,$$

whence $OP = PL$.

Therefore OL touches the hyperbola at P . [Props. 28, 30]

Case IV. Let the point O lie within one of the *exterior* angles made by the asymptotes.



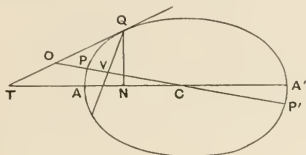
Join CO . Take any chord Qq parallel to CO , and let V be its middle point. Draw through V the diameter PP' . Then PP' is the diameter conjugate to CO . Now take on OC produced a point w such that $CO.Cw = \frac{1}{4}p$. $PP' [= CD^2]$, and draw through w the straight line wR parallel to PP' meeting the curve in R . Join OR . Then, since Rw is parallel to CP and Cw conjugate to it, while $CO.Cw = CD^2$, OR is the tangent at R . [Prop. 15]

Proposition 49. (Problem.)

[II. 49.]

To draw a tangent to an ellipse through any point on or outside the curve.

There are here two cases, (1) where the point is on the curve, and (2) where it is outside the curve; and the con-



structions correspond, *mutatis mutandis*, with Cases I. and II. of the hyperbola just given, depending as before on Prop. 13.

When the point is external to the ellipse, the construction gives, as before, the *two* tangents through the point.

Proposition 50. (Problem.)

[II. 50.]

To draw a tangent to a given conic making with the axis an angle equal to a given acute angle.

I. Let the conic be a *parabola*, and let DEF be the given acute angle. Draw DF perpendicular to EF , bisect EF at H , and join DH .

Now let AN be the axis of the parabola, and make the angle NAP equal to the angle DHF . Let AP meet the curve in P . Draw PN perpendicular to AN . Produce NA to T so that $AN = AT$, and join PT .

Then PT is a tangent, and we have to prove that

$$\angle PTN = \angle DEF.$$

Since

$$\angle DHF = \angle PAN,$$

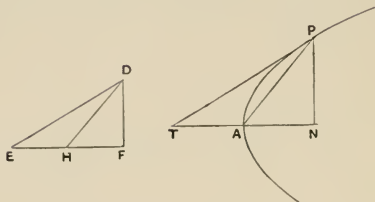
$$HF : FD = AN : NP.$$

$$\therefore 2HF : FD = 2AN : NP,$$

or

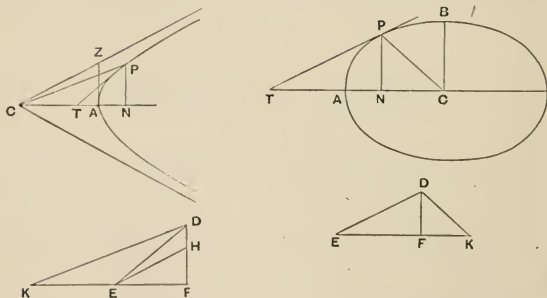
$$EF : FD = TN : NP.$$

$$\therefore \angle PTN = \angle DEF.$$



II. Let the conic be a central conic.

Then, for the *hyperbola*, it is a necessary condition of the possibility of the solution that the given angle DEF must be



greater than the angle between the axis and an asymptote, or half that between the asymptotes. If DEF be the given angle and DF be at right angles to EF , let H be so taken on DF that $\angle HEF = \angle ACZ$, or half the angle between the asymptotes. Let AZ be the tangent at A meeting an asymptote in Z .

We have then $CA^2 : AZ^2$ (or $CA^2 : CB^2$) $= EF^2 : FH^2$.

$$\therefore CA^2 : CB^2 > EF^2 : FD^2.$$

Take a point K on FE produced such that

$$CA^2 : CB^2 = KF \cdot FE : FD^2.$$

Thus

$$KF^2 : FD^2 > CA^2 : AZ^2.$$

Therefore, if DK be joined, the angle DKF is less than the angle ACZ . Hence, if the angle ACP be made equal to the angle DKF , CP must meet the hyperbola in some point P .

In the case of the *ellipse* K has to be taken on EF produced so that $CA^2 : CB^2 = KF \cdot FE : FD^2$, and from this point the constructions are similar for both the central conics, the angle ACP being made equal to the angle DKF in each case.

Draw now PN perpendicular to the axis, and draw the tangent PT . [Props. 48, 49]

$$\begin{aligned} \text{Then} \quad PN^2 : CN \cdot NT &= CB^2 : CA^2 && [\text{Prop. 14}] \\ &= FD^2 : KF \cdot FE, && \text{from above;} \end{aligned}$$

and, by similar triangles,

$$CN^2 : PN^2 = KF^2 : FD^2.$$

$$\therefore CN^2 : CN \cdot NT = KF^2 : KF \cdot FE,$$

$$\text{or} \quad CN : NT = KF : FE.$$

$$\text{And} \quad PN : CN = DF : KF.$$

$$\therefore PN : NT = DF : FE.$$

$$\text{Hence} \quad \angle PTN = \angle DEF.$$

Proposition 51.

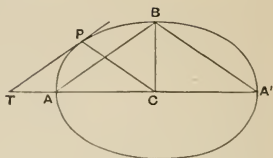
[II. 52.]

In an ellipse, if the tangent at any point P meet the major axis in T , the angle CPT is not greater than the angle ABA' (where B is one extremity of the minor axis).

Taking P in the quadrant AB , join PC .

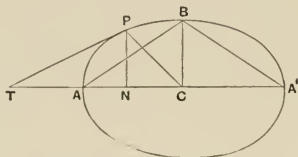
Then PC is either parallel to BA' or not parallel to it.

First, let PC be parallel to BA' . Then, by parallels, CP bisects AB . Therefore the tangent at P is parallel to AB , and $\angle CPT = \angle A'BA$.



Secondly, suppose that PC is not parallel to BA' , and we have in that case, drawing PN perpendicular to the axis,

$$\angle PCN \neq \angle BA'C, \text{ or } \angle BAC.$$



$$\therefore PN^2 : CN^2 \neq BC^2 : AC^2,$$

whence

$$PN^2 : CN^2 \neq PN^2 : CN \cdot NT.$$

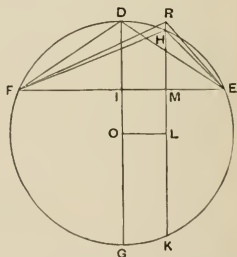
[Prop. 14]

$$\therefore CN \neq NT.$$

Let FDE be a segment in a circle containing an angle FDE equal to the angle ABA' , and let DG be the diameter of the circle bisecting FE at right angles in I . Divide FE in M so that

$$EM : MF = CN : NT,$$

and draw through M the chord HK at right angles to EF . From O , the centre of the circle, draw OL perpendicular to HK , and join EH , HF .



The triangles DFI , BAC are then similar, and

$$FI^2 : ID^2 = CA^2 : CB^2.$$

Now

$$OD : OI > LH : LM, \text{ since } OI = LM.$$

$$\therefore OD : DI < LH : HM$$

and, doubling the antecedents,

$$DG : DI < HK : HM,$$

whence

$$GI : ID < KM : MH.$$

But

$$GI : ID = FI^2 : ID^2 = CA^2 : CB^2 \\ = CN \cdot NT : PN^2.$$

$$\therefore CN \cdot NT : PN^2 < KM : MH \\ < KM \cdot MH : MH^2 \\ < EM \cdot MF : MH^2.$$

Let $CN \cdot NT : PN^2 = EM \cdot MF : MR^2,$

where R is some point on HK or HK produced.

It follows that $MR > MH$, and R lies on KH produced.
Join ER , RF .

Now $CN \cdot NT : EM \cdot MF = PN^2 : RM^2,$

and $CN^2 : EM^2 = CN \cdot NT : EM \cdot MF$

(since $CN : NT = EM : MF$).

$$\therefore CN : EM = PN : RM.$$

Therefore the triangles CPN , ERM are similar.

In like manner the triangles PTN , RFM are similar.

Therefore the triangles CPT , ERF are similar,

and $\angle CPT = \angle ERF;$

whence it follows that

$$\angle CPT \text{ is less than } \angle EHF, \text{ or } \angle ABA'.$$

Therefore, whether CP is parallel to BA' or not, the $\angle CPT$ is not greater than the $\angle ABA'$.

Proposition 52. (Problem.)

[II. 51, 53.]

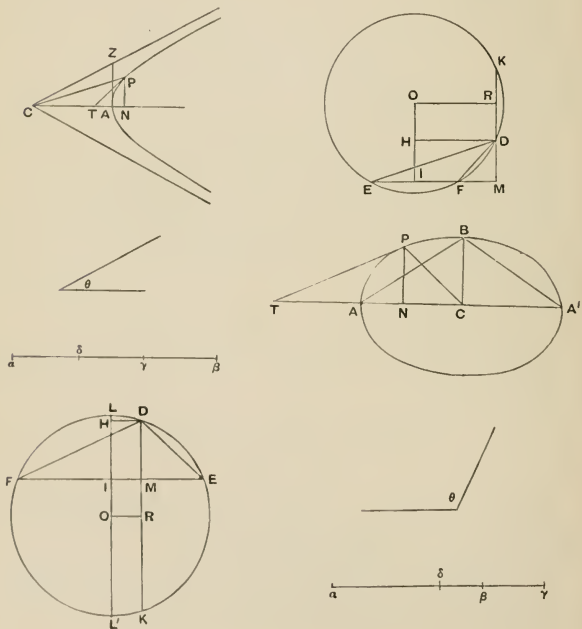
To draw a tangent to any given conic making a given angle with the diameter through the point of contact.

I. In the case of the *parabola* the given angle must be an acute angle, and, since any diameter is parallel to the axis, the problem reduces itself to Prop. 50 (1) above.

II. In the case of a central conic, the angle CPT must be acute for the *hyperbola*, and for the *ellipse* it must not be less than a right angle, nor greater than the angle ABA' , as proved in Prop. 51.

Suppose θ to be the given angle, and take first the particular case for the *ellipse* in which the angle θ is equal to the angle ABA' . In this case we have simply, as in Prop. 51, to draw CP parallel to BA' (or AB) and to draw through P a parallel to the chord AB (or $A'B$).

Next suppose θ to be any acute angle for the *hyperbola*, and for the *ellipse* any obtuse angle less than ABA' ; and suppose the problem solved, the angle CPT being equal to θ .



Imagine a segment of a circle taken containing an angle (EDF) equal to the angle θ . Then, if a point D on the circumference of the segment could be found such that, if DM be the perpendicular on the base EF , the ratio $EM.MF : DM^2$ is equal to the ratio $CA^2 : CB^2$, i.e. to the ratio $CN.NT : PN^2$, we should have

$$\angle CPT = \angle \theta = \angle EDF,$$

and $CN.NT : PN^2 = EM.MF : DM^2$,

and it would follow that triangles PCN , PTN are respectively similar to DEM , DFM *. Thus the angle DEM would be equal to the angle PCN .

The construction would then be as follows:

Draw CP so that the angle PCN is equal to the angle DEM , and draw the tangent at P meeting the axis AA' in T . Also let PN be perpendicular to the axis AA' .

Then $CN.NT : PN^2 = CA^2 : CB^2 = EM.MF : DM^2$,

and the triangles PCN , DEM are similar, whence it follows that the triangles PTN , DFM are similar, and therefore also the triangles CPT , EDF *.

$$\therefore \angle CPT = \angle EDF = \angle \theta.$$

It only remains to be proved for the *hyperbola* that, if the angle PCN be made equal to the angle DEM , CP must necessarily meet the curve, i.e. that the angle DEM is less than half the angle between the asymptotes. If AZ is perpendicular to the axis and meets an asymptote in Z , we have

$$EM.MF : DM^2 = CA^2 : CB^2 = CA^2 : AZ^2,$$

$$\therefore EM^2 : DM^2 > CA^2 : AZ^2,$$

and the angle DEM is less than the angle ZCA .

We have now shown that the construction reduces itself to finding the point D on the segment of the circle, such that

$$EM.MF : DM^2 = CA^2 : CB^2.$$

* These conclusions are taken for granted by Apollonius, but they are easily proved.

This is effected as follows :

Take lengths $\alpha\beta$, $\beta\gamma$ in one straight line such that

$$\alpha\beta : \beta\gamma = CA^2 : CB^2,$$

$\beta\gamma$ being measured towards α for the *hyperbola* and away from α for the *ellipse*; and let $\alpha\gamma$ be bisected in δ .

Draw OI from O , the centre of the circle, perpendicular to EF ; and on OI or OI produced take a point H such that

$$OH : HI = \delta\gamma : \gamma\beta,$$

(the points O , H , I occupying positions relative to one another corresponding to the relative positions of δ , γ , β).

Draw HD parallel to EF to meet the segment in D . Let DK be the chord through D at right angles to EF and meeting it in M .

Draw OR bisecting DK at right angles.

Then $RD : DM = OH : HI = \delta\gamma : \gamma\beta$.

Therefore, doubling the two antecedents,

$$KD : DM = \alpha\gamma : \gamma\beta;$$

so that

$$KM : DM = \alpha\beta : \beta\gamma.$$

Thus

$$KM \cdot MD : DM^2 = EM \cdot MF : DM^2 = \alpha\beta : \beta\gamma = CA^2 : CB^2.$$

Therefore the required point D is found.

In the particular case of the hyperbola where $CA^2 = CB^2$, *i.e.* for the *rectangular* hyperbola, we have $EM \cdot MF = DM^2$, or DM is the tangent to the circle at D .

Note. Apollonius proves incidentally that, in the second figure applying to the case of the ellipse, H falls between I and the middle point (L) of the segment as follows :

$$\angle FLI = \frac{1}{2} \angle CPT, \text{ which is less than } \frac{1}{2} \angle ABA';$$

$$\therefore \angle FLI \text{ is less than } \angle ABC,$$

whence $CA^2 : CB^2 > FI^2 : IL^2$
 $> L'I : IL.$

It follows that $\alpha\beta : \beta\gamma > L'I : IL,$
 so that $\alpha\gamma : \gamma\beta > L'L : IL,$

and, halving the antecedents,
 $\delta\gamma : \gamma\beta > OL : LI,$

so that $\delta\beta : \beta\gamma > OI : IL.$

Hence, if H be such a point that

$$\delta\beta : \beta\gamma = OI : IH,$$

IH is less than $IL.$

EXTENSIONS OF PROPOSITIONS 17—19.

Proposition 53.

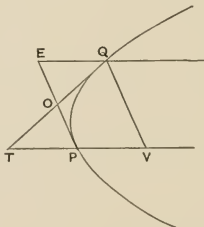
[III. 1, 4, 13.]

(1) *P, Q being any two points on a conic, if the tangent at P and the diameter through Q meet in E, and the tangent at Q and the diameter through P in T, and if the tangents intersect at O, then*

$$\triangle OPT = \triangle OQE.$$

(2) If P be any point on a hyperbola and Q any point on the conjugate hyperbola, and if T, E have the same significance as before, then $\Delta CPE = \Delta CQT$.

(1) Let QV be the ordinate from Q to the diameter through P .



Then for the *parabola* we have

$$TP = PV, \quad [\text{Prop. 12}]$$

so that

$$TV = 2PV,$$

and

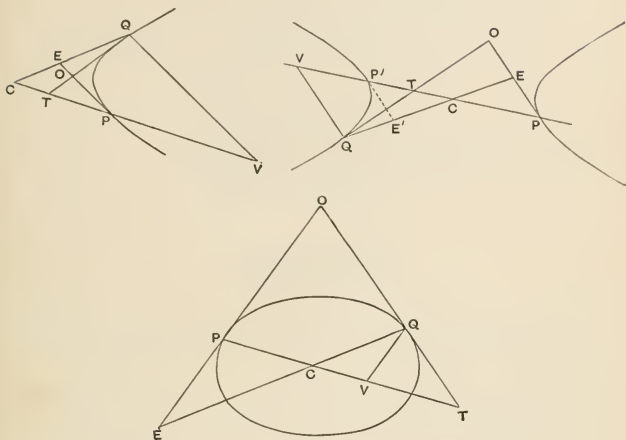
$$\square EV = \triangle QTV.$$

Subtracting the common area $OPVQ$,

$$\triangle OQE = \triangle OPT.$$

For the *central conic* we have

$$CV \cdot CT = CP^2,$$



or

$$CV : CT = CV^2 : CP^2;$$

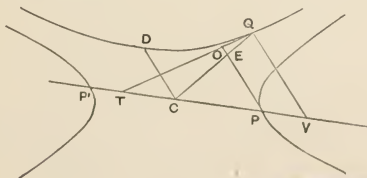
$$\therefore \triangle CQV : \triangle CQT = \triangle CQV : \triangle CPE;$$

$$\therefore \triangle CQT = \triangle CPE.$$

Hence the sums or differences of the area $OTCE$ and each triangle are equal, or

$$\triangle OPT = \triangle OQE.$$

(2) In the *conjugate hyperbolas* draw CD parallel to the



tangent at P to meet the conjugate hyperbola in D , and draw QV also parallel to PE meeting CP in V . Then CP, CD are conjugate diameters of both hyperbolas, and QV is drawn ordinate-wise to CP .

Therefore [Prop. 15]

$$CV \cdot CT = CP^2,$$

or

$$CP : CT = CV : CP$$

$$= CQ : CE;$$

$$\therefore CP \cdot CE = CQ \cdot CT.$$

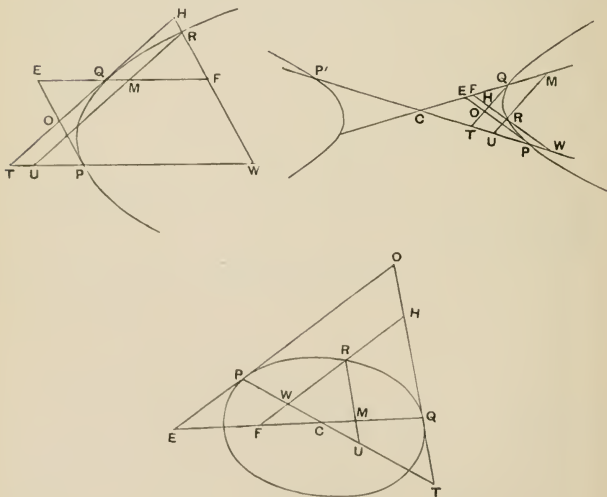
And the angles PCE, QCT are supplementary;

$$\therefore \triangle CQT = \triangle CPE.$$

Proposition 54.

[III. 2, 6.]

If we keep the notation of the last proposition, and if R be



any other point on the conic, let RU be drawn parallel to QT to meet the diameter through P in U , and let a parallel through R to the tangent at P meet QT and the diameters through Q , P in H , F , W respectively. Then

$$\triangle HQF = \text{quadrilateral } HTUR.$$

Let RU meet the diameter through Q in M . Then, as in Props. 22, 23, we have

$$\triangle RMF = \text{quadrilateral } QTUM;$$

$\therefore$, adding (or subtracting) the area HM ,

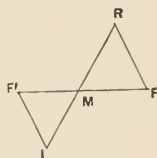
$$\triangle HQF = \text{quadrilateral } HTUR.$$

Proposition 55.

[III. 3, 7, 9, 10.]

If we keep the same notation as in the last proposition and take two points R' , R on the curve with points H' , F' , etc. corresponding to H , F , etc. and if, further, RU , $R'W'$ intersect in I and $R'U'$, RW in J , then the quadrilaterals $F'IRF$, $IUU'R'$ are equal, as also the quadrilaterals $FJR'F'$, $JU'UR$.

[N.B. It will be seen that in some cases (according to the positions of R , R') the quadrilaterals take a form like that in the margin, in which case $F'IRF$ must be taken as meaning the difference between the triangles $F'MI$, RMF .]



I. We have in figs. 1, 2, 3

$$\triangle HFQ = \text{quadrilateral } HTUR, \quad [\text{Prop. 54}]$$

$$\triangle H'F'Q = \text{quadrilateral } H'TU'R',$$

$$\therefore F'H'HF = H'TU'R' \sim HTUR$$

$$= IUU'R' \mp (IH);$$

whence, adding or subtracting IH ,

$$F'IRF = IUU'R' \dots\dots\dots (1).$$

and, adding (IJ) to both,

$$FJR'F' = JU'UR \dots \dots \dots (2).$$

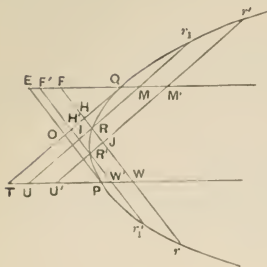


Fig. 1.

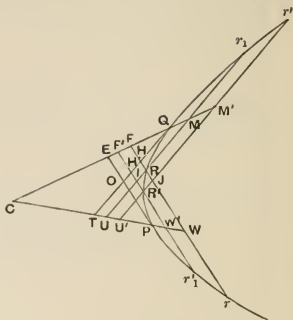


Fig. 2.

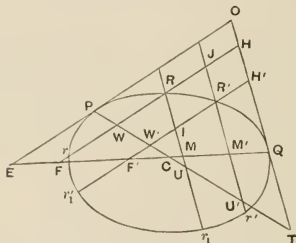


Fig. 3.

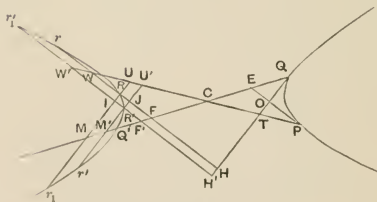


Fig. 4.

II. In figs. 4, 5, 6 we have

[Props. 18, 53]

$$\triangle R'U'W' = \triangle CF'W' - \triangle CQT,$$

so that

$$\triangle CQT = \text{quadrilateral } CU'R'F',$$

and, adding the quadrilateral $UF'H'T$, we have

$$\Delta H'F'Q = \text{quadrilateral } H'TU'R'.$$

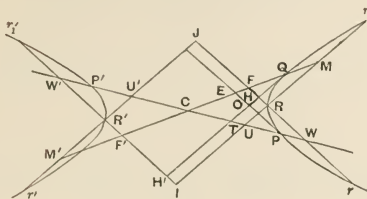


Fig. 5.

Similarly $\Delta HFQ = HTUR$;

and we deduce, as before,

$$F'IRF = IUU'R' \dots\dots\dots(1).$$

Thus e.g. in fig. 4,

$$\Delta H'F'Q' - \Delta HFQ = H'TU'R' - HTUR;$$

$$\therefore F'H'HF = (R'H) - (RU'),$$

and, subtracting each from (IH) ,

$$F'IRF = IUU'R'.$$

In fig. 6,

$$F'H'HF = H'TU'R' - \Delta HTW + \Delta RUW,$$

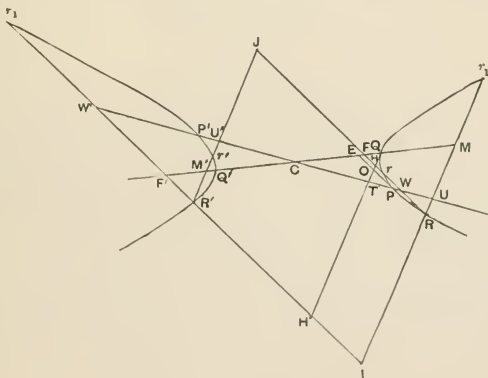


Fig. 6.

and, adding (IH) to each side,

$$\begin{aligned} F'IRF &= H'TU'R' + H'TUI \\ &= IUU'R' \dots\dots\dots (1). \end{aligned}$$

Then, subtracting (IJ) from each side in fig. 4, and subtracting each side from (IJ) in figs. 5, 6, we obtain

$$FJR'F' = JU'UR \dots\dots\dots (2),$$

(the quadrilaterals in fig. 6 being the differences between the triangles FJM' , $F'R'M'$ and between the triangles $JU'W$, $R UW$ respectively).

III. The same properties are proved in exactly the same manner in the case where P , Q are on opposite branches, and the quadrilaterals take the same form as in fig. 6 above.

COR. In the particular case of this proposition where R' coincides with P the results reduce to

$$\begin{aligned} EIRF &= \triangle PUI, \\ PJRU &= PJFE. \end{aligned}$$

Proposition 56.

[III. 8.]

If PP' , QQ' be two diameters and the tangents at P , P' , Q , Q' be drawn, the former two meeting QQ' in E , E' and the latter two meeting PP' in T , T' , and if the parallel through P' to the tangent at Q meets the tangent at P in K while the parallel through Q' to the tangent at P meets the tangent at Q in K' , then the quadrilaterals (EP') , (TQ') are equal, as also the quadrilaterals $(E'K)$, $(T'K')$.

Since the triangles CQT , CPE are equal [Prop. 53] and have a common vertical angle,

$$\begin{aligned} CQ \cdot CT &= CP \cdot CE; \\ \therefore CQ : CE &= CP : CT, \end{aligned}$$

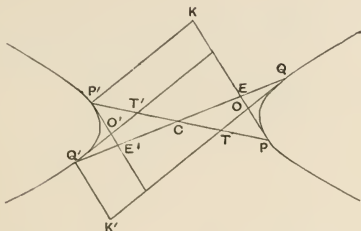
whence

$$QQ' : EQ = PP' : TP,$$

and the same proportion is true for the squares ;

$$\therefore \Delta QQ'K' : \Delta QEO = \Delta PP'K : \Delta PTO.$$

And the consequents are equal ;



$$\therefore \Delta QQ'K' = \Delta PP'K,$$

and, subtracting the equal triangles CQT , CPE , we obtain

$$(EP') = (TQ') \dots \dots \dots (1).$$

Adding the equal triangles $CP'E'$, $CQ'T'$ respectively, we have

$$(E'K) = (T'K') \dots \dots \dots (2).$$

Proposition 57.

[III. 5, 11, 12, 14.]

(Application to the case where the ordinates through R, R' , the points used in the last two propositions, are drawn to a *secondary* diameter.)

(1) Let Cv be the secondary diameter to which the ordinates are to be drawn. Let the tangent at Q meet it in t , and let the ordinate Rw meet Qt in h and CQ in f . Also let Ru , parallel to Qt , meet Cv in u .

Then [Prop. 19]

$$\Delta Ruw \sim \Delta Cfw = \Delta CQt \dots \dots \dots (A)$$

Proposition 58.

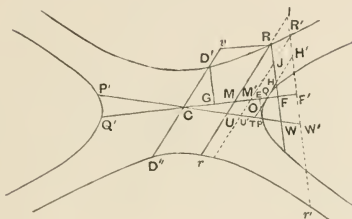
[III. 15.]

In the case where P, Q are on the original hyperbola and R on the conjugate hyperbola, the same properties as those formulated in Propositions 55, 57 still hold, viz.

$$\triangle RMF \sim \triangle CMU = \triangle CQT,$$

and

$$F'IRF = IUUR'.$$



Let $D'D''$ be the diameter of the conjugate hyperbola parallel to RU , and let QT be drawn; and from D' draw DG parallel to PE to meet CQ in G . Then $D'D''$ is the diameter conjugate to CQ .

Let p' be the parameter in the conjugate hyperbola corresponding to the transverse diameter $D'D''$, and let p be the parameter corresponding to the transverse diameter $Q'Q$ in the original hyperbola, so that

$$\frac{p}{2} \cdot CQ = CD'^2, \text{ and } \frac{p'}{2} \cdot CD' = CQ^2.$$

Now we have [Prop. 23]

$$OQ : QE = p : 2QT = \frac{p}{2} : QT;$$

$$\begin{aligned}
 \therefore D'C : CG &= \frac{p}{2} : QT \\
 &= \frac{p}{2} \cdot CQ : CQ \cdot QT \\
 &= CD'^2 : CQ \cdot QT.
 \end{aligned}$$

Hence $D'C \cdot CG = CQ \cdot QT,$

or $\triangle D'CG = \triangle CQT \dots \dots \dots (1).$

Again, $CM : MU = CQ : QT$

$$\begin{aligned}
 &= \left(CQ : \frac{p}{2} \right) \cdot (p : 2QT) \\
 &= (p' : D'D'') \cdot (OQ : QE) \\
 &= (p' : D'D'') \cdot (RM : MF) \dots \dots (2).
 \end{aligned}$$

Therefore the triangles CMU , RMF , $D'CG$, being respectively half of equiangular parallelograms on CM (or Rv), RM (or Cv), CD' , the last two of which are similar while the sides of the first two are connected by the relation (2), have the property of Prop. 16.

$$\therefore \triangle RMF \sim \triangle CMU = \triangle D'CG = \triangle CQT \dots \dots (3).$$

If R' be another point on the conjugate hyperbola, we have, by subtraction,

$$R'JFF' - RMM'J = MUU'M', \text{ or } RJFF' = RUU'J.$$

And, adding (IJ),

$$F'IRF = IUU'R' \dots \dots \dots (4)$$

RECTANGLES UNDER SEGMENTS OF INTERSECTING CHORDS.

Proposition 59.

[III. 16, 17, 18, 19, 20, 21, 22, 23.]

Case I. *If OP , OQ be two tangents to any conic and Rr , $R'r'$ two chords parallel to them respectively and intersecting in J , an internal or external point, then*

$$OP^2 : OQ^2 = RJ \cdot Jr : R'J \cdot Jr'.$$

(a) Let the construction and figures be the same as in Prop. 55.

We have then

$$RJ \cdot Jr = RW^2 \sim JW^2,$$

and $RW^2 : JW^2 = \triangle RUW : \triangle JU'W$;

$$\therefore RW^2 \sim JW^2 : RW^2 = JU'UR : \triangle RUW.$$

But $RW^2 : OP^2 = \triangle RUW : \triangle OPT$;

$$\therefore RJ \cdot Jr : OP^2 = JU'UR : \triangle OPT \dots\dots\dots(1).$$

Again $R'J \cdot Jr' = R'M'^2 \sim JM'^2$

and $R'M'^2 : JM'^2 = \triangle R'F'M' : \triangle JFM'$,

or $R'M'^2 \sim JM'^2 : R'M'^2 = FJR'F' : \triangle R'F'M'$.

But $R'M'^2 : OQ^2 = \triangle R'F'M' : \triangle OQE$;

$$\therefore R'J \cdot Jr' : OQ^2 = FJR'F' : \triangle OQE \dots\dots\dots(2).$$

Comparing (1) and (2), we have

$$JU'UR = FJR'F', \text{ by Prop. 55.}$$

and $\triangle OPT = \triangle OQE, \text{ by Prop. 53.}$

Thus $RJ.Jr : OP^2 = R'J.Jr' : OQ^2,$

or $OP^2 : OQ^2 = RJ.Jr : R'J.Jr'.$

(b) If we had taken the chords $R'r'_1, Rr_1$ parallel respectively to OP, OQ and intersecting in I , an internal or external point, we should have established in the same manner that

$$OP^2 : OQ^2 = R'I.Ir'_1 : R'I.Ir_1.$$

Hence the proposition is completely demonstrated.

[COR. If I , or J , which may be any internal or external point be assumed (as a particular case) to be the centre, we have the proposition that the rectangles under the segments of intersecting chords in fixed directions are as the squares of the parallel semi-diameters.]

Case II. *If P be a point on the conjugate hyperbola and the tangent at Q meet CP in t ; if further qq' be drawn through t parallel to the tangent at P , and $Rr, R'r'$ be two chords parallel respectively to the tangents at Q, P , and intersecting at i , then*

$$tQ^2 : tq^2 = Ri.ir : R'i.ir'.$$

Using the figure of Prop. 57, we have

$$Ri.ir = Mi^2 \sim MR^2,$$

and $Mi^2 : MR^2 = \triangle Mf'i : \triangle MfR.$

Hence $Ri.ir : MR^2 = f'iRf : \triangle MfR.$

Therefore, if QC, qq' (both produced) meet in L ,

$$Ri.ir : tQ^2 = f'iRf : \triangle QtL \dots\dots\dots (1).$$

Similarly, $R'i.ir' : R'w'^2 = iu'u'R' : \triangle R'u'w' :$

$$\therefore R'i.ir' : tq^2 = iu'u'R' : \triangle tqK \dots\dots\dots (2),$$

where qK is parallel to Qt and meets Ct produced in K .

But, comparing (1) and (2), we have

$$f'iRf = iuu'R', \quad [\text{Prop. 57}]$$

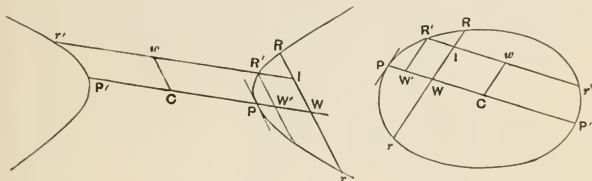
and $\triangle tqK = \triangle CLt + \triangle CQt = \triangle QtL. \quad [\text{Prop. 19}]$

$$\therefore Ri.ir : tQ^2 = R'i.ir' : tq^2,$$

or $tQ^2 : tq^2 = Ri.ir : R'i.ir'.$

Case III. If PP' be a diameter and $Rr, R'r'$ be chords parallel respectively to the tangent at P and the diameter PP' and intersecting in I , then

$$RI.Ir : R'I.Ir' = p : PP'.$$



If $RW, R'W'$ are ordinates to PP' ,

$$p : PP' = RW^2 : CW^2 \sim CP^2 \quad [\text{Prop. 8}]$$

$$= R'W'^2 : CW'^2 \sim CP'^2$$

$$= RW^2 \sim R'W'^2 : CW^2 \sim CW'^2$$

$$= RI.Ir : R'I.Ir'.$$

Case IV. If OP, OQ be tangents to a hyperbola and $Rr, R'r'$ be two chords of the conjugate hyperbola parallel respectively to OQ, OP , and meeting in I , then

$$OQ^2 : OP^2 = RI.Ir : R'I.Ir'.$$

Using the figure of Prop. 58, we have

$$\begin{aligned} OQ^2 : \triangle OQE &= RM^2 : \triangle RMF \\ &= MI^2 : \triangle MIF' \\ &= RI.Ir : \triangle RMF \sim \triangle MIF' \\ &= RI.Ir : F'IRF', \end{aligned}$$

and, in the same way,

$$\begin{aligned} OP^2 : \triangle OPT &= R'I . Ir' : \triangle R'U'W' \sim \triangle IUW' \\ &= R'I . Ir' : IUU'R'; \end{aligned}$$

whence, by Props. 53 and 58, as before,

$$OQ^2 : RI . Ir = OP^2 : R'I . Ir',$$

or

$$OQ^2 : OP^2 = RI . Ir : R'I . Ir'.$$

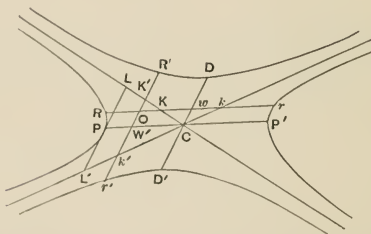
Proposition 60.

[III. 24, 25, 26.]

If Rr , $R'r'$ be chords of conjugate hyperbolus meeting in O and parallel respectively to conjugate diameters PP' , DD' , then

$$RO . Or + \frac{CP^2}{CD^2} . R'O . Or' = 2CP^2$$

$$\left[Or \cdot \frac{RO . Or}{CP^2} \pm \frac{R'O . Or'}{CD^2} = 2 \right].$$



Let Rr , $R'r'$ meet the asymptotes in K , k ; K' , k' , and CD , CP in w , W' respectively. Draw LPL' , the tangent at P , meeting the asymptotes in L , L' , so that $PL = PL'$.

Then

$$LP . PL' = CD^2,$$

and

$$LP . PL' : CP^2 = CD^2 : CP^2.$$

Now

$$LP : CP = K'O : OK,$$

$$PL' : CP = Ok' : Ok;$$

$$\therefore CD^2 : CP^2 = K'O . Ok' : KO . Ok.$$

[From this point Apollonius distinguishes five cases: (1) where O is in the angle LCL' , (2) where O is on one of the asymptotes, (3) where O is in the angle LCK or its opposite, (4) where O is within one of the branches of the original hyperbola, (5) where O lies within one of the branches of the conjugate hyperbola. The proof is similar in all these cases, and it will be sufficient to take case (1), that represented in the accompanying figure.]

We have therefore

$$\begin{aligned}
 CD^2 : CP^2 &= K'O . Ok' + CD^2 : KO . Ok + CP^2 \\
 &= K'O . Ok' + K'R' . R'k' : KO . Ok + CP^2 \\
 &= K'W'^2 - OW'^2 + R'W'^2 - K'W'^2 : Ow^2 - Kw^2 + CP^2 \\
 &= R'W'^2 - OW'^2 : Rw^2 - Kw^2 - Rw^2 + Ow^2 + CP^2 \\
 &= R'O . Or' : RK . Kr + CP^2 - RO . Or \\
 &= R'O . Or' : 2CP^2 - RO . Or \text{ (since } Kr = Rk),
 \end{aligned}$$

whence $RO . Or + \frac{CP^2}{CD^2} . R'O . Or' = 2CP^2,$

or $\frac{RO . Or}{CP^2} + \frac{R'O . Or'}{CD^2} = 2.$

[The following proof serves for all the cases : we have

$$R'W'^2 - CD^2 : CW'^2 = CD^2 : CP^2$$

and $Cw^2 : Rw^2 - CP^2 = CD^2 : CP^2;$

$$\therefore R'W'^2 - Cw^2 - CD^2 : CP^2 - (Rw^2 - CW'^2) = CD^2 : CP^2,$$

so that $\pm R'O . Or' - CD^2 : CP^2 \pm RO . Or = CD^2 : CP^2,$

whence $\pm R'O . Or' : 2CP^2 \pm RO . Or = CD^2 : CP^2$

or $\frac{R'O . Or'}{CD^2} \pm \frac{RO . Or}{CP^2} = 2.]$

Proposition 61.

[III. 27, 28, 29.]

If in an ellipse or in conjugate hyperbolas two chords Rr , $R'r'$ be drawn meeting in O and parallel respectively to two conjugate diameters PP' , DD' , then

(1) for the ellipse

$$RO^2 + Or^2 + \frac{CP^2}{CD^2} (R'O^2 + Or'^2) = 4CP^2,$$

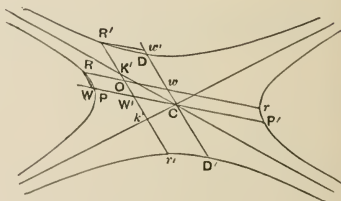
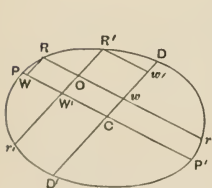
or
$$\frac{RO^2 + Or^2}{CP^2} + \frac{R'O^2 + Or'^2}{CD^2} = 4,$$

and for the hyperbolas

$$RO^2 + Or^2 : R'O^2 + Or'^2 = CP^2 : CD^2.$$

Also, (2) if $R'r'$ in the hyperbolas meet the asymptotes in K' , k' , then

$$K'O^2 + Ok'^2 + 2CD^2 : RO^2 + Or^2 = CD^2 : CP^2.$$



(1) We have for both curves

$$CP^2 : CD^2 = PW \cdot WP' : RW^2$$

$$= R'w'^2 : Dw' \cdot w'D'$$

$$= CP^2 + PW \cdot WP' \pm R'w'^2 : CD^2 + RW^2 \pm Dw' \cdot w'D',$$

(taking the upper sign for the hyperbolas and the lower for the ellipse);

$$\therefore CP^2 : CD^2 = CP^2 \pm CW'^2 + PW.WP' : CD^2 + Cw^2 \pm Dw'.w'D',$$

whence, for the *hyperbolas*,

$$\begin{aligned} CP^2 : CD^2 &= CW'^2 + CW^2 : Cw^2 + Cw'^2 \\ &= \frac{1}{2}(RO^2 + Or'^2) : \frac{1}{2}(R'O^2 + Or'^2), \end{aligned}$$

$$\text{or} \quad RO^2 + Or'^2 : R'O^2 + Or'^2 = CP^2 : CD^2 \dots\dots\dots (A),$$

while, for the *ellipse*,

$$\begin{aligned} CP^2 : CD^2 &= 2CP^2 - (CW'^2 + CW^2) : Cw'^2 + Cw^2 \\ &= 4CP^2 - (RO^2 + Or'^2) : (R'O^2 + Or'^2), \end{aligned}$$

$$\text{whence} \quad \frac{RO^2 + Or'^2}{CP^2} + \frac{R'O^2 + Or'^2}{CD^2} = 4 \dots\dots\dots (B).$$

(2) We have to prove that, in the *hyperbolas*,

$$R'O^2 + Or'^2 = K'O^2 + Ok'^2 + 2CD^2.$$

$$\text{Now} \quad R'O^2 - K'O^2 = R'K'^2 + 2R'K'.K'O,$$

$$\begin{aligned} \text{and} \quad Or'^2 - Ok'^2 &= r'k'^2 + 2r'k'.k'O \\ &= R'K'^2 + 2R'K'.k'O. \end{aligned}$$

Therefore, by addition,

$$\begin{aligned} R'O^2 + Or'^2 - K'O^2 - Ok'^2 &= 2R'K' (R'K' + K'O + Ok') \\ &= 2R'K' . R'k' \\ &= 2CD^2. \end{aligned}$$

$$\therefore R'O^2 + Or'^2 = K'O^2 + Ok'^2 + 2CD^2,$$

$$\text{whence} \quad K'O^2 + Ok'^2 + 2CD^2 : RO^2 + Or'^2 = CD^2 : CP^2,$$

by means of (A) above.

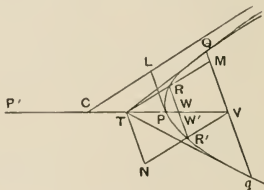
HARMONIC PROPERTIES OF POLES AND POLARS.

Proposition 62.

[III. 30, 31, 32, 33, 34.]

TQ, Tq being tangents to a hyperbola, if V be the middle point of Qq, and if TM be drawn parallel to an asymptote meeting the curve in R and Qq in M, while VN parallel to an asymptote meets the curve in R' and the parallel through T to the chord of contact in N, then

$$\begin{aligned} TR &= RM, \\ VR' &= R'N^*. \end{aligned}$$



I. Let CV meet the curve in P , and draw the tangent PL , which is therefore parallel to Qq . Also draw the ordinates RW , $R'W'$ to CP .

Then, since the triangles CPL , TWR are similar,

$$\begin{aligned} RW^2 : TW^2 &= PL^2 : CP^2 = CD^2 : CP^2 \\ &= RW^2 : PW \cdot WP'; \\ \therefore TW^2 &= PW \cdot WP'. \end{aligned}$$

* It will be observed from this proposition and the next that Apollonius begins with two particular cases of the general property in Prop. 64, namely (a) the case where the transversal is parallel to an asymptote, (b) the case where the chord of contact is parallel to an asymptote, i.e. where one of the tangents is an asymptote, or a tangent at infinity.

Also $CV \cdot CT = CP^2$;

$\therefore PW \cdot WP' + CP^2 = CV \cdot CT + TW^2$,

or $CW^2 = CV \cdot CT + TW^2$,

whence $CT(CW + TW) = CV \cdot CT$,

and $TW = WV$.

It follows by parallels that $TR = RM$(1).

Again $CP^2 : PL^2 = W'V^2 : W'R'^2$;

$\therefore W'V^2 : W'R'^2 = PW' \cdot W'P' : W'R'^2$,

so that $PW' \cdot W'P' = W'V^2$.

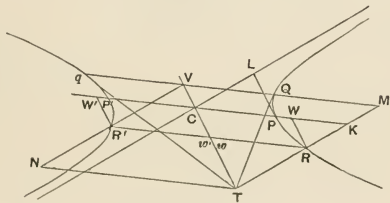
And $CV \cdot CT = CP^2$;

$\therefore CW'^2 = CV \cdot CT + W'V^2$,

whence, as before, $TW' = W'V$,

and $NR' = R'V$(2).

II. Next let Q, q be on opposite branches, and let $P'P$ be the diameter parallel to Qq . Draw the tangent PL , and the ordinates from R, R' , as before.



Let TM, CP intersect in K .

Then, since the triangles CPL, KWR are similar,

$CP^2 : PL^2 = KW^2 : WR^2$,

and $CP^2 : CD^2 = PW \cdot WP' : WR^2$;

$\therefore KW^2 = PW \cdot WP'$.

Hence, adding CP^2 ,

$CW^2 [= R w^2] = KW^2 + CP^2$.

But $R w^2 : KW^2 + CP^2 = T w^2 : R W^2 + PL^2$,

by similar triangles.

Therefore

$T w^2 = R W^2 + CD^2$
 $= C w^2 + CV \cdot CT$,

whence $Tw - Cw = CV$, or $Tw = wV$;

$$\therefore TR = RM \dots\dots\dots(1).$$

$$\begin{aligned} \text{Again } CP^2 : PL^2 &= PW' \cdot W'P' : R'W'^2 \\ &= PW' \cdot W'P' + CP'^2 : R'W'^2 + CD^2 \\ &= CW'^2 : Cw'^2 + CV \cdot CT. \end{aligned}$$

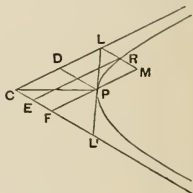
$$\begin{aligned} \text{Also } CP^2 : PL^2 &= R'w'^2 : w'V^2; \\ \therefore w'V^2 &= Cw'^2 + CV \cdot CT, \end{aligned}$$

whence, as before, $Tw' = w'V$,

$$\text{and, by parallels, } NR' = R'V \dots\dots\dots(2).$$

III. The particular case in which one of the tangents is a tangent at infinity, or an asymptote, is separately proved as follows.

Let LPL' be the tangent at P . Draw PD , LM parallel to CL' , and let LM meet the curve in R and the straight line PF drawn through P parallel to CL in M . Also draw RE parallel to CL .



$$\text{Now } LP = PL';$$

$$\therefore PD = CF = FL', \quad FP = CD = DL.$$

$$\text{And } FP \cdot PD = ER \cdot RL. \quad [\text{Prop. 34}]$$

$$\text{But } ER = LC = 2CD = 2FP;$$

$$\therefore PD = 2LR,$$

or

$$LR = RM.$$

Proposition 63.

[III. 35, 36.]

If PL , the tangent to a hyperbola at P , meet the asymptote in L , and if PO be parallel to that asymptote, and any straight line $LQOQ'$ be drawn meeting the hyperbola in Q , Q' and PO in O , then

$$LQ' : LQ = Q'O : OQ.$$

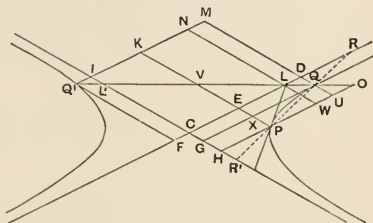
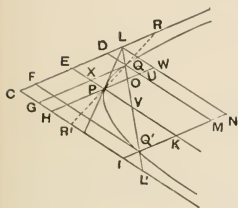
We have, drawing parallels through L, Q, P, Q' to both asymptotes as in the figures,

$LQ = Q'L'$; whence, by similar triangles, $DL = IQ' = CF$

$$\therefore CD = FL,$$

and

$$\begin{aligned} CD : DL &= FL : LD \\ &= Q'L : LQ \\ &= MD : DQ. \end{aligned}$$



Hence $(HD) : (DW) = (MC) : (CQ)$
 $= (MC) : (EW),$

since $(CQ) = (CP) = (EW).$ [Prop. 34]

Therefore

$$\begin{aligned} (MC) : (EW) &= (MC) \pm (HD) : (EW) \pm (DW) \\ &= (MH) : (EU) \dots \dots \dots (1). \end{aligned}$$

Now $(DG) = (HE).$ [Prop. 34]

Therefore, subtracting CX from both,

$$(DX) = (XH),$$

and, adding (XU) to each, $(EU) = (HQ).$

Hence, from (1), since $(EW) = (CQ),$

$$(MC) : (CQ) = (MH) : (HQ),$$

or

$$LQ' : LQ = Q'O : OQ.$$

[Apollonius gives separate proofs of the above for the two cases in which Q, Q' are (1) on the same branch, and (2) on opposite branches, but the second proof is omitted for the sake of brevity.

Eutocius gives two simpler proofs, of which the following is one.

Join PQ and produce it both ways to meet the asymptotes in R, R' . Draw PV parallel to CR' meeting QQ' in V .

Then

$$LV = VL'.$$

But

$$QL = Q'L'; \therefore QV = VQ'.$$

Now

$$QV : VL' = QP : PR'$$

$$= PQ : QR$$

$$= OQ : QL.$$

$$\therefore 2QV : 2VL' = OQ : QL,$$

or

$$QQ' : OQ = LL' : QL;$$

$$\therefore Q'O : OQ = LQ' : LQ.]$$

Proposition 64.

[III. 37, 38, 39, 40.]

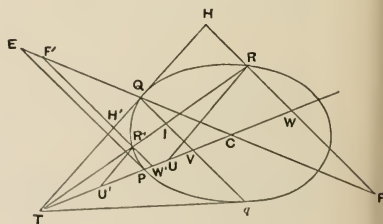
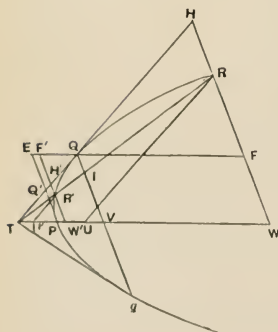
(1) If TQ, Tq be tangents to a conic and any straight line be drawn through T meeting the conic and the chord of contact, the straight line is divided harmonically;

(2) If any straight line be drawn through V , the middle point of Qq , to meet the conic and the parallel through T to Qq [or the polar of the point V], this straight line is also divided harmonically;

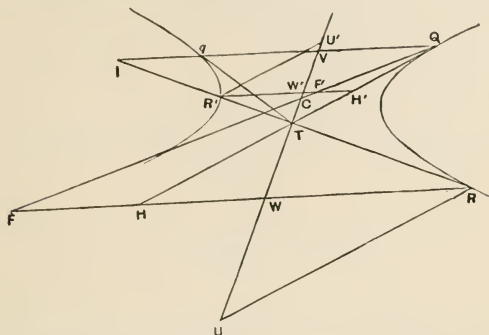
i.e. in the figures drawn below

$$(1) \quad RT : TR' = RI : IR',$$

$$(2) \quad RO : OR' = RV : VR'.$$



Let TP be the diameter bisecting Qq in V . Draw as usual $HRFW$, $H'R'F'W'$, EP ordinate-wise to the diameter TP ; and draw RU , $R'U'$ parallel to QT meeting TP in U , U' .



(1) We have then

$$\begin{aligned} R'I^2 : IR^2 &= H'Q^2 : HQ^2 \\ &= \triangle H'F'Q : \triangle HFQ \\ &= H'TU'R' : HTUR. \quad [\text{Props. 54, 55}] \end{aligned}$$

$$\begin{aligned} \text{Also } R'T^2 : TR^2 &= R'U'^2 : RU^2 \\ &= \triangle R'U'W' : \triangle RUW; \end{aligned}$$

and at the same time

$$\begin{aligned} R'T^2 : TR^2 &= TW'^2 : TW^2 \\ &= \triangle TH'W' : \triangle THW; \end{aligned}$$

$$\begin{aligned} \therefore R'T^2 : TR^2 &= \triangle R'U'W' \sim \triangle TH'W' : \triangle RUW \sim \triangle THW \\ &= H'TU'R' : HTUR \\ &= R'I^2 : IR^2, \text{ from above.} \\ \therefore RT : TR' &= RI : IR'. \end{aligned}$$

(2) We have in this case (it is unnecessary to give more than two figures)

$$\begin{aligned} RV^2 : VR'^2 &= RU^2 : R'U'^2 \\ &= \triangle RUW : \triangle R'U'W'. \end{aligned}$$

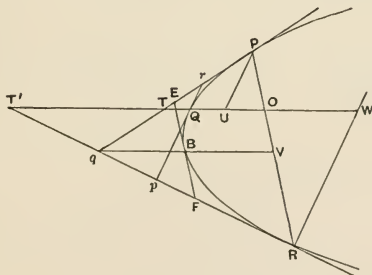
INTERCEPTS MADE ON TWO TANGENTS BY A THIRD.

Proposition 65.

[III. 41.]

If the tangents to a parabola at three points P, Q, R form a triangle pqr , all three tangents are divided in the same proportion, or

$$Pr : rq = rQ : Qp = qp : pR.$$



Let V be the middle point of PR , and join qV , which is therefore a diameter. Draw $T'TQW$ parallel to it through Q , meeting Pq in T and qR in T' . Then QW is also a diameter. Draw the ordinates to it from P, R , viz. PU, RW , which are therefore parallel to pQr .

Now, if qV passes through Q , the proposition is obvious, and the ratios will all be ratios of equality.

If not, we have, by the properties of tangents, drawing EBF the tangent at the point B where qV meets the curve,

$$TQ = QU, \quad T'Q = QW, \quad qB = BV,$$

whence, by parallels,

$$Pr = rT, \quad Tp = pR, \quad qF = FR.$$

Then (1) $rP : PT = EP : Pq = 1 : 2,$

and, alternately, $rP : PE = TP : Pq$
 $= OP : PV,$

whence, doubling the consequents,

$$rP : Pq = OP : PR,$$

and

$$Pr : rq = PO : OR \dots\dots\dots (1).$$

(2) $rQ : Qp = PU : RW,$

since $PU = 2rQ$, and $RW = 2pQ$;

$$\therefore rQ : Qp = PO : OR \dots\dots\dots (2).$$

(3) $FR : Rq = pR : RT',$

and, alternately,

$$FR : Rp = qR : RT' \\ = VR : RO.$$

Therefore, doubling the antecedents,

$$qR : Rp = PR : RO,$$

whence

$$qp : pR = PO : OR \dots\dots\dots (3).$$

It follows from (1), (2) and (3) that

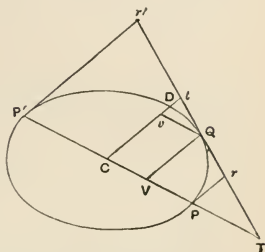
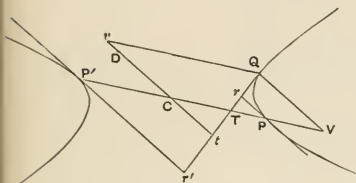
$$Pr : rq = rQ : Qp = qp : pR.$$

Proposition 66.

[III. 42.]

If the tangents at the extremities of a diameter PP' of a central conic be drawn, and any other tangent meet them in r, r' respectively, then

$$Pr \cdot P'r' = CD^2.$$



Draw the ordinates QV, Qv to the conjugate diameters PP' and DD' ; and let the tangent at Q meet the diameters in T, t respectively.

If now, in the case of an *ellipse or circle*, CD pass through Q , the proposition is evident, since in that case $rP, CD, r'P'$ will all be equal.

If not, we have for all three curves

$$CT \cdot CV = CP^2,$$

so that

$$CT : CP = CP : CV$$

$$= CT \sim CP : CP \sim CV$$

$$= PT : PV;$$

$$\therefore CT : CP' = PT : PV,$$

whence

$$CT : P'T = PT : VT.$$

Hence, by parallels, $Ct : P'r' = Pr : QV$

$$= Pr : Cv;$$

$$\therefore Pr \cdot P'r' = Cv \cdot Ct = CD^2.$$

Proposition 67.

[III. 43.]

If a tangent to a hyperbola, LPL' , meet the asymptotes in L, L' , the triangle LCL' has a constant area, or the rectangle $LC \cdot CL'$ is constant.

Draw PD, PF parallel to the asymptotes (as in the third figure of Prop. 62).

Now $LP = PL'$;

$$\therefore CL = 2CD = 2PF,$$

$$CL' = 2CF = 2PD.$$

$$\therefore LC \cdot CL' = 4DP \cdot PF,$$

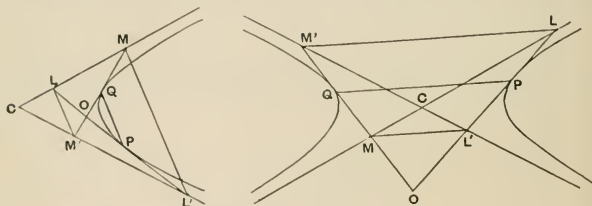
which is constant for all positions of P .

[Prop. 34]

Proposition 68.

[III. 44.]

If the tangents at P, Q to a hyperbola meet the asymptotes respectively in $L, L'; M, M'$, then $LM', L'M$ are each parallel to PQ , the chord of contact.



Let the tangents meet at O .

We have then [Prop. 67]

$$LC \cdot CL' = MC \cdot CM',$$

so that

$$LC : CM' = MC : CL';$$

$\therefore LM', L'M$ are parallel.

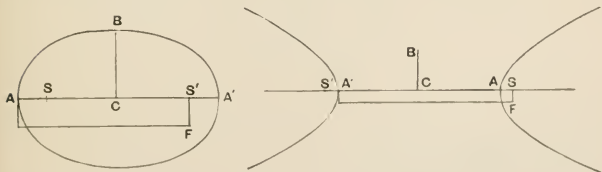
It follows that $OL : LL' = OM' : M'M$,
or, halving the consequents,

$$OL : LP = OM' : M'Q;$$

$\therefore LM', PQ$ are parallel.

FOCAL PROPERTIES OF CENTRAL CONICS.

The foci are not spoken of by Apollonius under any equivalent of that name, but they are determined as the two points on the axis of a central conic (lying in the case of the ellipse between the vertices, and in the case of the hyperbola within each branch, or on the axis produced) such that the rectangles $AS \cdot SA'$, $AS' \cdot S'A'$ are each equal to "one-fourth part of the figure of the conic," i.e. $\frac{1}{4}p_a \cdot AA'$ or CB^2 . The shortened expression by which S , S' are denoted is τὰ ἐκ τῆς παραβολῆς γινόμενα σημεῖα, "the points arising out of the application." The meaning of this will appear from the full description of the method by which they are arrived at, which is as follows: ἐὰν τῷ τετάρτῳ μέρει τοῦ εἵδους ἴσον παρὰ τὸν ἄξονα παραβληθῇ ἐφ' ἑκάτερα ἐπὶ μὲν τῆς ὑπερβολῆς καὶ τῶν ἀντικειμένων ὑπερβάλλον εἶδει τετραγώνῳ, ἐπὶ δὲ τῆς ἐλλείψεως ἐλλείπον, "if there be applied along the axis in each direction [a rectangle] equal to one-fourth part of the figure, in the case of the hyperbola and opposite branches exceeding, and in the case of



the ellipse falling short, by a square figure." This determines two points, which are accordingly τὰ ἐκ τῆς παραβολῆς γενηθέντα

$$\begin{aligned}
 \text{Therefore} \quad A'r' : r'R &= S'r' : r'O \\
 &= Sr : rO, \text{ by similar triangles,} \\
 &= Ar : rR,
 \end{aligned}$$

because the triangles ArS , RrO are similar;

$$\begin{aligned}
 \therefore r'R : Rr &= A'r' : Ar \\
 &= A'T : TA \dots \dots \dots (1).
 \end{aligned}$$

Again, if PN be drawn perpendicular to the axis, we have
[Prop. 13] $A'T : TA = A'N : NA$

$$= r'P : Pr, \text{ by parallels.}$$

Hence, from (1), $r'R : Rr = r'P : Pr$,
and therefore R coincides with P .

It follows that OP is perpendicular to the tangent at P .

Proposition 71.

[III. 48.]

The focal distances of P make equal angles with the tangent at that point.

In the above figures, since the angles rSO , OPr are right
[Props. 69, 70] the points O , P , r , S are concyclic;

$$\therefore \angle SPr = \angle SOr, \text{ in the same segment.}$$

In like manner $\angle S'Pr' = \angle S'Or'$,
and the angles SOr , $S'Or'$ are equal, being the same or opposite angles.

$$\text{Therefore} \quad \angle SPr = \angle S'Pr'.$$

Proposition 72.

[III. 49, 50.]

(1) *If, from either focus, as S , SY be drawn perpendicular to the tangent at any point P , the angle AYA' will be a right angle, or the locus of Y is a circle on the axis AA' as diameter.*

(2) *The line drawn through C parallel to either of the focal distances of P to meet the tangent will be equal in length to CA , or CA' .*

Draw SY perpendicular to the tangent, and join AY , YA' .
Let the rest of the construction be as in the foregoing propositions.

We have then

(1) the angles rAS , rYS are right ;

$\therefore A, r, Y, S$ are concyclic, and

$$\begin{aligned}\angle AYS &= \angle Ars \\ &= \angle r'SA', \text{ since } \angle rSr' \text{ is right} \\ &= \angle r'YA', \text{ in the same segment,}\end{aligned}$$

S, Y, r', A' being concyclic ;

$\therefore$, adding the angle SYA' , or subtracting each angle from it,

$$\angle AYA' = \angle SYr' = \text{a right angle.}$$

Therefore Y lies on the circle having AA' for diameter.

Similarly for Y' .

(2) Draw CZ parallel to SP meeting the tangent in Z , and draw $S'K$ also parallel to SP , meeting the tangent in K .

$$\text{Now} \quad AS \cdot SA' = AS' \cdot S'A',$$

whence $AS = S'A'$, and therefore $CS = CS'$.

Therefore, by parallels, $PZ = ZK$.

Again $\angle S'KP = \angle SPY$, since $SP, S'K$ are parallel,

$$= \angle S'PK ; \quad [\text{Prop. 71}]$$

$$\therefore S'P = S'K.$$

And $PZ = ZK ;$

$\therefore S'Z$ is at right angles to the tangent, or Z coincides with Y' .

But Y' is on the circle having AA' for diameter ;

$$\therefore CY' = CA, \text{ or } CA'.$$

And similarly for CY .

Proposition 73.

[III. 51, 52.]

In an ellipse the sum, and in a hyperbola the difference, of the focal distances of any point is equal to the axis AA' .

We have, as in the last proposition, if SP , CY' , $S'K$ are parallel, $S'K = S'P$. Let $S'P$, CY' meet in M .

Then, since

$$SC = CS',$$

$$SP = 2CM,$$

$$S'P = S'K = 2MY';$$

$$\begin{aligned}\therefore SP \pm S'P &= 2(CM \pm MY') \\ &= 2CY' \\ &= AA'.\end{aligned}$$

[Prop. 72]

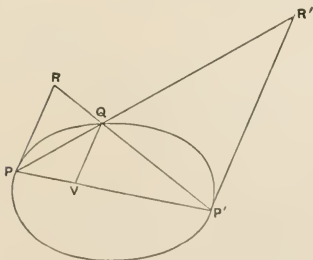
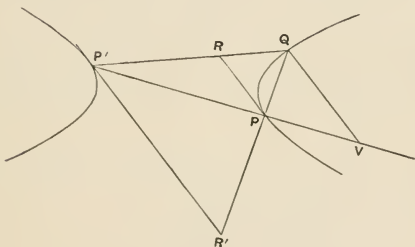
THE LOCUS WITH RESPECT TO THREE LINES &c.

Proposition 74.

[III. 53.]

If PP' be a diameter of a central conic, and Q any other point on it, and if $PQ, P'Q$ respectively meet the tangents at P', P in R', R , then

$$PR \cdot P'R' = DD'^2.$$



Draw the ordinate QV to PP' .

Now $p : PP' = QV^2 : PV \cdot P'V$ [Prop. 8]

$$= (QV : PV) \cdot (QV : P'V)$$

$$= (P'R' : PP') \cdot (PR : PP'), \text{ by similar triangles ;}$$

Hence $p : PP' = PR \cdot P'R' : PP'^2$.

Therefore $PR \cdot P'R' = p \cdot PP'$

$$= DD'^2.$$

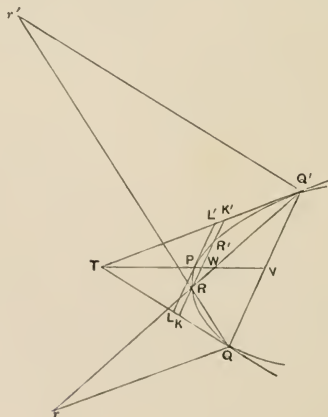
Proposition 75.

[III. 54, 56.]

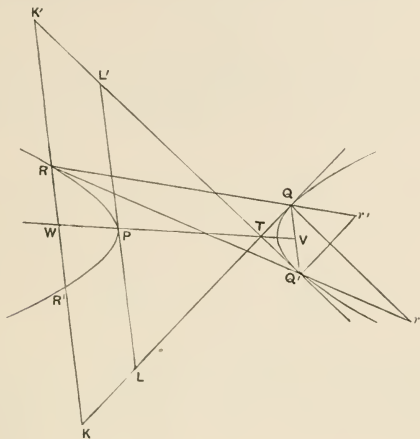
TQ, TQ' being two tangents to a conic, and R any other point on it, if $Qr, Q'r'$ be drawn parallel respectively to TQ', TQ , and if $Qr, Q'R$ meet in r and $Q'r', QR$ in r' , then

$$Qr \cdot Q'r' : QQ'^2 = (PV^2 : PT^2) \times (TQ \cdot TQ' : QV^2),$$

where P is the point of contact of a tangent parallel to QQ' .



Draw through R the ordinate RW (parallel to QQ') meeting the curve again in R' and meeting TQ, TQ' in K, K' respectively; also let the tangent at P meet TQ, TQ' in L, L' . Then, since PV bisects QQ' , it bisects LL', KK', RR' also.



$$\begin{aligned} \text{Now } QL^2 : LP \cdot PL' &= QL^2 : LP^2 \\ &= QK^2 : RK \cdot KR' \quad [\text{Prop. 59}] \\ &= QK^2 : RK \cdot RK'. \end{aligned}$$

$$\text{But } QL \cdot Q'L' : QL^2 = QK \cdot Q'K' : QK^2.$$

Therefore, *ex aequali*,

$$\begin{aligned} QL \cdot Q'L' : LP \cdot PL' &= QK \cdot Q'K' : RK \cdot RK' \\ &= (Q'K' : K'R) \cdot (QK : KR) \\ &= (Qr : QQ') \cdot (Q'r' : QQ') \\ &= Qr \cdot Q'r' : QQ'^2; \end{aligned}$$

$$\begin{aligned} \therefore Qr \cdot Q'r' : QQ'^2 &= QL \cdot Q'L' : LP \cdot PL' \\ &= (QL \cdot Q'L' : LT \cdot TL') \cdot (LT \cdot TL' : LP \cdot PL') \\ &= (PV^2 : PT^2) \cdot (TQ \cdot TQ' : QV^2). \end{aligned}$$

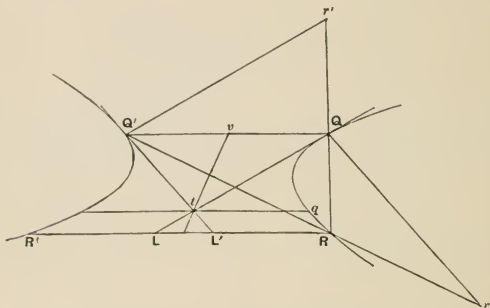
Proposition 76.

[III. 55.]

If the tangents are tangents to opposite branches and meet in t , and if tq is half the chord through t parallel to QQ' , while R, r, r' have the same meaning as before, then

$$Qr \cdot Q'r' : QQ'^2 = tQ \cdot tQ' : tq^2.$$

Let RR' be the chord parallel to QQ' drawn through R , and let it meet tQ, tQ' in L, L' . Then QQ', RR', LL' are all bisected by tv .



$$\begin{aligned} \text{Now} \quad tq^2 : tQ^2 &= R'L \cdot LR : LQ^2 & [\text{Prop. 59}] \\ &= L'R \cdot RL : LQ^2. \end{aligned}$$

$$\text{But} \quad tQ^2 : tQ \cdot tQ' = LQ^2 : LQ \cdot L'Q'.$$

Therefore, *ex aequali*,

$$\begin{aligned} tq^2 : tQ \cdot tQ' &= L'R \cdot RL : LQ \cdot L'Q' \\ &= (L'R : L'Q') \cdot (RL : LQ) \\ &= (QQ' : Qr) \cdot (QQ' : Q'r') = QQ'^2 : Qr \cdot Q'r'. \end{aligned}$$

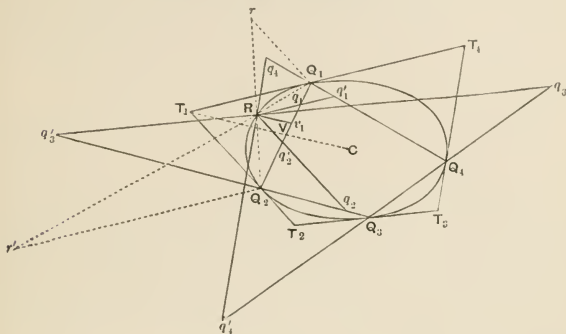
$$\text{Thus} \quad Qr \cdot Q'r' : QQ'^2 = tQ \cdot tQ' : tq^2.$$

[It is easy to see that the last two propositions give the property of the *three-line locus*. For, since the two tangents and the chord of contact are fixed while the position of R alone varies, the result may be expressed thus,

$$Qr \cdot Q'r' = (\text{const.}).$$

Now suppose Q_1, Q_2, T_1 in the accompanying figure substituted for Q, Q', T respectively in the first figure of Prop. 75, and we have

$$Q_1r \cdot Q_2r' = (\text{const.})$$



Draw Rq_1, Rq_2' parallel respectively to T_1Q_1, T_1Q_2 and meeting Q_1Q_2 in q_1, q_2' . Also let Rv_1 be drawn parallel to the diameter CT_1 and meeting Q_1Q_2 in v_1 .

Then, by similar triangles,

$$Q_1r : Rq_2' = Q_1Q_2 : Q_2q_2',$$

$$Q_2r' : Rq_1 = Q_1Q_2 : Q_1q_1.$$

$$\text{Hence } Q_1r \cdot Q_2r' : Rq_1 \cdot Rq_2' = Q_1Q_2^2 : Q_1q_1 \cdot Q_2q_2'.$$

But $Rq_1 \cdot Rq_2' : Rv_1^2 = T_1Q_1 \cdot T_1Q_2 : T_1V^2$, by similar triangles

$$\therefore Rq_1 \cdot Rq_2' : Rv_1^2 = (\text{const.}).$$

Also $Q_1Q_2^2$ is constant, and $Q_1r \cdot Q_2r'$ is constant, as proved.

It follows that

$$Rv_1^2 : Q_1q_1 \cdot Q_2q_2' = (\text{const.}).$$

But Rv_1 is the distance of R from Q_1Q_2 , the chord of contact measured in a fixed direction (parallel to CT_1); and Q_1q_1, Q_2q_2' are equal to the distances of R from the tangents T_1Q_1, T_1Q_2 respectively, measured in a fixed direction (parallel to the chord of contact). If the distances are measured in any

other fixed directions, they will be similarly related, and the constant value of the ratio will alone be changed.

Hence R is such a point that, if three straight lines be drawn from it to meet three fixed straight lines at given angles, the rectangle contained by two of the straight lines so drawn bears a constant ratio to the square on the third. In other words, a conic is a "three-line locus" where the three lines are any two tangents and the chord of contact.

The *four-line locus* can be easily deduced from the three-line locus, as presented by Apollonius, in the following manner.

If $Q_1Q_2Q_3Q_4$ be an inscribed quadrilateral, and the tangents at Q_1, Q_2 meet at T_1 , the tangents at Q_2, Q_3 at T_2 and so on, suppose Rq_2, Rq_3' drawn parallel to the tangents at Q_2, Q_3 respectively and meeting Q_2Q_3 in q_2, q_3' (in the same way as Rq_1, Rq_2' were drawn parallel to the tangents at Q_1, Q_2 to meet Q_1Q_2), and let similar pairs of lines Rq_3, Rq_4' and Rq_4, Rq_1' be drawn to meet Q_3Q_4 and Q_4Q_1 respectively.

Also suppose Rv_2 drawn parallel to the diameter CT_2 , meeting Q_2Q_3 in v_2 , and so on.

Then we have

$$\left. \begin{aligned} Q_1q_1 \cdot Q_2q_2' &= k_1 \cdot Rv_1^2 \\ Q_2q_2 \cdot Q_3q_3' &= k_2 \cdot Rv_2^2 \\ Q_3q_3 \cdot Q_4q_4' &= k_3 \cdot Rv_3^2 \\ Q_4q_4 \cdot Q_1q_1' &= k_4 \cdot Rv_4^2 \end{aligned} \right\} \begin{array}{l} \text{where } k_1, k_2, k_3, k_4 \text{ are} \\ \text{constants.} \end{array}$$

Hence we derive

$$\frac{Rv_1^2 \cdot Rv_3^2}{Rv_2^2 \cdot Rv_4^2} = k \cdot \frac{Q_1q_1}{Q_1q_1'} \cdot \frac{Q_3q_3}{Q_3q_3'} \cdot \frac{Q_2q_2'}{Q_2q_2} \cdot \frac{Q_4q_4'}{Q_4q_4}$$

where k is some constant.

But the triangles $Q_1q_1q_1', Q_2q_2q_2'$ etc. are given in species, as all their sides are in fixed directions. Hence all the ratios

$\frac{Q_1q_1}{Q_1q_1'}$, etc. are constant;

$$\therefore \frac{Rv_1 \cdot Rv_3}{Rv_2 \cdot Rv_4} = (\text{const.}).$$

But Rv_1, Rv_2, Rv_3, Rv_4 are straight lines drawn in fixed directions (parallel to CT_1 , etc.) to meet the sides of the inscribed quadrilateral $Q_1Q_2Q_3Q_4$.

Hence the conic has the property of the four-line locus with respect to the sides of any inscribed quadrilateral.]

The beginning of Book IV. of Apollonius' work contains a series of propositions, 23 in number, in which he proves the converse of Propositions 62, 63, and 64 above for a great variety of different cases. The method of proof adopted is the *reductio ad absurdum*, and it has therefore been thought unnecessary to reproduce the propositions.

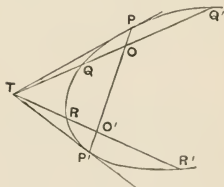
It may, however, be observed that one of them [IV. 9] gives a method of drawing two tangents to a conic from an external point.

Draw any two straight lines through T each cutting the conic in two points as Q, Q' and R, R' . Divide QQ' in O and RR' in O' so that

$$TQ : TQ' = QO : OQ',$$

$$TR : TR' = RO' : O'R'.$$

Join OO' , and produce it both ways to meet the conic in P, P' . Then P, P' are the points of contact of the two tangents from T .



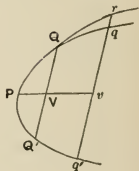
INTERSECTING CONICS.

Proposition 77.

[IV. 24.]

No two conics can intersect in such a way that part of one of them is common to both, while the rest is not.

If possible, let a portion $q'Q'PQ$ of a conic be common to two, and let them diverge at Q . Take Q' any other point on the common portion and join QQ' . Bisect QQ' in V and draw the diameter PV . Draw $rvvq'$ parallel to QQ' .



Then the line through P parallel to QQ' will touch both curves and we shall have in one of them $qv = vq'$, and in the other $rv = vq'$;

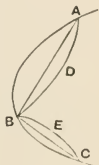
$\therefore rv = qv$, which is impossible.

There follow a large number of propositions with regard to the number of points in which two conics can meet or touch each other, but to give all these propositions in detail would require too much space. They have accordingly been divided into five groups, three of which can be combined in a general enunciation and are accordingly given as Props. 78, 79 and 80, while indications are given of the proofs by which each particular case under all the five groups is established. The terms "conic" and "hyperbola" in the various enunciations do not (except when otherwise stated) include the double-branch hyperbola but only the single branch. The term "conic" must be understood as including a circle.

Group I. Propositions depending on the more elementary considerations affecting conics.

1. Two conics having their concavities in opposite directions will not meet in more than two points. [IV. 35.]

If possible, let ABC , $ADBEC$ be two such conics meeting in three points, and draw the chords of contact AB , BC . Then AB , BC contain an angle towards the same parts as the concavity of ABC . And for the same reason they contain an angle towards the same parts as the concavity of $ADBEC$.



Therefore the concavity of the two curves is in the same direction: which is contrary to the hypothesis.

2. If a conic meet one branch of a hyperbola in two points, and the concavities of the conic and the branch are in the same direction, the part of the conic produced beyond the chord of contact will not meet the opposite branch of the hyperbola. [IV. 36.]

The chord joining the two points of intersection will cut both the lines forming one of the angles made by the asymptotes of the double hyperbola. It will not therefore fall within the opposite angle between the asymptotes and so cannot meet the opposite branch. Therefore neither can the part of the conic more remote than the said chord.

3. If a conic meet one branch of a hyperbola, it will not meet the other branch in more points than two. [IV. 37.]

The conic, being a one-branch curve, must have its concavity in the opposite direction to that of the branch which it meets in two points, for otherwise it could not meet the opposite branch in a third point [by the last proposition]. The proposition therefore follows from (1) above. The same is true if the conic touches the first branch.

4. A conic touching one branch of a hyperbola with its concave side will not meet the opposite branch. [IV. 39.]

Both the conic and the branch which it touches must be on the same side of the common tangent and therefore will be



separated by the tangent from the opposite branch. Whence the proposition follows.

5. If one branch of a hyperbola meet one branch of another hyperbola with concavity in the opposite direction in two points, the opposite branch of the first hyperbola will not meet the opposite branch of the second. [IV. 41.]

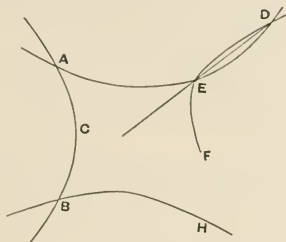


The chord joining the two points of concurrence will fall across one asymptotal angle in each hyperbola. It will not therefore fall across the opposite asymptotal angle and therefore will not meet either of the opposite branches. Therefore neither will the opposite branches themselves meet, being separated by the chord referred to.

6. If one branch of a hyperbola meet both branches of another hyperbola, the opposite branch of the former will not meet either branch of the second in two points. [IV. 42.]

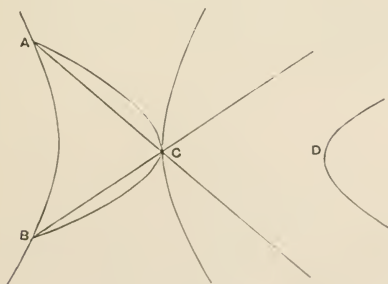
For, if possible, let the second branch of the former meet one branch of the latter in D, E . Then, joining DE , we use

the same argument as in the last proposition. For DE crosses one asymptotal angle of each hyperbola, and it will therefore not meet either of the branches opposite to the branches DE . Hence those branches are separated by DE and therefore cannot meet one another: which contradicts the hypothesis.



Similarly, if the two branches DE touch, the result will be the same, an impossibility.

7. If one branch of a hyperbola meet one branch of another hyperbola with concavity in the same direction, and if it also meet the other branch of the second hyperbola in one point, then the opposite branch of the first hyperbola will not meet either branch of the second. [IV. 45.]



A, B being the points of meeting with the first branch and

C that with the opposite branch, by the same principle as before, neither AC nor BC will meet the branch opposite to ACB . Also they will not meet the branch C opposite to AB in any other point than C , for, if either met it in two points, it would not meet the branch AB , which, however, it does, by hypothesis.

Hence D will be within the angle formed by AC , BC produced and will not meet C or AB .

8. If a hyperbola touch one of the branches of a second hyperbola with its concavity in the opposite direction, the opposite branch of the first will not meet the opposite branch of the second. [IV. 54.]

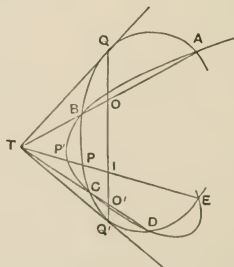
The figure is like that in (6) above except that in this case D and E are two consecutive points; and it is seen in a similar manner that the second branches of the two hyperbolas are separated by the common tangent to the first branches, and therefore the second branches cannot meet.

Group II. containing propositions capable of being expressed in one general enunciation as follows:

Proposition 78.

No two conics (including under the term a hyperbola with two branches) can intersect in more than four points.

1. Suppose the double-branch hyperbola to be alone excluded. [IV. 25.]



If possible, let there be five points of intersection A, B, C, D, E , being *successive* intersections, so that there are no others between. Join AB, DC and produce them. Then

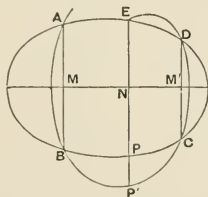
(a) if they meet, let them meet at T . Let O, O' be taken on AB, DC such that TA, TD are harmonically divided. If OO' be joined and produced it will meet each conic, and the lines joining the intersections to T will be tangents to the conics. Then TE cuts the two conics in different points P, P' , since it does not pass through any common point except E .

Therefore $ET : TP = EI : IP$ }
and $ET : TP' = EI : IP'$ }

where OO', TE intersect at I .

But these ratios cannot hold simultaneously; therefore the conics do not intersect in a fifth point E .

(b) If AB, DC are parallel, the conics will be either ellipses or circles. Bisect AB, DC at M, M' ; MM' is then



a diameter. Draw $ENPP'$ through E parallel to AB or DC , meeting MM' in N and the conics in P, P' . Then, since MM' is a diameter of both,

$$NP = NE = NP',$$

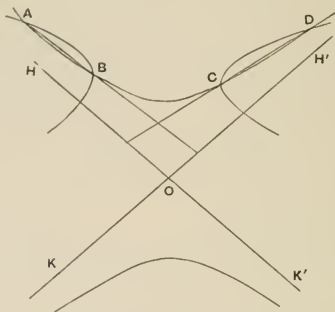
which is impossible.

Thus the conics do not intersect in more than four points.

2. A conic section not having two branches will not meet a double-branch hyperbola in more than four points. [IV. 38.]

This is clear from the fact that [Group I. 3] the conic meeting one branch will not meet the opposite branch in more points than two.

3. If one branch of a hyperbola cut each branch of a second hyperbola in two points, the opposite branch of the first hyperbola will not meet either branch of the second. [IV. 43.]



The text of the proof in Apollonius is corrupt, but Eutocius gives a proof similar to that in Group I. 5 above. Let HOH' be the asymptotal angle containing the one branch of the first hyperbola, and KOK' that containing the other branch. Now AB , meeting one branch of the second hyperbola, will not meet the other, and therefore AB separates the latter from the asymptote OK' . Similarly DC separates the former branch from OK . Therefore the proposition follows.

4. If one branch of a hyperbola cut one branch of a second in four points, the opposite branch of the first will not meet the opposite branch of the second. [IV. 44.]

The proof is like that of 1 (a) above. If E is the supposed fifth point and T is determined as before, ET meets the intersecting branches in separate points, whence the harmonic property produces an absurdity.

5. If one branch of a hyperbola meet one branch of a second in three points, the other branch of the first will not meet the other branch of the second in more than one point. [IV. 46.]

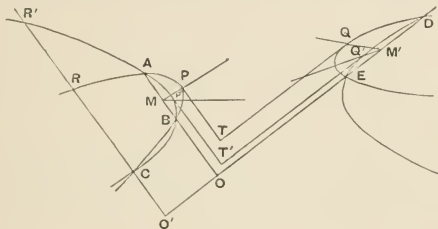
Let the first two branches intersect in A, B, C , and (if possible) the other two in D, E . Then

(a) if AB, DE be parallel, the line joining their middle points will be a diameter of both conics, and the parallel chord through C in both conics will be bisected by the diameter; which is impossible.

(b) If AB, DE be not parallel, let them meet in O .

Bisect AB, DE in M, M' , and draw the diameters MP, MP' and $M'Q, M'Q'$ in the respective hyperbolas. Then the tangents at P, P' will be parallel to AO , and the tangents at Q, Q' parallel to BO .

Let the tangents at P, Q and P', Q' meet in T, T' .



Let CRR' be parallel to AO and meet the hyperbolas in R, R' , and DO in O' .

Then
$$TP^2 : TQ^2 = AO \cdot OB : DO \cdot OE$$

$$= T'P'^2 : T'Q'^2. \quad [\text{Prop. 59}]$$

It follows that

$$RO' \cdot O'C : DO' \cdot O'E = R'O' \cdot O'C : D'O' \cdot O'E,$$

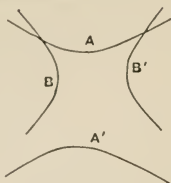
whence
$$RO' \cdot O'C = R'O' \cdot O'C;$$

which is impossible.

Therefore, etc.

6. The two branches of a hyperbola do not meet the two branches of another hyperbola in more points than four. [IV. 55.]

Let A, A' be the two branches of the first hyperbola and B, B' the two branches of the second.



Then (a) if A meet B, B' each in two points, the proposition follows from (3) above;

(b) if A meet B in two points and B' in one point, A' cannot meet B' at all [Group I. 5], and it can only meet B in one point, for if A' met B in two points A could not have met B' (which it does);

(c) if A meet B in two points and A' meet B , A' will not meet B' [Group I. 5], and A' cannot meet B in more points than two [Group I. 3];

(d) if A meet B in one point and B' in one point, A' will not meet either B or B' in two points [Group I. 6];

(e) if the branches A, B have their concavities in the same direction, and A cut B in four points, A' will not cut B' [case (4) above] nor B [case (2) above];

(f) if A meet B in three points, A' will not meet B' in more than one point [case (5) above].

And similarly for all possible cases.

Group III. being particular cases of

Proposition 79.

Two conics (including double hyperbolas) which touch at one point cannot intersect in more than two other points.

1. The proposition is true of all conics excluding hyperbolas with two branches. [IV. 26.]

The proof follows the method of Prop. 78 (1) above.

2. If one branch of a hyperbola touch one branch of another in one point and meet the other branch of the second hyperbola in two points, the opposite branch of the first will not meet either branch of the second. [IV. 47.]

The text of Apollonius' proof is corrupt, but the proof of Prop. 78 (3) can be applied.

3. If one branch of a hyperbola touch one branch of a second in one point and cut the same branch in two other points, the opposite branch of the first does not meet either branch of the second. [IV. 48.]

Proved by the harmonic property like Prop. 78 (4).

4. If one branch of a hyperbola touch one branch of a second hyperbola in one point and meet it in one other point, the opposite branch of the first will not meet the opposite branch of the second in more than one point. [IV. 49.]

The proof follows the method of Prop. 78 (5).

5. If one branch of a hyperbola touch one branch of another hyperbola (having its concavity in the same direction), the opposite branch of the first will not meet the opposite branch of the second in more than two points. [IV. 50.]

The proof follows the method of Prop. 78 (5), like the last case (4).

6. If a hyperbola with two branches touch another hyperbola with two branches in one point, the hyperbolas will not meet in more than two other points. [IV. 56.]

The proofs of the separate cases follow the methods employed in Group I. 3, 5, and 8.

Group IV. merging in

Proposition 80.

No two conics touching each other at two points can intersect at any other point.

1. The proposition is true of all conics excluding hyperbolas with two branches. [IV. 27, 28, 29.]

Suppose the conics touch at A, B . Then, if possible, let them also cut at C .

(a) If the tangents are not parallel and C does not lie between A and B , the proposition is proved from the harmonic property;

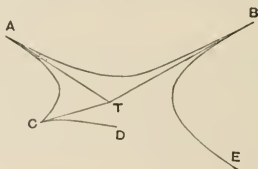
(b) if the tangents are parallel, the absurdity is proved by the bisection of the chord of each conic through C by the chord of contact which is a diameter;

(c) if the tangents are not parallel, and C is between A and B , draw TV from the point of intersection of the tangents to the middle point of AB . Then TV cannot pass through C , for then the parallel through C to AB would touch both conics, which is absurd. And the bisection of the chords parallel to AB through C in each conic results in an absurdity.

2. If a single-branch conic touch each branch of a hyperbola, it will not intersect either branch in any other point. [IV. 40.]

This follows by the method employed in Group I. 4.

3. If one branch of a hyperbola touch each branch of a second hyperbola, the opposite branch of the first will not meet either branch of the second. [IV. 51.]



Let the branch AB touch the branches AC, BE in A, B . Draw the tangents at A, B meeting in T . If possible, let CD , the opposite branch to AB , meet AC in C . Join CT .

Then T is within the asymptotes to AB , and therefore CT falls within the angle ATB . But BT , touching BE , cannot meet the opposite branch AC . Therefore BT falls on the side of CT remote from the branch AC , or CT passes through the angle adjacent to ATB ; which is impossible, since it falls within the angle ATB .

4. If one branch of one hyperbola touch one branch of another in one point, and if also the other branches touch in one point, the concavities of each pair being in the same direction, there are no other points of intersection. [IV. 52.]

This is proved at once by means of the bisection of chords parallel to the chord of contact.

5. If one branch of a hyperbola touch one branch of another in two points, the opposite branches do not intersect. [IV. 53.]

This is proved by the harmonic property.

6. If a hyperbola with two branches touch another hyperbola with two branches in two points, the hyperbolas will not meet in any other point. [IV. 57.]

The proofs of the separate cases follow those of (3), (4), (5) above and Group I. 8.

Group V. Propositions respecting double contact between conics.

1. A parabola cannot touch another parabola in more points than one. [IV. 30.]

This follows at once from the property that $TP = PV$.

2. A parabola, if it fall outside a hyperbola, cannot have double contact with the hyperbola. [IV. 31.]

For the hyperbola

$$\begin{aligned} CV : CP &= CP : CT \\ &= CV - CP : CP - CT \\ &= PV : PT. \end{aligned}$$

Therefore

$$PV > PT.$$

And for the parabola $P'V = P'T$: therefore the hyperbola falls outside the parabola, which is impossible.

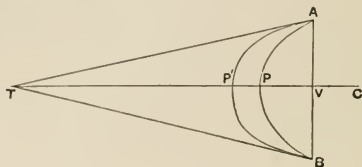
3. A parabola cannot have internal double contact with an ellipse or circle. [IV. 32.]

The proof is similar to the preceding.

4. A hyperbola cannot have double contact with another hyperbola having the same centre. [IV. 33.]

Proved by means of CV . $CT = CP^2$.

5. If an ellipse have double contact with an ellipse or a circle having the same centre, the chord of contact will pass through the centre. [IV. 34.]



Let (if possible) the tangents at A, B meet in T , and let V be the middle point of AB . Then TV is a diameter. If possible, let C be the centre.

Then $CP^2 = CV \cdot CT = CP'^2$, which is absurd. Therefore the tangents at A, B do not meet, i.e. they are parallel. Therefore AB is a diameter and accordingly passes through the centre.

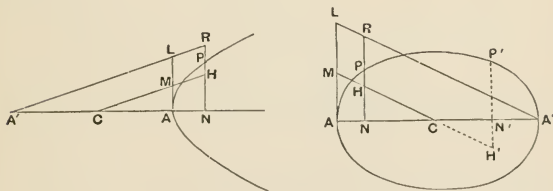
NORMALS AS MAXIMA AND MINIMA.

Proposition 81. (Preliminary.)

[V. 1, 2, 3.]

If in an ellipse or a hyperbola AM be drawn perpendicular to the axis AA' and equal to one-half its parameter, and if CM meet the ordinate PN of any point P on the curve in H , then

$$PN^2 = 2 (\text{quadrilateral } MANH).$$



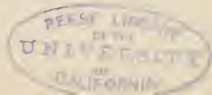
Let AL be twice AM , i.e. let AL be the latus rectum or parameter. Join $A'L$ meeting PN in R . Then $A'L$ is parallel to CM . Therefore $HR = LM = AM$.

Now $PN^2 = AN \cdot NR$; [Props. 2, 3]

$$\therefore PN^2 = AN (AM + HN)$$

$$= 2 (\text{quadrilateral } MANH).$$

In the particular case where P is between C and A' in the



ellipse, the quadrilateral becomes the difference between two triangles, and

$$P'N'^2 = 2 (\triangle CAM - \triangle CN'H').$$

Also, if P be the end of the minor axis of the ellipse, the quadrilateral becomes the triangle CAM , and

$$BC^2 = 2 \triangle CAM.$$

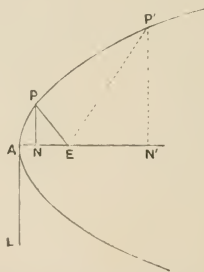
[The two last cases are proved by Apollonius in separate propositions. Cf. the note on Prop. 23 above, p. 40.]

Proposition 82.

[V. 4.]

*In a parabola, if E be a point on the axis such that AE is equal to half the latus rectum, then the **minimum** straight line from E to the curve is AE ; and, if P be any other point on the curve, PE increases as P moves further from A on either side. Also for any point*

$$PE^2 = AE^2 + AN^2.$$



Let AL be the parameter or latus rectum.

$$\begin{aligned} \text{Then} \quad PN^2 &= AL \cdot AN \\ &= 2AE \cdot AN. \end{aligned}$$

Adding EN^2 , we have

$$\begin{aligned} PE^2 &= 2AE \cdot AN + EN^2 \\ &= 2AE \cdot AN + (AE - AN)^2 \\ &= AE^2 + AN^2. \end{aligned}$$

Thus $PE^2 > AE^2$ and increases with AN , i.e. as P moves further and further from A .

Also the *minimum* value of PE is AE , or AE is the shortest straight line from E to the curve.

[In this proposition, as in the succeeding propositions, Apollonius takes three cases, (1) where N is between A and E , (2) where N coincides with E and PE is therefore perpendicular to the axis, (3) where AN is greater than AE , and he proves the result separately for each. The three cases will for the sake of brevity be compressed, where possible, into one.]

Proposition 83.

[V. 5, 6.]

If E be a point on the axis of a hyperbola or an ellipse such that AE is equal to half the latus rectum, then AE is the least of all the straight lines which can be drawn from E to the curve; and, if P be any other point on it, PE increases as P moves further from A on either side, and

$$PE^2 = AE^2 + AN^2 \cdot \frac{AA' \pm p_a}{AA'} [= AE^2 + e^2 \cdot AN^2]$$

(where the upper sign refers to the hyperbola)*.

Also in the ellipse EA' is the maximum straight line from E to the curve.

Let AL be drawn perpendicular to the axis and equal to the parameter; and let AL be bisected at M , so that $AM = AE$.

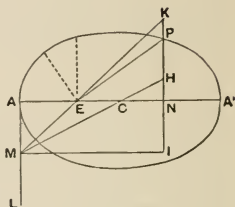
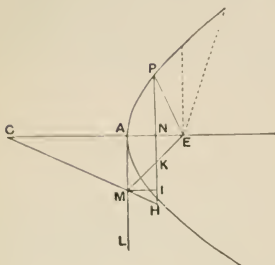
Let P be any point on the curve, and let PN (produced if necessary) meet CM in H and EM in K . Join EP , and draw MI perpendicular to HK . Then, by similar triangles,

$$MI = IK, \text{ and } EN = NK.$$

* The area represented by the second term on the right-hand side of the equation is of course described, in Apollonius' phrase, as the rectangle on the base AN similar to that contained by the axis (as base) and the sum (or difference) of the axis and its parameter. A similar remark applies to the similar expression on the next page.

Now $PN^2 = 2$ (quadrilateral $MANH$), [Prop. 81]
 and $EN^2 = 2 \triangle ENK$;

$$\begin{aligned}\therefore PE^2 &= 2 (\triangle EAM + \triangle MHK) \\ &= AE^2 + MI \cdot HK \\ &= AE^2 + MI \cdot (IK \pm IH) \\ &= AE^2 + MI \cdot (MI \pm IH) \dots \dots \dots (1).\end{aligned}$$



Now $MI : IH = CA : AM = AA' : p_a$.

Therefore $MI \cdot (MI \pm IH) : AA' \cdot (AA' \pm p_a) = MI^2 : AA'^2$,

$$\begin{aligned}\text{or } MI \cdot (MI \pm IH) &= \frac{MI^2}{AA'^2} \cdot AA' \cdot (AA' \pm p_a) \\ &= MI^2 \cdot \frac{AA' \pm p_a}{AA'} \\ &= AN^2 \cdot \frac{AA' \pm p_a}{AA'},\end{aligned}$$

whence, by means of (1),

$$PE^2 = AE^2 + AN^2 \cdot \frac{AA' \pm p_a}{AA'}.$$

It follows that AE is the *minimum* value of PE , and that PE increases with AN , i.e. as the point P moves further from A .

Also in the ellipse the *maximum* value of PE^2 is

$$\begin{aligned}AE^2 + AA' (AA' - p_a) &= AE^2 + AA'^2 - 2AE \cdot AA' \\ &= EA'^2.\end{aligned}$$

Proposition 84.

[V. 7.]

If any point O be taken on the axis of any conic such that $AO < \frac{1}{2}p_a$, then OA is the **minimum** straight line from O to the curve, and OP (if P is any other point on it) increases as P moves further and further from A .

Let AE be set off along the axis equal to half the parameter, and join PE , PO , PA .

Then [Props. 82, 83] $PE > AE$,

so that $\angle PAE > \angle APE$;

and *a fortiori*

$$\angle PAO > \angle APO,$$

so that $PO > AO$.

And, if P' be another point more remote from A ,

$$P'E > PE.$$

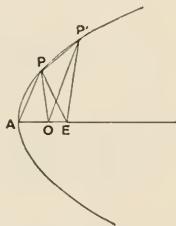
$$\therefore \angle EPP' > \angle EP'P;$$

and *a fortiori*

$$\angle OPP' > \angle OP'P.$$

$$\therefore OP' > OP,$$

and so on.


Proposition 85.

[V. 8.]

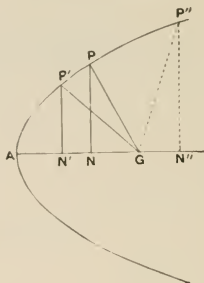
In a parabola, if G be a point on the axis such that $AG > \frac{1}{2}p_a$, and if N be taken between A and G such that

$$NG = \frac{p_a}{2},$$

then, if NP is drawn perpendicular to the axis meeting the curve in P , PG is the **minimum** straight line from G to the curve [or the normal at P].

If P' be any other point on the curve, $P'G$ increases as P' moves further from P in either direction.

Also $P'G^2 = PG^2 + NN'^2$.



We have $P'N'^2 = p_a \cdot AN'$
 $= 2NG \cdot AN'$.

Also $N'G^2 = NN'^2 + NG^2 \pm 2NG \cdot NN'$
 (according to the position of N').

Therefore, adding,

$$\begin{aligned} P'G^2 &= 2NG \cdot AN + NN'^2 + NG^2 \\ &= PN^2 + NG^2 + NN'^2 \\ &= PG^2 + NN'^2. \end{aligned}$$

Thus it is clear that PG is the *minimum* straight line from G to the curve [or the normal at P].

And $P'G$ increases with NN' , i.e. as P' moves further from P in either direction.

Proposition 86.

[V. 9, 10, 11.]

In a hyperbola or an ellipse, if G be any point on AA' (within the curve) such that $AG > \frac{p_a}{2}$, and if GN be measured towards the nearer vertex A so that

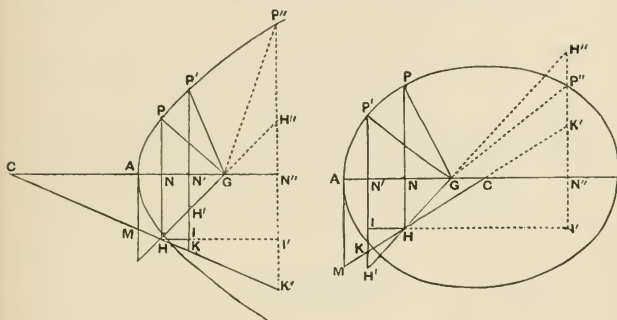
$$NG : CN = p_a : AA' [= CB^2 : CA^2],$$

then, if the ordinate through N meet the curve in P , PG is the **minimum** straight line from G to the curve [or PG is the normal at P]; and, if P' be any other point on the curve, $P'G$ increases as P' moves further from P on either side.

$$\text{Also} \quad P'G^2 - PG^2 = NN'^2 \cdot \frac{AA' \pm p_a}{AA'}$$

$$[= e^2 \cdot NN'^2],$$

where $P'N'$ is the ordinate from P' .



Draw AM perpendicular to the axis and equal to half the parameter. Join CM meeting PN in H and $P'N'$ in K . Join GH meeting $P'N'$ in H' .

Then since, by hypothesis,

$$NG : CN = p_a : AA',$$

and, by similar triangles,

$$\begin{aligned} NH : CN &= AM : AC \\ &= p_a : AA', \end{aligned}$$

it follows that

$$NH = NG,$$

whence also

$$N'H' = N'G.$$

$$\text{Now} \quad PN^2 = 2 \text{ (quadrilateral } MANH), \quad [\text{Prop. 81}]$$

$$NG^2 = 2\Delta HNG.$$

Therefore, by addition, $PG^2 = 2 \text{ (quadrilateral } AMHG).$

$$\begin{aligned}\text{Also } P'G^2 &= P'N'^2 + N'G^2 = 2 (\text{quadr. } AMKN') + 2 \triangle H'N'G \\ &= 2 (\text{quadr. } AMHG) + 2 \triangle HH'K.\end{aligned}$$

$$\begin{aligned}\therefore P'G^2 - PG^2 &= 2 \triangle HH'K \\ &= HI \cdot (H'I \pm IK) \\ &= HI \cdot (HI \pm IK) \\ &= HI^2 \cdot \frac{CA \pm AM}{CA} = NN'^2 \cdot \frac{AA' \pm p_a}{AA'}.\end{aligned}$$

Thus it follows that PG is the *minimum* straight line from G to the curve, and $P'G$ increases with NN' as P' moves further from P in either direction.

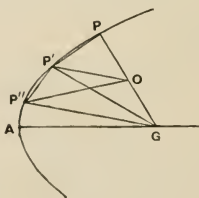
In the ellipse GA' will be the *maximum* straight line from G to the curve, as is easily proved in a similar manner.

COR. In the particular case where G coincides with C , the centre, the two minimum straight lines are proved in a similar manner to be CB , CB' , and the two maxima CA , CA' , and CP increases continually as P moves from B to A .

Proposition 87.

[V. 12.]

If G be a point on the axis of a conic and GP be the minimum straight line from G to the curve [or the normal at P], and if O be any point on PG , then OP is the minimum straight line from O to the curve, and OP' continually increases as P' moves from P to A [or to A'].



Since

$$\begin{aligned}P'G &> PG, \\ \angle GPP' &> \angle GPP.\end{aligned}$$

Therefore, *a fortiori*,

$$\angle OPP' > \angle OP'P,$$

or

$$OP' > OP.$$

Similarly

$$OP'' > OP' \text{ [&c. as in Prop. 84].}$$

[There follow three propositions establishing for the three curves, by *reductio ad absurdum*, the converse of the propositions 85 and 86 just given. It is also proved that the normal makes with the axis towards the nearer vertex an acute angle.]

Proposition 88.

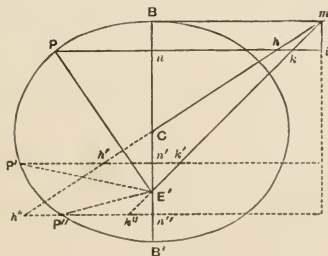
[V. 16, 17, 18.]

If E' be a point on the minor axis of an ellipse at a distance from B equal to half the parameter of BB' [or $\frac{CA^2}{CB}$], then $E'B$ is the **maximum** straight line from E to the curve; and, if P be any other point on it, $E'P$ diminishes as P moves further from B on either side.

$$\text{Also } E'B^2 - E'P^2 = Bn^2 \cdot \frac{pb - BB'}{BB'} \left[= Bn^2 \cdot \frac{CA^2 - CB^2}{CB^2} \right].$$

Apollonius proves this separately for the cases (1) where $\frac{pb}{2} < BB'$, (2) where $\frac{pb}{2} = BB'$, and (3) where $\frac{pb}{2} > BB'$.

The method of proof is the same for all three cases, and only the first case of the three is given here.



By Prop. 81 (which is applicable to either axis) we have, if $Bm = \frac{p_b}{2} = BE'$, and Pn meets Cm , $E'm$ in h , k respectively,

$$Pn^2 = 2(\text{quadrilateral } mBnh).$$

Also $nE'^2 = 2\triangle n k E'.$

$$\therefore PE'^2 = 2\triangle mBE' - 2\triangle m h k.$$

But $BE'^2 = 2\triangle mBE'.$

$$\begin{aligned} \therefore BE'^2 - PE'^2 &= 2\triangle m h k \\ &= mi \cdot (hi - ki) = mi \cdot (hi - mi) \\ &= mi^2 \cdot \frac{mB - CB}{CB} \\ &= Bn^2 \cdot \frac{p_b - BB'}{BB'}, \end{aligned}$$

whence the proposition follows.

Proposition 89.

[V. 19.]

*If BE' be measured along the minor axis of an ellipse equal to half the parameter $\left[\text{or } \frac{CA^2}{CB} \right]$ and any point O be taken on the minor axis such that $BO > BE'$, then OB is the **maximum** straight line from O to the curve; and, if P be any other point on it, OP diminishes continually as P moves in either direction from B to B' .*

The proof follows the method of Props. 84, 87.

Proposition 90.

[V. 20, 21, 22.]

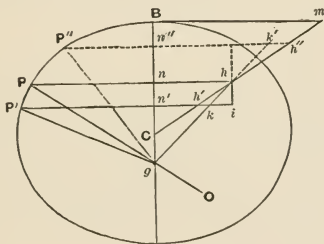
If g be a point on the minor axis of an ellipse such that $Bg > BC$ and $Bg < \frac{1}{2}pb$ [or $\frac{CA^2}{CB}$], and if Cn be measured towards B so that

$$Cn : ng = BB' : p_b [= CB^2 : CA^2],$$

then the perpendicular through n to BB' will meet the curve in two points P such that Pg is the **maximum** straight line from g to the curve.

Also, if P' be any other point on the curve, $P'g$ diminishes as P' moves further from P on either side to B or B' , and

$$Pg^2 - P'g^2 = nn'^2 \cdot \frac{pb - BB'}{BB'} \left[= nn'^2 \cdot \frac{CA^2 - CB^2}{CB^2} \right].$$



Draw Bm perpendicular to BB' and equal to half its parameter p_b . Join Cm meeting Pn in h and $P'n'$ in h' , and join gh meeting $P'n'$ in k .

Then since, by hypothesis,

$$Cn : ng = BB' : p_b = BC : Bm,$$

and

$Cn : nh = BC : Bm$, by similar triangles,

it follows that $ng = nh$. Also $gn' = n'h$, and $hi = ik$, where hi is perpendicular to $P'n'$.

Now $Pn^2 = 2$ (quadrilateral $mBnh$),

$$ng^2 = 2 \triangle hng;$$

$$\therefore Pg^2 = 2 (mBnh + \triangle hng).$$

Similarly $P'g^2 = 2 (mBn'h' + \triangle kn'g).$

By subtraction,

$$\begin{aligned} Pg^2 - P'g^2 &= 2 \triangle hh'k \\ &= hi \cdot (h'i - ki) \\ &= hi \cdot (h'i - hi) \\ &= hi^2 \cdot \left(\frac{Bm - BC}{BC} \right) \\ &= nn'^2 \cdot \frac{pb - BB'}{BB'}; \end{aligned}$$

whence it follows that Pg is the *maximum* straight line from g to the curve, and the difference between Pg^2 and $P'g^2$ is the area described.

COR. 1. It follows from the same method of proof as that used in Props. 84, 87, 89 that, if O be any point on Pg produced beyond the minor axis, PO is the *maximum* straight line that can be drawn from O to the same part of the ellipse in which Pg is a maximum, i.e. to the semi-ellipse BPB' , and if OP' be drawn to any other point on the semi-ellipse, OP' diminishes as P' moves from P to B or B' .

COR. 2. In the particular case where g coincides with the centre C , the maximum straight line from C to the ellipse is perpendicular to BB' , viz. CA or CA' . Also, if g be not the centre, the angle PgB must be acute if Pg is a *maximum*; and, if Pg is a maximum [or a normal],

$$Cn : ng = CB^2 : CA^2.$$

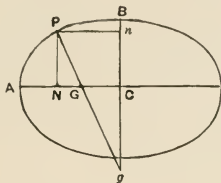
[This corollary is proved separately by *reductio ad absurdum*.]

Proposition 91.

[V. 23.]

*If g be on the minor axis of an ellipse, and gP is a maximum straight line from g to the curve, and if gP meet the major axis in G , GP is a **minimum** straight line from G to the curve.*

[In other words, the minimum from G and the maximum from g determine one and the same normal.]



We have

$$Cn : ng = BB' : p_b$$

[Prop. 90]

$$[= CB^2 : CA^2]$$

$$= p_a : AA'.$$

Also

$$Cn : ng = PN : ng$$

$$= NG : Pn, \text{ by similar triangles.}$$

$$= NG : CN.$$

$$\therefore NG : CN = p_a : AA',$$

or PG is the normal determined as the *minimum* straight line from G . [Prop. 86]

Proposition 92.

[V. 24, 25, 26.]

Only one normal can be drawn from any one point of a conic, whether such normal be regarded as the minimum straight line from the point in which it meets AA' , or as the maximum straight line from the point in which (in the case of an ellipse) it meets the minor axis.

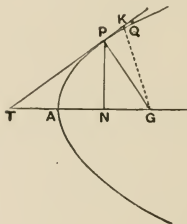
This is at once proved by *reductio ad absurdum* on assuming that PG , PH (meeting the axis AA' in G , H) are minimum straight lines from G and H to the curve, and on a similar assumption for the minor axis of an ellipse.

Proposition 93.

[V. 27, 28, 29, 30.]

The normal at any point P on a conic, whether regarded as a minimum straight line from its intersection with the axis AA' or as a maximum from its intersection with BB' (in the case of an ellipse), is perpendicular to the tangent at P .

Let the tangent at P meet the axis of the parabola, or the axis AA' of a hyperbola or an ellipse, in T . Then we have to prove that TPG is a right angle.



(1) For the *parabola* we have

$$AT = AN, \text{ and } NG = \frac{p_a}{2};$$

$$\therefore NG : p_a = AN : NT,$$

so that

$$\begin{aligned} TN \cdot NG &= p_a \cdot AN \\ &= PN^2. \end{aligned}$$

And the angle at N is a right angle ;

$$\therefore \angle TPG \text{ is a right angle.}$$

(2) For the *hyperbola* or *ellipse*

$$PN^2 : CN \cdot NT$$

$$= p_a : AA'$$

[Prop. 14]

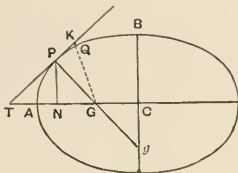
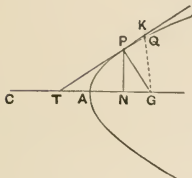
$$= NG : CN, \text{ by the property of the minimum,}$$

[Prop. 86]

$$= TN \cdot NG : CN \cdot NT.$$

$\therefore PN^2 = TN \cdot NG$, while the angle at N is right ;

$\therefore \angle TPG$ is a right angle.



(3) If Pg be the maximum straight line from g on the minor axis of an ellipse, and if Pg meet AA' in G , PG is a minimum from G , and the result follows as in (2).

[Apollonius gives an alternative proof applicable to all three conics. If GP is not perpendicular to the tangent, let GK be perpendicular to it.

Then $\angle GKP > \angle GPK$, and therefore $GP > GK$.

Hence *a fortiori* $GP > GQ$, where Q is the point in which GK cuts the conic; and this is impossible because GP is a *minimum*. Therefore &c.]

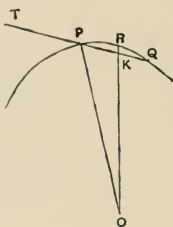
Proposition 94.

[V. 31, 33, 34.]

(1) In general, if O be any point within a conic and OP be a maximum or a minimum straight line from O to the conic, a straight line PT drawn at right angles to PO will touch the conic at P .

(2) If O' be any point on OP produced outside the conic, then, of all straight lines drawn from O' to meet the conic in one point but not produced so as to meet it in a second point, $O'P$ will be the minimum; and of the rest that which is nearer to it will be less than that which is more remote.

(1) First, let OP be a maximum. Then, if TP does not touch the conic, let it cut it again at Q , and draw OK to meet PQ in K and the curve in R .

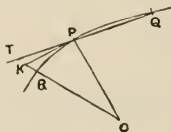


Then, since the angle OPK is right, $\angle OPK > \angle OKP$.

Therefore $OK > OP$, and *a fortiori* $OR > OP$: which is impossible, since OP is a maximum.

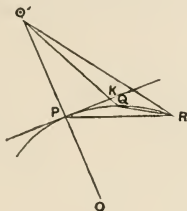
Therefore TP must touch the conic at P .

Secondly, let OP be a minimum. If possible, let TP cut the curve again in Q . From any point between TP and the curve draw a straight line to P and draw ORK perpendicular to this



line meeting it at K and the curve in R . Then the angle OKP is a right angle. Therefore $OP > OK$, and *a fortiori* $OP > OR$: which is impossible, since OP is a minimum. Therefore TP must touch the curve.

(2) Let O' be any point on OP produced. Draw the tangent at P , as PK , which is therefore at right angles to OP . Then draw $O'Q$, $O'R$ to meet the curve in one point only, and let $O'Q$ meet PK in K .



Then $O'K > O'P$. Therefore *a fortiori* $O'Q > O'P$, and $O'P$ is a minimum.

Join RP , RQ . Then the angle $O'QR$ is obtuse, and therefore the angle $O'RK$ is acute. Therefore $O'R > O'Q$, and so on.

Proposition 95.

[V. 35, 36, 37, 38, 39, 40.]

(1) If the normal at P meet the axis of a parabola or the axis AA' of a hyperbola or ellipse in G , the angle PGA increases as P or G moves further and further from A , but in the hyperbola the angle PGA will always be less than the complement of half the angle between the asymptotes.

(2) Two normals at points on the same side of the axis AA' will meet on the opposite side of that axis.

(3) Two normals at points on the same quadrant of an ellipse, as AB , will meet at a point within the angle ACB' .

(1) Suppose P' is further from the vertex than P . Then, since PG , $P'G'$ are minimum straight lines from G , G' to the curve, we have

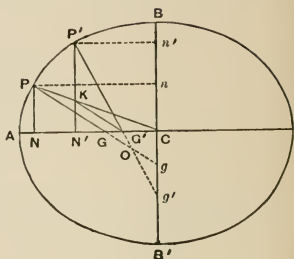
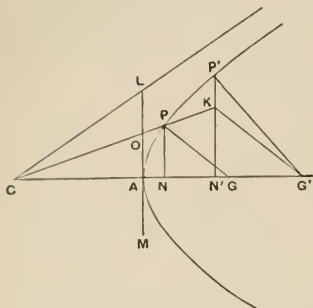
(a) For the *parabola*

$$NG = \frac{p_a}{2} = N'G',$$

and

$$P'N' > PN;$$

$$\therefore \angle P'G'A > \angle PGA.$$



(b) For the *hyperbola* and *ellipse*, joining CP and producing it if necessary to meet $P'N'$ in K , and joining KG' , we have

$$N'G' : CN' = p_a : AA' \quad [\text{Prop. 86}]$$

$$= NG : CN;$$

$$\therefore N'G' : NG = CN' : CN$$

$$= KN' : PN, \text{ by similar triangles.}$$

Therefore the triangles PNG , $KN'G'$ are similar, and

$$\angle KG'N' = \angle PGN.$$

Therefore

$$\angle P'G'N' > \angle PGN.$$

(c) In the *hyperbola*, let AL be drawn perpendicular to AA' to meet the asymptote in L and CP in O . Also let AM be equal to $\frac{p_a}{2}$.

$$\text{Now } AA' : p_a = CA : AM = CN : NG,$$

and

$$OA : CA = PN : CN, \text{ by similar triangles;}$$

therefore, *ex aequali*, $OA : AM = PN : NG$.

Hence

$$AL : AM > PN : NG.$$

But $AL : AM = CA : AL$, [Prop. 28]

$$\therefore CA : AL > PN : NG;$$

$$\therefore \angle PGN \text{ is less than } \angle CLA.$$

(2) It follows at once from (1) that two normals at points on one side of AA' will meet on the other side of AA' .

(3) Regard the two normals as the *maximum* straight lines from g, g' , the points where they meet the minor axis of the ellipse.

Then $Cn' : n'g' = BB' : p_b$ [Prop. 90]

$$= Cn : ng;$$

$$\therefore Cn' : Cg' = Cn : Cg.$$

But $Cn' > Cn$; $\therefore Cg' > Cg$,

whence it follows that $Pg, P'g'$ must cross at a point O before cutting the minor axis. Therefore O lies on the side of BB' towards A .

And, by (2) above, O lies below AC ; therefore O lies within the $\angle ACB'$.

Proposition 96.

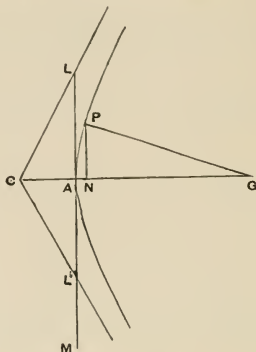
[V. 41, 42, 43.]

(1) *In a parabola or an ellipse any normal PG will meet the curve again.*

(2) *In the hyperbola (a), if AA' be not greater than p_a , no normal can meet the curve in a second point on the same branch; but (b), if $AA' > p_a$, some normals will meet the same branch again and others not.*

(1) For the *ellipse* the proposition is sufficiently obvious, and in the *parabola*, since PG meets a diameter (the axis), it will meet another diameter, viz. that through the point of contact of the tangent parallel to PG , i.e. the diameter bisecting it. Therefore it will meet the curve again.

(2) (a) Let CL , CL' be the asymptotes, and let the tangent at A meet them in L , L' . Take AM equal to $\frac{p_a}{2}$. Let PG be any normal and PN the ordinate.



Then, by hypothesis, $CA \nmid AM$,
and $CA : AM = CA^2 : AL^2$; [Prop. 28]

$$\therefore CA \nmid AL;$$

hence the angle CLA is not greater than ACL or ACL' .

But $\angle CLA > \angle PGN$; [Prop. 95]

$$\therefore \angle ACL' > \angle PGN.$$

It follows that the angle ACL' together with the angle adjacent to PGN will be greater than two right angles.

Therefore PG will not meet CL' towards L' and therefore will not meet the branch of the hyperbola again.

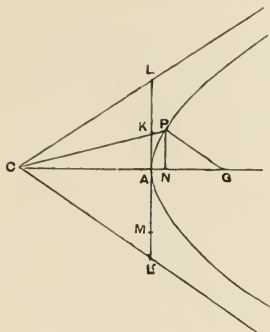
(b) Suppose $CA > AM$ or $\frac{p_a}{2}$. Then

$$LA : AM > LA : AC.$$

Take a point K on AL such that

$$KA : AM = LA : AC.$$

Join CK , and produce it to meet the hyperbola in P , and let PN be the ordinate, and PG the normal, at P .



PG is then the *minimum* from G to the curve, and

$$\begin{aligned} NG : CN &= p_a : AA' \\ &= AM : AC. \end{aligned}$$

Also $CN : PN = AC : AK$, by similar triangles.

Therefore, *ex aequali*, $NG : PN = AM : AK$
 $= CA : AL$, from above.

Hence $\angle ACL' = \angle ACL = \angle PGN$;

$\therefore PG, CL'$ are parallel and do not meet.

But the normals at points between A and P make with the axis angles less than the angle PGN , and normals at points beyond P make with the axis angles greater than PGN .

Therefore normals at points between A and P will not meet the asymptote CL' , or the branch of the hyperbola, again; but normals beyond P will meet the branch again.

Proposition 97.

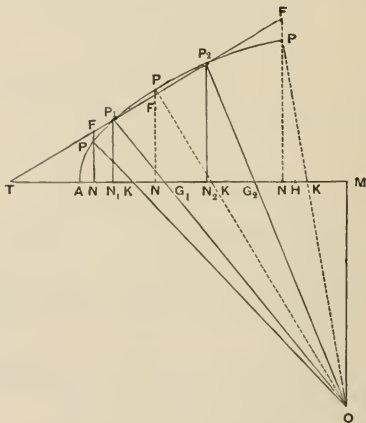
[V. 44, 45, 46, 47, 48.]

If P_1G_1 , P_2G_2 be normals at points on one side of the axis of a conic meeting in O , and if O be joined to any other point P on the conic (it being further supposed in the case of the ellipse that all three lines OP_1 , OP_2 , OP cut the same half of the axis), then

- (1) OP cannot be a normal to the curve ;
- (2) if OP meet the axis in K , and PG be the normal at P ,

$AG < AK$ when P is intermediate between P_1 and P_2 ,
and $AG > AK$ when P does not lie between P_1 and P_2 .

I. First let the conic be a PARABOLA.



Let P_1P_2 meet the axis in T , and draw the ordinates P_1N_1 , P_2N_2 .

Draw OM perpendicular to the axis, and measure MH towards the vertex equal to $\frac{p_a}{2}$.

$$\text{Then} \quad MH = N_2 G_2,$$

$$\text{and} \quad N_2 H = G_2 M.$$

$$\begin{aligned} \text{Therefore} \quad MH : HN_2 &= N_2 G_2 : G_2 M \\ &= P_2 N_2 : MO, \text{ by similar triangles.} \end{aligned}$$

$$\begin{aligned} \text{Therefore} \quad HM \cdot MO &= P_2 N_2 \cdot N_2 H \} \\ \text{Similarly} \quad HM \cdot MO &= P_1 N_1 \cdot N_1 H \} \dots\dots\dots (A). \end{aligned}$$

$$\begin{aligned} \text{Therefore} \quad HN_1 : HN_2 &= P_2 N_2 : P_1 N_1 \\ &= TN_2 : TN_1, \end{aligned}$$

$$\text{whence} \quad N_1 N_2 : HN_1 = N_1 N_2 : TN_2;$$

$$\begin{aligned} \therefore TN_2 &= HN_1 \} \\ \text{and} \quad TN_1 &= HN_2 \} \dots\dots\dots (B). \end{aligned}$$

If P be a variable point and PN the ordinate*, we have now three cases:

$$\begin{aligned} TN < TN_1 \text{ or } HN_2 &\dots\dots\dots (1), \\ TN > TN_1 \text{ or } HN_2, \text{ but } < TN_2 \text{ or } HN_1 &\dots\dots (2), \\ TN > TN_2 \text{ or } HN_1 &\dots\dots\dots (3). \end{aligned}$$

Thus, denoting the several cases by the numbers (1), (2), (3), we have

$$\begin{aligned} N_2 N : TN > N_2 N : HN_2 &\dots\dots\dots (1), \\ < N_2 N : HN_2 &\dots\dots\dots (2), \\ < N_2 N : HN_2 &\dots\dots\dots (3), \end{aligned}$$

and we derive respectively

$$\begin{aligned} TN_2 : TN > HN : HN_2 &\dots\dots\dots (1), \\ < HN : HN_2 &\dots\dots\dots (2), \\ > HN : HN_2 &\dots\dots\dots (3). \end{aligned}$$

* It will be observed that there are three sets of points P, N, K , in the figure denoted by the same letters. This is done in order to exhibit the three different cases; and it is only necessary to bear in mind that attention must be confined to one at a time as indicated in the course of the proof.

If NP meet P_1P_2 in F , we have, by similar triangles,

$$P_2N_2 : FN > HN : HN_2 \dots\dots (1) \text{ and } (3), \\ < HN : HN_2 \dots\dots\dots (2).$$

But in (1) and (3) $FN > PN$, and in (2) $FN < PN$.

Therefore, *a fortiori* in all the cases,

$$P_2N_2 : PN > HN : HN_2 \dots\dots (1) \text{ and } (3), \\ < HN : HN_2 \dots\dots\dots (2).$$

$$\text{Thus } P_2N_2 \cdot N_2H > PN \cdot NH \dots\dots\dots (1) \text{ and } (3), \\ < PN \cdot NH \dots\dots\dots (2).$$

$$\text{Hence } HM \cdot MO > PN \cdot NH \dots (1) \text{ and } (3) \left\{ \begin{array}{l} < PN \cdot NH \dots\dots\dots (2) \end{array} \right\}, \text{ by (A) above.}$$

$$\text{Therefore } MO : PN > NH : HM \dots\dots\dots (1) \text{ and } (3), \\ < NH : HM \dots\dots\dots (2)$$

$$\text{and } MO : PN = MK : NK.$$

$$\text{Therefore } MK : NK > NH : HM \dots\dots (1) \text{ and } (3), \\ < NH : HM \dots\dots\dots (2),$$

$$\text{whence we obtain } MN : NK > MN : HM \dots\dots (1) \text{ and } (3), \\ < MN : HM \dots\dots\dots (2),$$

$$\text{so that } HM \text{ or } NG > NK \text{ in } (1) \text{ and } (3), \\ \text{and } < NK \text{ in } (2).$$

Thus the proposition is proved.

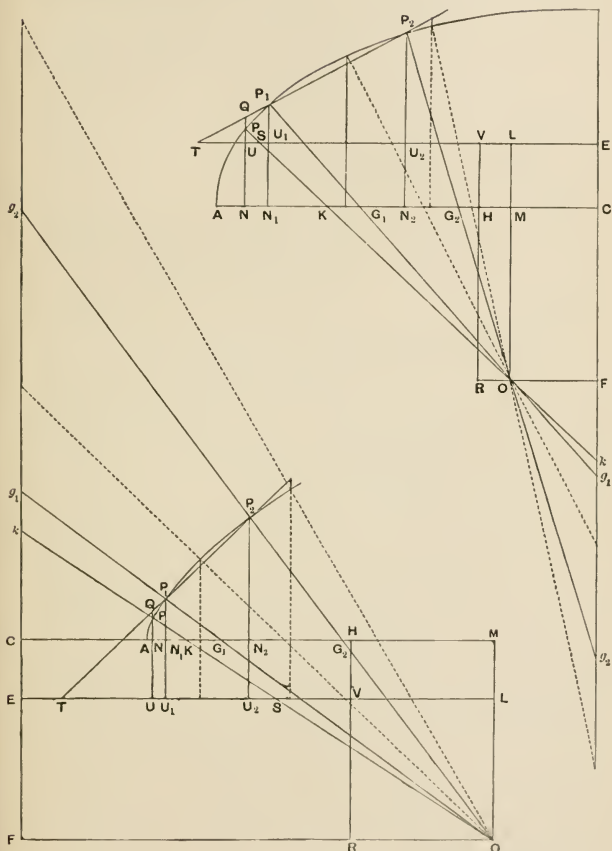
II. Let the conic be a HYPERBOLA or an ELLIPSE.

Let the normals at P_1, P_2 meet at O , and draw OM perpendicular to the axis. Divide CM in H (internally for the hyperbola and externally for the ellipse) so that

$$CH : HM = AA' : p_a [\text{or } CA^2 : CB^2],$$

and let OM be similarly divided at L . Draw HVR parallel to OM and LVE, ORF parallel to CM .

Suppose P_2P_1 produced to meet EL in T , and let P_1N_1 , P_2N_2 meet it in U_1 , U_2 .



Take any other point P on the curve. Join OP meeting the axes in K , k , and let PN meet P_1P_2 in Q and EL in U .

Now $CN_2 : N_2G_2 = AA' : p_a = CH : HM$.

Therefore, *componendo* for the hyperbola and *dividendo* for the ellipse,

$$\begin{aligned} CM : CH &= CG_2 : CN_2 \\ &= CG_2 \sim CM : CN_2 \sim CH \\ &= MG_2 : HN_2 \\ &= MG_2 : VU_2 \dots\dots\dots (A). \end{aligned}$$

Next

$$\begin{aligned} FE : EC &= AA' : p_a = CN_2 : N_2G_2, \\ \text{so that } FC : CE &= CG_2 : N_2G_2. \\ \text{Thus } FC : N_2U_2 &= CG_2 : N_2G_2 \\ &= Cg_2 : P_2N_2, \text{ by similar triangles,} \\ &= FC \pm Cg_2 : N_2U_2 \pm P_2N_2 \\ &= Fg_2 : P_2U_2 \dots\dots\dots (B). \end{aligned}$$

Again

$$\begin{aligned} FC . CM : EC . CH &= (FC : CE) . (CM : CH) \\ &= (Fg_2 : P_2U_2) . (MG_2 : VU_2), \\ &\qquad\qquad\qquad \text{from (A) and (B),} \end{aligned}$$

$$\text{and } FC . CM = Fg_2 . MG_2, \quad \therefore Fg_2 : CM = FC : MG_2.$$

$$\therefore EC . CH = P_2U_2 . U_2V,$$

$$\begin{aligned} \text{or } CE . EV &= P_2U_2 . U_2V \\ &= P_1U_1 . U_1V, \text{ in like manner;} \end{aligned}$$

$$\begin{aligned} \therefore U_1V : U_2V &= P_2U_2 : P_1U_1 \\ &= TU_2 : TU_1, \text{ by similar triangles,} \end{aligned}$$

$$\text{whence } U_1U_2 : U_1V = U_1U_2 : TU_2;$$

$$\begin{aligned} \therefore TU_2 &= VU_1 \} \\ \text{and } TU_1 &= VU_2 \} \dots\dots\dots (C). \end{aligned}$$

Now suppose (1) that $AN < AN_1$;

then $U_2V > TU$, from (C) above;

$$\therefore UU_2 : TU > UU_2 : U_2V;$$

$$\text{hence } TU_2 : TU > UV : U_2V;$$

$$\therefore P_2U_2 : QU > UV : U_2V,$$

by similar triangles.

$$\text{Therefore } P_2U_2 . U_2V > QU . UV,$$

$$\text{and a fortiori } > PU . UV.$$

But $P_2 U_2 . U_2 V = CE . EV$, from above,
 $= LO . OR$, $\therefore CE : LO = OR : EV$;
 $\therefore LO . OR > PU . UV$.

Suppose (2) that $AN > AN_1$ but $< AN_2$.

Then $TU_1 < UV$;

$$\therefore U_1 U : TU_1 > U_1 U : UV,$$

whence $TU : TU_1 > U_1 V : UV$;

$$\therefore QU : P_1 U_1 > U_1 V : UV,$$

by similar triangles.

Therefore (*a fortiori*) $PU . UV > P_1 U_1 . U_1 V$
 $> LO . OR$.

Lastly (3) let AN be $> AN_2$.

Then $TU_1 > UV$;

$$\therefore U_1 U : TU_1 < U_1 U : UV,$$

whence $TU : TU_1 < U_1 V : UV$,

or $QU : P_1 U_1 < U_1 V : UV$;

$$\therefore P_1 U_1 . U_1 V > QU . UV,$$

and *a fortiori* $> PU . UV$;

$$\therefore LO . OR > PU . UV,$$

as in (1) above.

Thus we have for cases (1) and (3)

$$LO . OR > PU . UV,$$

and for (2) $LO . OR < PU . UV$.

That is, we shall have, supposing the upper symbol to refer to (1) and (3) and the lower to (2),

$$LO : PU \gtrless UV : OR,$$

i.e. $LS : SU \gtrless UV : LV$;

$$\therefore LU : US \gtrless LU : LV,$$

and $LV \gtrless US$.

It follows that

$$FO : LV \leq FO : SU, \text{ or } Fk : PU,$$

or

$$CM : MH \leq Fk : PU;$$

$$\therefore FC : CE \leq Fk : PU$$

$$\leq Fk \mp FC : PU \mp CE$$

$$\leq Ck : PN$$

$$\leq CK : NK.$$

Therefore, *componendo* or *dividendo*,

$$FE : EC \leq CN : NK,$$

or

$$CN : NK \geq FE : EC,$$

i.e.

$$\geq AA' : p_a.$$

But

$$CN : NG = AA' : p_a;$$

$$\therefore NK \leq NG;$$

i.e. when P is not between P_1 and P_2 , $NK < NG$, and when P lies between P_1 and P_2 , $NK > NG$, whence the proposition follows.

COR. 1. In the particular case of a quadrant of an ellipse where P_2 coincides with B , i.e. where O coincides with g_1 , it follows that no other normal besides P_1g_1 , Bg_1 can be drawn through g_1 to the quadrant, and, if P be a point between A and P_1 , while Pg_1 meets the axis in K , $NG > NK$.

But if P lie between P_1 and B , $NG < NK$.

[This is separately proved by Apollonius from the property in Prop. 95 (3).]

COR. 2. Three normals at points on one quadrant of an ellipse cannot meet at one point.

This follows at once from the preceding propositions.

COR. 3. *Four normals at points on one semi-ellipse bounded by the major axis cannot meet at one point.*

For, if four such normals cut the major axis and meet in one point, the centre must (1) separate one normal from the three others, or (2) must separate two from the other two, or (3) must lie on one of them.

In cases (1) and (3) a contradiction of the preceding proposition is involved, and in case (2) a contradiction of Prop. 95 (3) which requires two points of intersection, one on each side of the minor axis.

Proposition 98.

[V. 49, 50.]

In any conic, if M be any point on the axis such that AM is not greater than half the latus rectum, and if O be any point on the perpendicular to the axis through M , then no straight line drawn to any point on the curve on the side of the axis opposite to O and meeting the axis between A and M can be a normal.

Let OP be drawn to the curve meeting the axis in K , and let PN be the ordinate at P .

We have in the *parabola*, since $AM \nless \frac{p_a}{2}$,

$$NM < \frac{p_a}{2}, \quad \text{i.e.} < NG.$$

Therefore, *a fortiori*, $NK < NG$.

For the *hyperbola* and *ellipse* $AA' : p_a$ is not greater than $CA : AM$,

and $CN : NM > CA : AM$;

$$\begin{aligned} \therefore CN : NM &> AA' : p_a \\ &> CN : NG; \end{aligned}$$

$$\therefore NM < NG,$$

and *a fortiori* $NK < NG$.

Therefore OP is not a normal.

PROPOSITIONS LEADING IMMEDIATELY TO THE DETERMINATION OF THE *EVOLUTE*.

Proposition 99.

[V. 51, 52.]

If AM measured along the axis be greater than $\frac{p_a}{2}$ (but in the case of the ellipse less than AC), and if MO be drawn perpendicular to the axis, then a certain length [y] can be assigned such that

(a) *if $OM > y$, no normal can be drawn through O which cuts the axis; but, if OP be any straight line drawn to the curve cutting the axis in K, $NK < NG$, where PN is the ordinate and PG the normal at P;*

(b) *if $OM = y$, only one normal can be so drawn through O, and, if OP be any other straight line drawn to the curve and meeting the axis in K, $NK < NG$, as before;*

(c) *if $OM < y$, two normals can be so drawn through O, and, if OP be any other straight line drawn to the curve, NK is less or greater than NG according as OP is not, or is, intermediate between the two normals.*

I. Suppose the conic is a PARABOLA.

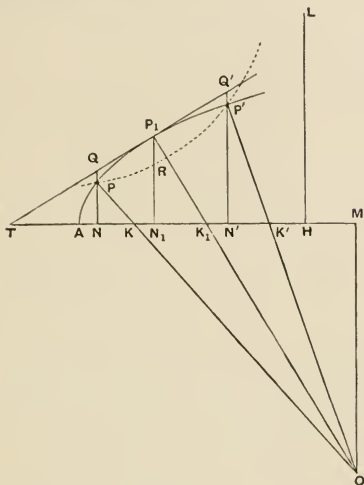
Measure MH towards the vertex equal to $\frac{p_a}{2}$, and divide AH at N_1 so that $HN_1 = 2N_1A$.

Take a length y such that

$$y : P, N_1 = N_1 H : HM,$$

where $P_i N_i$ is the ordinate passing through N_i .

(a) Suppose $OM > y$.



Join OP , meeting the axis in K .

Then $y : P_1 N_1 = N_1 H : HM$;

$$\therefore OM : P_1N_1 > N_1H : HM,$$

or $MK_1 : K_1N_1 > N_1H : HM$;

hence $MN_1 : N_1K_1 > MN_1 : HM$,

so that $N_1 K_1 < HM$,

i.e. $N_1 K_1 < \frac{p_a}{2}$.

Therefore OP_1 is not a normal, and $N_1K_1 < N_1G_1$.

Next let P be any other point. Join OP meeting the axis in K , and let the ordinate PN meet the tangent at P , in Q .

Then, if $AN < AN_1$, we have, | If $AN > AN_1$,
 since $N_1T = 2AN_1 = N_1H$, | $N_1T > NH$;
 $N_1H > NT$; | $\therefore N_1N : NH > N_1N : N_1T$,
 $\therefore N_1N : NT > N_1N : HN_1$; | whence
 thus $TN_1 : TN > HN : HN_1$, | $HN_1 : HN > TN : TN_1$,
 or $P_1N_1 : QN > HN : HN_1$, | $> QN : P_1N_1$,
 and *a fortiori* | $> PN : P_1N_1$,
 $P_1N_1 : PN > HN : HN_1$, | *a fortiori*
 or $P_1N_1 \cdot N_1H > PN \cdot NH$; | $\therefore P_1N_1 \cdot N_1H > PN \cdot NH$.
 But $OM \cdot MH > P_1N_1 \cdot N_1H$, by hypothesis;
 $\therefore OM \cdot MH > PN \cdot NH$,
 or $OM : PN > NH : HM$,
 i.e. $MK : KN > NH : HM$,
 by similar triangles.

Therefore, *componendo*, $MN : NK > MN : HM$,

whence $NK < HM$ or $\frac{pa}{2}$.

Therefore OP is not a normal, and $NK < NG$.

(b) Suppose $OM = y$, and we have in this case

$$MN_1 : N_1K_1 = MN_1 : HM,$$

or $N_1K_1 = HM = \frac{pa}{2} = N_1G_1$,

and P_1O is a normal.

If P is any other point, we have, as before,

$$P_1N_1 \cdot N_1H > PN \cdot NH,$$

and $P_1N_1 \cdot N_1H$ is in this case equal to $OM \cdot MH$.

Therefore $OM \cdot MH > PN \cdot NH$,

and it follows as before that OP is not normal, and $NK < NG$.

(c) Lastly, if $OM < y$,

$$OM : P_1N_1 < N_1H : HM,$$

or $OM \cdot MH < P_1N_1 \cdot N_1H$.

Let N_1R be measured along N_1P_1 so that

$$OM \cdot MH = RN_1 \cdot N_1H.$$

Thus R lies within the curve.

Let HL be drawn perpendicular to the axis, and with AH , HL as asymptotes draw a hyperbola passing through R . This hyperbola will therefore cut the parabola in two points, say P , P' .

Now, by the property of the hyperbola,

$$\begin{aligned} PN \cdot NH &= RN_1 \cdot N_1H \\ &= OM \cdot MH, \text{ from above;} \end{aligned}$$

$$\therefore OM : PN = NH : HM,$$

or $MK : KN = NH : HM,$

and, *componendo*, $MN : NK = MN : HM;$

$$\therefore NK = HM = \frac{p_a}{2} = NG,$$

and PO is normal.

Similarly $P'O$ is normal.

Thus we have two normals meeting in O , and the rest of the proposition follows from Prop. 97.

[It is clear that in the second case where $OM = y$, O is the intersection of two consecutive normals, i.e. is the centre of curvature at the point P_1 .

If then x , y be the coordinates of O , so that $AM = x$, and if $4a = p_a$,

$$HM = 2a,$$

$$N_1H = \frac{2}{3}(x - 2a),$$

$$AN_1 = \frac{1}{3}(x - 2a).$$

Also $y^2 : P_1N_1^2 = N_1H^2 : HM^2,$

or $y^2 : 4a \cdot AN_1 = N_1H^2 : 4a^2;$

$$\therefore ay^2 = AN_1 \cdot N_1H^2$$

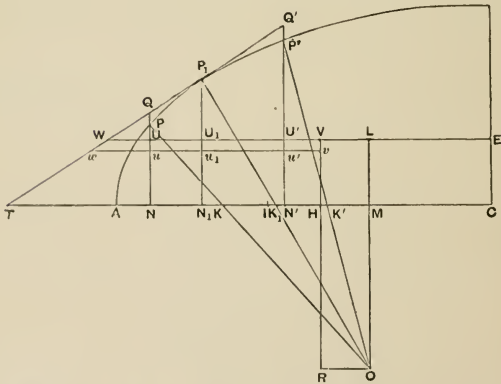
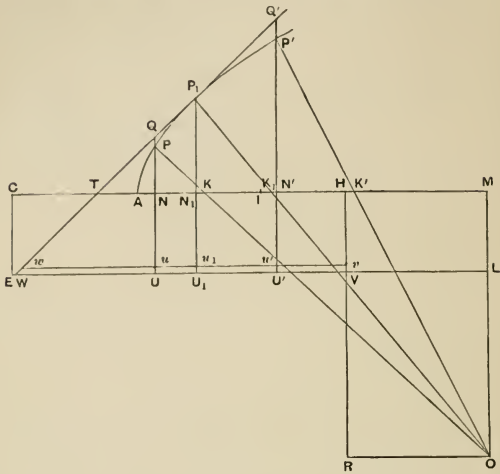
$$= \frac{4}{27}(x - 2a)^3,$$

or $27ay^2 = 4(x - 2a)^3,$

which is the Cartesian equation of the **evolute** of a parabola.]

II. Let the curve be a HYPERBOLA or an ELLIPSE.

We have $AM > \frac{p_a}{2}$, so that $CA : AM < AA' : p_a$.



Therefore, if H be taken on AM such that $CH : HM = AA' : p_a$, H will fall between A and M .

Take two mean proportionals CN_1 , CI between CA and CH^* , and let P_1N_1 be the ordinate through N_1 .

Take a point L on OM (in the hyperbola) or on OM produced (in the ellipse) such that $OL : LM = AA' : p_a$. Draw LVE , OR both parallel to the axis, and CE , HVR both perpendicular to the axis. Let the tangent at P_1 meet the axis in T and EL in W , and let P_1N_1 meet EL in U_1 . Join OP_1 , meeting the axis in K_1 .

Let now y be such a length that

$$y : P_1N_1 = (CM : MH) \cdot (HN_1 : N_1C).$$

(a) Suppose first that $OM > y$;

$$\therefore OM : P_1N_1 > y : P_1N_1.$$

But

$$\begin{aligned} OM : P_1N_1 &= (OM : ML) \cdot (ML : P_1N_1) \\ &= (OM : ML) \cdot (N_1U_1 : P_1N_1), \end{aligned}$$

and

$$\begin{aligned} y : P_1N_1 &= (CM : MH) \cdot (HN_1 : N_1C) \\ &= (OM : ML) \cdot (HN_1 : N_1C); \end{aligned}$$

$$\therefore N_1U_1 : P_1N_1 > HN_1 : N_1C \dots\dots\dots (1),$$

or

$$P_1N_1 \cdot N_1H < CN_1 \cdot N_1U_1.$$

Adding or subtracting the rectangle $U_1N_1 \cdot N_1H$, we have

$$\begin{aligned} P_1U_1 \cdot U_1V &< CH \cdot HV \\ &< LO \cdot OR, \because CH : HM = OL : LM. \end{aligned}$$

But, for a normal at P_1 , we must have [from the proof of Prop. 97]

$$P_1U_1 \cdot U_1V = LO \cdot OR.$$

Therefore P_1O is not a normal, and [as in the proof of Prop. 97]

$$N_1K_1 < N_1G_1.$$

* For Apollonius' method of finding two mean proportionals see the Introduction.

Next let P be any other point than P_1 , and let U, N, K have the same relation to P that U_1, N_1, K_1 have to P_1 .

Also, since $U_1N_1 : N_1P_1 > HN_1 : N_1C$ by (1) above, let u_1 be taken on U_1N_1 such that

$$u_1N_1 : N_1P_1 = HN_1 : N_1C \dots \dots \dots (2),$$

and draw wu_1v parallel to WU_1V .

Now $CN_1 \cdot CT = CA^2$, so that $CN_1 : CA = CA : CT$;

$\therefore CT$ is a third proportional to CN_1, CA .

But CN_1 is a third proportional to CH, CI ,

and $CN_1 : CA = CI : CN_1 = CH : CI$;

$$\therefore CH : CN_1 = CN_1 : CT$$

$$= CH \sim CN_1 : CN_1 \sim CT$$

$$= HN_1 : N_1T.$$

And

$$CH : CN_1 = P_1u_1 : P_1N_1,$$

since $u_1N_1 : N_1P_1 = HN_1 : N_1C$, from (2) above;

$$\therefore HN_1 : N_1T = P_1u_1 : P_1N_1$$

$$= u_1w : N_1T;$$

thus

$$u_1w = HN_1 = u_1v.$$

If $AN < AN_1$,

$$wu < u_1v,$$

and $u_1u : uw > u_1u : u_1v$,

whence $u_1w : uw > uv : u_1v$.

$$\therefore P_1u_1 : Qu > uv : u_1v$$

(where PN meets P_1T in Q);

thus $P_1u_1 \cdot u_1v > Qu \cdot uv$

$$> Pu \cdot uv,$$

a fortiori.

But, since

$$HN_1 : N_1C = u_1N_1 : P_1N_1,$$

$$P_1N_1 \cdot N_1H = CN_1 \cdot N_1u_1,$$

and, adding or subtracting the rectangle $u_1N_1 \cdot N_1H$,

If $AN > AN_1$,

$$wu_1 > uv;$$

$$\therefore uu_1 : uv > uu_1 : wu_1,$$

whence

$$vu_1 : vu > wu : wu_1$$

$$> Qu : P_1u_1;$$

thus $P_1u_1 \cdot u_1v > Qu \cdot uv$

$$> Pu \cdot uv,$$

a fortiori,

and the proof proceeds as in the first column, leading to the same result,

$$PU \cdot UV < LO \cdot OR.$$

$$P_1 u_1 . u_1 v = CH . Hv ;$$

$$\therefore CH . Hv > Pu . uv,$$

and, adding or subtracting the rectangle $uU . UV$,

$$PU . UV < CH . Hv + uU . UV$$

for the hyperbola,

or

$$PU . UV < CH . Hv - uU . UV$$

for the ellipse,

$\therefore$ in either case, *a fortiori*,

$$PU . UV < CH . HV,$$

or $PU . UV < LO . OR.$

Therefore, as in the proof of Prop. 97, PO is not a normal, but $NK < NG$.

(b) Next suppose $OM = y$, so that $OM : P_1 N_1 = y : P_1 N_1$, and we obtain in this case

$$U_1 N_1 : N_1 P_1 = HN_1 : N_1 C ;$$

$$\therefore CN_1 . N_1 U_1 = P_1 N_1 . N_1 H.$$

Adding or subtracting $U_1 N_1 . N_1 H$, we have

$$P_1 U_1 . U_1 V = CH . HV = LO . OR,$$

and this [Prop. 97] is the property of the normal at P_1 .

Therefore one normal can be drawn from O .

If P be any other point on the curve, it will be shown as before that $U_1 W = U_1 V$, because in this case the lines WV , wv coincide; also

$UU_1 : UW > UU_1 : U_1 V$ in the case where $UW < U_1 V$,
and

$UU_1 : UV > UU_1 : U_1 W$ in the case where $U_1 W > UV$,
whence, exactly as before, we derive that

$$\begin{aligned} P_1 U_1 . U_1 V &> QU . UV \\ &> PU . UV, \text{ a fortiori,} \end{aligned}$$

and thence that $PU . UV < LO . OR$.

Therefore PO is not a normal, and $NK < NG$.

and we shall derive

$$LO \cdot OR < P_1U_1 \cdot U_1V.$$

Let S be taken on P_1N_1 such that $LO \cdot OR = SU_1 \cdot U_1V$, and through S describe a hyperbola whose asymptotes are VW and VH produced. This hyperbola will therefore meet the conic in two points P, P' , and by the property of the hyperbola

$$PU \cdot UV = P'U' \cdot U'V = SU_1 \cdot U_1V = LO \cdot OR,$$

so that $PO, P'O$ are both normals.

The rest of the proposition follows at once from Prop. 97.

[It is clear that in case (b) O is the point of intersection of two consecutive normals, or the centre of the circle of curvature at P .

To find the Cartesian equation of the **evolute** we have

$$\left. \begin{aligned} x &= CM, \\ \frac{CH}{HM} &= \frac{a^2}{b^2}, \text{ or } \frac{CH}{x \sim CH} = \frac{a^2}{b^2} \end{aligned} \right\} \dots\dots\dots (1).$$

Also
$$\frac{y}{P_1N_1} = \frac{CM}{MH} \cdot \frac{HN_1}{N_1C} \dots\dots\dots (2),$$

and
$$\frac{CN_1^2}{a^2} \mp \frac{P_1N_1^2}{b^2} = 1 \dots\dots\dots (3),$$

where the upper sign refers to the hyperbola.

And, lastly, $a : CN_1 = CN_1 : CI = CI : CH \dots\dots\dots (4).$

From (4) $CN_1^2 = a \cdot CI,$

and
$$CN_1 = \frac{a \cdot CH}{CI};$$

$$\therefore CN_1^3 = a^2 \cdot CH \dots\dots\dots (5).$$

Now, from (2),

$$\begin{aligned} \frac{y}{P_1N_1} &= \frac{CM}{MH} \cdot \frac{HN_1}{N_1C} \\ &= \frac{a^2 \pm b^2}{b^2} \cdot \frac{CH \sim CN_1}{CN_1}, \text{ by aid of (1),} \\ &= \frac{a^2 \pm b^2}{b^2} \cdot \frac{CN_1^2 \sim a^2}{a^2}, \text{ by (5),} \\ &= \frac{a^2 \pm b^2}{b^2} \cdot \frac{P_1N_1^2}{b^2}, \text{ by (3).} \end{aligned}$$

Thus $P_1 N_1^3 = \frac{b^4 y}{a^2 \pm b^2},$

whence $P_1 N_1^2 = b^2 \cdot \left(\frac{by}{a^2 \pm b^2} \right)^{\frac{2}{3}} \dots\dots\dots (6).$

But, from (1), $CH = \frac{a^2 x}{a^2 \pm b^2}.$

Therefore, by (5), $CN_1^3 = \frac{a^4 x}{a^2 \pm b^2},$

whence $CN_1^2 = a^2 \cdot \left(\frac{ax}{a^2 \pm b^2} \right)^{\frac{2}{3}} \dots\dots\dots (7).$

Thus, from (6) and (7), by the aid of (3),

$$\left(\frac{ax}{a^2 \pm b^2} \right)^{\frac{2}{3}} \mp \left(\frac{by}{a^2 \pm b^2} \right)^{\frac{2}{3}} = 1,$$

or $(ax)^{\frac{2}{3}} \mp (by)^{\frac{2}{3}} = (a^2 \pm b^2)^{\frac{2}{3}}.]$

Proposition 100.

[V. 53, 54.]

If O be a point on the minor axis of an ellipse, then

(a) *if $OB : BC \nless AA' : p_a$, and P be any point on either of the quadrants BA, BA' except the point B, and if OP meet the major axis in K,*

PO cannot be a normal, but $NK < NG$;

(b) *if $OB : BC < AA' : p_a$, one normal only besides OB can be drawn to either of the two quadrants as OP, and, if P' be any other point, $N'K'$ is less or greater than $N'G'$ according as P' is further from, or nearer to, the minor axis than P.*

[This proposition follows at once as a particular case of the preceding, but Apollonius proves it separately thus.]

(a) We have $OB : BC < On : nC$;

$\therefore On : nC$, or $CN : NK > AA' : p_a$,

whence $CN : NK > CN : NG$,

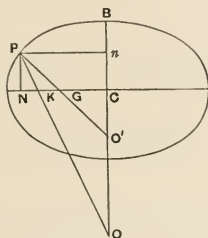
and $NK < NG.$

(b) Suppose now that

$$O'B : BC < AA' : p_a.$$

Take a point n on $O'B$ such that

$$O'n : nC = AA' : p_a.$$



Therefore $CN : NK_1 = AA' : p_a$,

where N is the foot of the ordinate of P , the point in which nP drawn parallel to the major axis meets the ellipse, and K_1 is the point in which $O'P$ meets the major axis;

$\therefore NK_1 = NG$, and PO' is a normal.

PO' , BO' are then two normals through O' , and the rest of the proposition follows from Prop. 97.

CONSTRUCTION OF NORMALS.

Proposition 101.

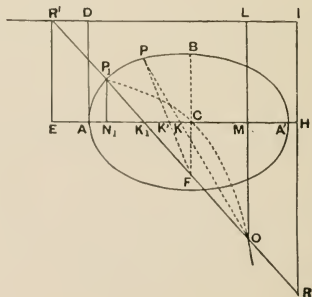
[V. 55, 56, 57.]

If O is any point below the axis AA' of an ellipse, and $AM > AC$ (where M is the foot of the perpendicular from O on the axis), then one normal to the ellipse can always be drawn through O cutting the axis between A and C , but never more than one such normal.

Produce OM to L and CM to H so that

$$OL : LM = CH : HM = AA' : p_a,$$

and draw LI , IH parallel and perpendicular to the axis respectively. Then with IL , IH as asymptotes describe a [rectangular] hyperbola passing through O .



This will meet the ellipse in some point P_1 . For, drawing AD , the tangent at A , to meet IL produced in D , we have

$$AH : HM > CH : HM$$

$$> AA' : p_a$$

$$> OL : LM;$$

$$\therefore AH \cdot LM > OL \cdot HM,$$

or

$$AD \cdot DI > OL \cdot LI.$$

Thus, from the property of the hyperbola, it must meet AD between A and D , and therefore must meet the ellipse in some point P_1 .

Produce OP_1 both ways to meet the asymptotes in R, R' , and draw $R'E$ perpendicular to the axis.

Therefore $OR = P_1R'$, and consequently $EN_1 = MH$.

$$\text{Now} \quad AA' : p_a = OL : LM$$

$$= ME : EK_1, \text{ by similar triangles.}$$

$$\text{Also} \quad AA' : p_a = CH : HM;$$

$$\therefore AA' : p_a = ME - CH : EK_1 - MH$$

$$= CN_1 : N_1K_1,$$

since

$$EN_1 = MH.$$

Therefore $N_1K_1 = N_1G_1$, and P_1O is a normal.

Let P be any other point such that OP meets AC in K .

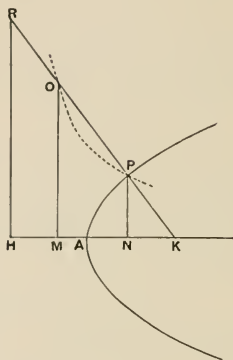
Produce BC to meet OP_1 in F , and join FP , meeting the axis in K' .

Then, since two normals [at P_1, B] meet in F , FP is not a normal, but $NK' > NG$. Therefore, *a fortiori*, $NK > NG$. And, if P is between A and P_1 , $NK < NG$. [Prop. 97, Cor. 1.]

Proposition 102.

[V. 58, 59, 60, 61.]

If O be any point outside a conic, but not on the axis whose extremity is A , we can draw a normal to the curve through O .



For the *parabola* we have only to measure MH in the direction of the axis produced outside the curve, and of length equal to $\frac{p_a}{2}$, to draw HR perpendicular to the axis on the same side as O , and, with HR , HA as asymptotes, to describe a [rectangular] hyperbola through O . This will meet the curve in a point P , and, if OP be joined and produced to meet the axis in K and HR in R , we have at once $HM = NK$.

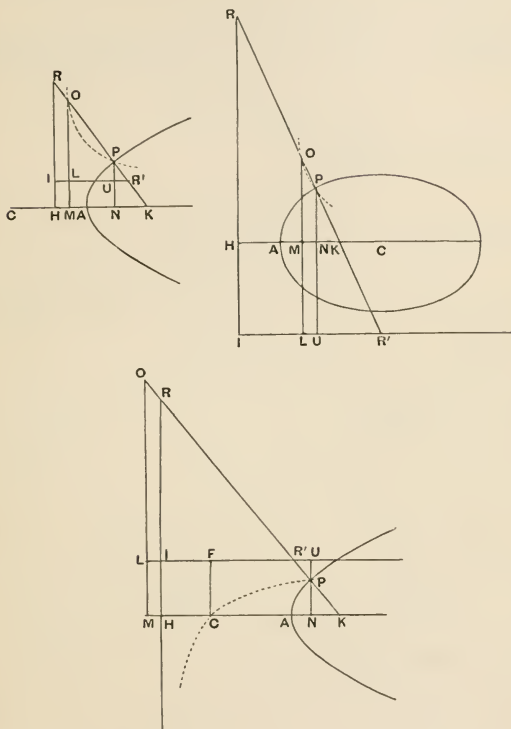
Therefore $NK = \frac{p_a}{2}$,

and PK is a normal.

In the *hyperbola* or *ellipse* take H on CM or on CM produced, and L on OM or OM produced, so that

$$CH : HM = OL : LM = AA' : p_a.$$

Then draw HIR perpendicular to the axis, and ILR' through L parallel to the axis.



(1) If M falls on the side of C towards A , draw with asymptotes IR, IL , and through O , a [rectangular] hyperbola cutting the curve in P .

(2) If M falls on the side of C further from A in the *hyperbola*, draw a [rectangular] hyperbola with IH, IR' as asymptotes and through C , the centre, cutting the curve in P .

Then OP will be a normal.

For we have (1) $MK : HN = MK : LR'$,

since $OR = PR'$, and therefore $IL = UR'$.

Therefore $MK : HN = MO : OL$, by similar triangles,
 $= MC : CH$,

$$\therefore CH : HM = OL : LM.$$

Therefore, alternately,

$$MK : MC = NH : HC \dots\dots\dots (\Lambda).$$

In case (2) $OL : LM = CH : HM$,

or $OL \cdot LI = CH \cdot HI$,

[so that O, C are on opposite branches of the same rectangular hyperbola].

Therefore $PU : OL = LI : IU$,

or, by similar triangles,

$$UR' : R'L = LI : IU,$$

whence

$$R'L = IU = HN;$$

$$\begin{aligned} \therefore MK : HN &= MK : R'L \\ &= MO : OL \\ &= MC : CH, \end{aligned}$$

and $MK : MC = NH : HC$, as before (Λ).

Thus, in either case, we derive

$$CK : CM = CN : CH,$$

and hence, alternately,

$$CN : CK = CH : CM,$$

so that

$$\begin{aligned} CN : NK &= CH : HM \\ &= AA' : p_a; \end{aligned}$$

$$\therefore NK = NG,$$

and OP is the normal at P .

(3) For the *hyperbola*, in the particular case where M coincides with C , or O is on the conjugate axis, we need only divide OC in L , so that

$$OL : LC = AA' : p_a,$$

and then draw LP parallel to AA' to meet the hyperbola in P . P is then the foot of the normal through O , for

$$\begin{aligned} AA' : p_a &= OL : LC \\ &= OP : PK \\ &= CN : NK, \end{aligned}$$

and

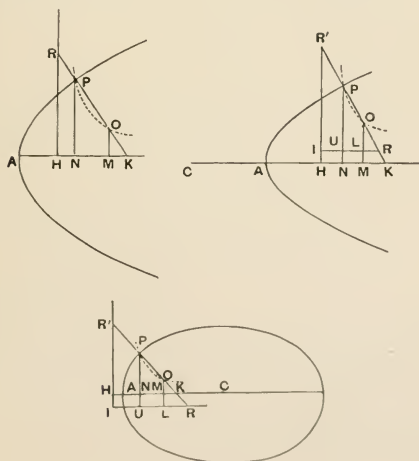
$$NK = NG.$$

[The particular case is that in which the hyperbola used in the construction reduces to two straight lines.]

Proposition 103.

[V. 62, 63.]

If O be an internal point, we can draw through O a normal to the conic.



The construction and proof proceed as in the preceding proposition, *mutatis mutandis*.

The case of the *parabola* is obvious; and for the *hyperbola* or *ellipse*

$$MK : HN = OM : OL$$

$$= CM : CH.$$

$$\therefore CM : CH = CM \pm MK : CH \pm HN$$

$$= CK : CN;$$

$$\therefore NK : CN = HM : CH$$

$$= p_a : AA';$$

$$\therefore NK = NG,$$

and PO is a normal.

OTHER PROPOSITIONS RESPECTING MAXIMA AND MINIMA.

Proposition 104.

[V. 64, 65, 66, 67.]

*If O be a point below the axis of any conic such that either no normal, or only one normal, can be drawn to the curve through O which cuts the axis (between A and C in the case of the ellipse), then OA is the **least** of the lines OP cutting the axis, and that which is nearer to OA is less than that which is more remote.*

If OM be perpendicular to the axis, we must have

$$AM > \frac{p_a}{2},$$

and also OM must be either greater than or equal to y , where

(a) in the case of the *parabola*

$$y : P_1 N_1 = N_1 H : HM;$$

(b) in the case of the *hyperbola* or *ellipse*

$$y : P_1 N_1 = (CM : MH) \cdot (HN_1 : N_1 C),$$

with the notation of Prop. 99.

In the case where $OM > y$, we have proved in Prop. 99 for all three curves that, for any straight line OP drawn from O to the curve and cutting the axis in K , $NK < NG$;

but, in the case where $OM = y$, $NK < NG$ for any point P between A and P_1 except P_1 itself, for which $N_1 K_1 = N_1 G_1$.

It only remains therefore to prove that

(a) $OP_1 >$ any straight line OP between OA and OP_1 ,

(b) $OP_1 <$ any straight line OP' beyond OP_1 .

(a) Suppose first, if possible, that $OP = OP_1$, and let Q be any point between them, so that, by the preceding proof, $OQ > OP$. Measure along OQ a length Oq such that Oq is greater than OP_1 and less than OQ . With O as centre and Oq as radius describe a circle meeting OP_1 produced in p_1 . This circle must then meet the conic in an intermediate point R .

Thus, by the preceding proof, OQ is less than OR , and therefore is less than Oq : which is absurd.

Therefore OP is not equal to OP_1 .

Again suppose, if possible, that $OP > OP_1$. Then, by taking on OP_1 a length Op_1 greater than OP_1 and less than OP , an absurdity is proved in the same manner.

Therefore, since OP is neither equal to nor greater than OP_1 ,

$$OP < OP_1.$$

(b) If OP' lies more remote from OA than OP_1 , an exactly similar proof will show that $OP_1 < OP'$.

Thus the proposition is completely established.

Proposition 105. (Lemma.)

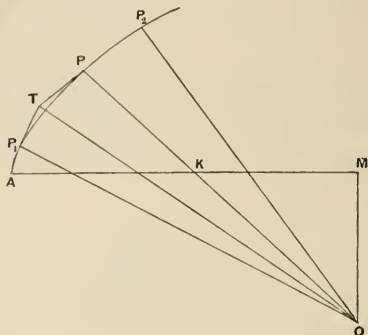
[V. 68, 69, 70, 71.]

If two tangents at points Q, Q' on one side of the axis of a conic meet in T , and if Q be nearer to the axis than Q' , then $TQ < TQ'$.

The proposition is proved at once for the parabola and hyperbola and for the case where Q, Q' are on one quadrant of an ellipse: for the angle TVQ' is greater than the angle TVQ , and $QV = VQ'$.

(1) if P lies between A and P_1 , OP_1 is the **greatest** of all the lines OP , and that which is nearer to OP_1 on each side is greater than that which is more remote;

(2) if P lies between P_1 and P_2 , or beyond P_2 , OP_2 is the **least** of all the lines OP , and the nearer to OP_2 is less than the more remote.



By Prop. 99, if P is between A and P_1 , OP is not a normal, but $NK < NG$. Therefore, by the same proof as that employed in Prop. 104, we find that OP increases continually as P moves from A towards P_1 .

We have therefore to prove that OP diminishes continually as P moves from P_1 to P_2 . Let P be any point between P_1 and P_2 , and let the tangents at P_1 , P meet in T . Join OT .

Then, by Prop. 105, $TP_1 < TP$.

Also $TP_1^2 + OP_1^2 > TP^2 + OP^2$,

since $AK > AG$, and consequently the angle OPT is obtuse.

Therefore $OP < OP_1$.

Similarly it can be proved that, if P' is a point between P and P_2 , $OP' < OP$.

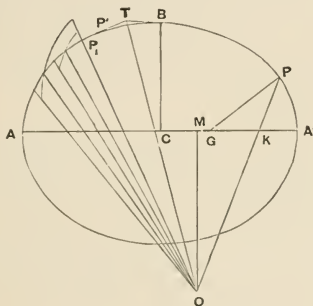
That OP increases continually as P moves from P_2 further away from A and P_1 is proved by the method of Prop. 104.

Thus the proposition is established.

Proposition 107.

[V. 73.]

If O be a point below the major axis of an ellipse such that it is possible to draw through O one normal only to the whole of the semi-ellipse ABA' , then, if OP_1 be that normal and P_1 is on the quadrant AB , OP_1 will be the **greatest** of all the straight lines drawn from O to the semi-ellipse, and that which is nearer to OP_1 will be greater than that which is more remote. Also OA' will be the **least** of all the straight lines drawn from O to the semi-ellipse.



It follows from Props. 99 and 101 that, if OM be perpendicular to the axis, M must lie between C and A' , and that OM must be greater than the length y determined as in Prop. 99.

Thus for all points P between A' and B , since K is nearer to A' than G is, it is proved by the method of Prop. 10+ that OA' is the *least* of all such lines OP , and OP increases continually as P passes from A' to B .

For any point P' between B and P_1 we use the method of Prop. 106, drawing the tangents at P' and B , meeting in T .

Thus we derive at once that $OB < OP'$, and similarly that OP' increases continually as P' passes from B to P_1 .

For the part of the curve between P_1 and A we employ the method of *reductio ad absurdum* used in the second part of Prop. 104.

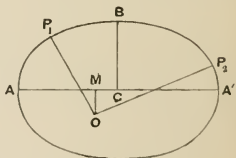
Proposition 108.

[V. 74.]

If O be a point below the major axis of an ellipse such that two normals only can be drawn through it to the whole semi-ellipse ABA' , then that normal, OP_2 , which cuts the minor axis is the **greatest** of all straight lines from O to the semi-ellipse, and that which is nearer to it is greater than that which is more remote. Also OA , joining O to the nearer vertex A , is the **least** of all such straight lines.

It follows from Prop. 99 that, if O be nearer to A than to A' , then P_1 , the point at which O is the centre of curvature, is on the quadrant AB , and that OP_1 is one of the only two possible normals, while P_2 , the extremity of the other, is on the quadrant BA' ; also $OM = y$ determined as in Prop. 99.

In this case, since only one normal can be drawn to the quadrant AB , we prove that OP increases as P moves from A to P_1 by the method of Prop. 104, as also that OP increases as P moves from P_1 to B .



That OP increases as P moves from B to P_2 , and diminishes as it passes from P_2 to A' , is established by the method employed in the last proposition.

Proposition 109.

[V. 75, 76, 77.]

If O be a point below the major axis of an ellipse such that three normals can be drawn to the semi-ellipse ABA' at points P_1, P_2, P_3 , where P_1, P_2 are on the quadrant AB and P_3 on the quadrant BA' , then (if P_1 be nearest to the vertex A),

(1) OP_3 is the **greatest** of all lines drawn from O to points on the semi-ellipse between A' and P_2 , and the nearer to OP_3 on either side is greater than the more remote;

(2) OP_1 is the **greatest** of all lines from O to points on the semi-ellipse from A to P_2 , and the nearer to OP_1 on either side is greater than the more remote.

(3) of the two **maxima**, $OP_3 > OP_1$.

Part (2) of this proposition is established by the method of Prop. 106.

Part (1) is proved by the method of Prop. 107.

It remains to prove (3).

We have

$$CN_1 : N_1G_1 = AA' : p_a = CN_3 : N_3G_3;$$

$$\therefore MN_1 : N_1G_1 < CN_3 : N_3G_3$$

$$< MN_3 : N_3G_3, \text{ a fortiori,}$$

$$\text{whence} \quad MG_1 : N_1G_1 < MG_3 : N_3G_3;$$

and, by similar triangles,

$$OM : P_1N_1 < OM : P_3N_3,$$

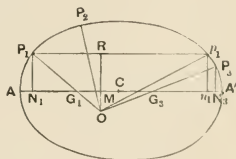
$$\text{or} \quad P_1N_1 > P_3N_3.$$

If then P_1p_1 be parallel to the axis, meeting the curve in p_1 , we have at once, on producing OM to R ,

$$p_1R > P_1R,$$

$$\text{so that} \quad Op_1 > OP_1;$$

$$\therefore \text{ a fortiori } OP_3 > OP_1.$$



As particular cases of the foregoing propositions we have

(1) If O be on the minor axis, and *no* normal except OB can be drawn to the ellipse, OB is *greater* than any other straight line from O to the curve, and the nearer to it is greater than the more remote.

(2) If O be on the minor axis, and *one* normal (besides OB) can be drawn to either quadrant as OP_1 , then OP_1 is the *greatest* of all straight lines from O to the curve, and the nearer to it is greater than the more remote.

EQUAL AND SIMILAR CONICS.

DEFINITIONS.

1. Conic sections are said to be **equal** when one can be applied to the other in such a way that they everywhere coincide and nowhere cut one another. When this is not the case they are **unequal**.

2. Conics are said to be **similar** if, the same number of ordinates being drawn to the axis at proportional distances from the vertex, all the ordinates are respectively proportional to the corresponding abscissae. Otherwise they are **dissimilar**.

3. The straight line subtending a segment of a circle or a conic is called the **base** of the segment.

4. The **diameter** of the segment is the straight line which bisects all chords in it parallel to the base, and the point where the diameter meets the segment is the **vertex** of the segment.

5. **Equal** segments are such that one can be applied to the other in such a way that they everywhere coincide and nowhere cut one another. Otherwise they are **unequal**.

6. Segments are **similar** in which the angles between the respective bases and diameters are equal, and in which, parallels to the base being drawn from points on each segment to meet the diameter at points proportionally distant from the vertex, each parallel is respectively proportional to the corresponding abscissa in each.

Proposition 110.

[VI. 1, 2.]

(1) *In two parabolas, if the ordinates to a diameter in each are inclined to the respective diameters at equal angles, and if the corresponding parameters are equal, the two parabolas are equal.*

(2) *If the ordinates to a diameter in each of two hyperbolas or two ellipses are equally inclined to the respective diameters, and if the diameters as well as the corresponding parameters are equal respectively, the two conics are equal, and conversely.*

This proposition is at once established by means of the fundamental properties

(1) $QV^2 = PL \cdot PV$ for the parabola, and

(2) $QV^2 = PV \cdot VR$ for the hyperbola or ellipse
proved in Props. 1—3.

Proposition 111.

[VI. 3.]

Since an ellipse is limited, while a parabola and a hyperbola proceed to infinity, an ellipse cannot be equal to either of the other curves. Also a parabola cannot be equal to a hyperbola.

For, if a parabola be equal to a hyperbola, they can be applied to one another so as to coincide throughout. If then equal abscissae AN , AN' be taken along the axes in each we have for the parabola

$$AN : AN' = PN^2 : P'N'^2.$$

Therefore the same holds for the hyperbola: which is impossible, because

$$PN^2 : P'N'^2 = AN \cdot A'N : AN' \cdot A'N'.$$

Therefore a parabola and hyperbola cannot be equal.

[Here follow six easy propositions, chiefly depending upon the symmetrical form of a conic, which need not be reproduced.]

Proposition 112.

[VI. 11, 12, 13.]

(1) *All parabolas are similar.*

(2) *Hyperbolas, or ellipses, are similar to one another when the "figure" on a diameter of one is similar to the "figure" on a diameter of the other and the ordinates to the diameters in each make equal angles with the diameters respectively.*

(1) The result is derived at once from the property

$$PN^2 = p_a \cdot AN.$$

(2) Suppose the diameters to be *axes* in the first place (conjugate axes for *hyperbolas*, and both major or both minor axes for *ellipses*) so that the ordinates are at right angles to the diameters in both.

Then the ratio $p_a : AA'$ is the same in both curves. Therefore, using capital letters for one conic and small letters for the other, and making $AN : an$ equal to $AA' : aa'$, we have at the same time

$$PN^2 : AN \cdot NA' = pn^2 : an \cdot na'.$$

But

$$AN \cdot NA' : AN^2 = an \cdot na' : an^2,$$

because

$$A'N : AN = a'n : an;$$

$$\therefore PN^2 : AN^2 = pn^2 : an^2,$$

or

$$PN : AN = pn : an,$$

and the condition of similarity is satisfied (Def. 2).

Again, let PP' , pp' be diameters in two hyperbolas or two ellipses, such that the corresponding ordinates make equal angles with the diameters, and the ratios of each diameter to its parameter are equal.

Draw tangents at P , p meeting the axes in T , t respectively. Then the angles CPT , cpt are equal. Draw AH , ah perpendicular to the axes and meeting CP , cp in H , h ; and on CH , ch as diameters describe circles, which therefore pass respectively through A , a . Draw QAR , qar through A , a parallel respectively to the tangents at P , p and meeting the circles just described in R , r .

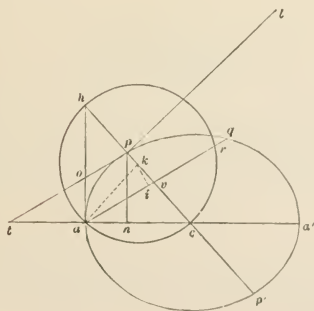
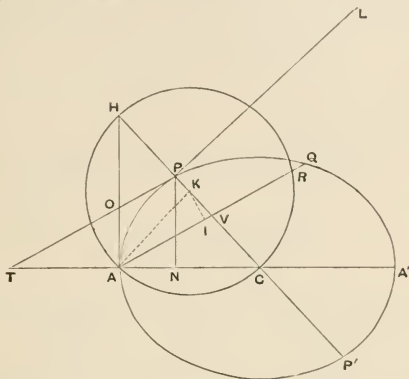
[For, if K, k be the centres of the circles, and I, i the middle points of AR, ar , we derive from (α)

$$VA : AI = va : ai;$$

and, since

$$\angle KVI = \angle kvi,$$

the triangles KVI, kvi are similar.



Therefore, since VI, vi are divided at A, a in the same ratio' the triangles KVA, kva are similar;

$$\therefore \angle AKV = \angle akv;$$

hence the halves of these angles, or of their supplements, are equal, or

$$\angle KCA = \angle kca.]$$

Therefore, since the angles at P, p are also equal, the triangles CPT, cpt are similar.

Draw PN, pn perpendicular to the axes, and it will follow that

$$PN^2 : CN \cdot NT = pn^2 : cn \cdot nt,$$

whence the ratio of AA' to its parameter and that of aa' to its parameter are equal. [Prop. 14]

Therefore (by the previous case) the conics are similar.

Proposition 113.

[VI. 14, 15.]

A parabola is neither similar to a hyperbola nor to an ellipse; and a hyperbola is not similar to an ellipse.

[Proved by *reductio ad absurdum* from the ordinate properties.]

Proposition 114.

[VI. 17, 18.]

(1) *If PT, pt be tangents to two similar conics meeting the axes in T, t respectively and making equal angles with them; if, further, PV, pv be measured along the diameters through P, p so that*

$$PV : PT = pv : pt,$$

and if QQ', qq' be the chords through V, v parallel to PT, pt respectively: then the segments QPQ', qpq' are similar and similarly situated.

(2) *And, conversely, if the segments are similar and similarly situated, $PV : PT = pv : pt$, and the tangents are equally inclined to the axes.*

Similarly, if V' , v' be points on PV , pv such that

$$PV : PV' = pv : pv',$$

and therefore $PL : PV' = pl : pv'$,

it follows that the ordinates passing through V' , v' are in the same ratio to their respective abscissae.

Therefore the segments are similar. (Def. 6.)

(2) If the segments are similar and similarly situated, we have to prove that

$$\angle PTA = \angle pta,$$

and $PV : PT = pv : pt$.

Now the tangents at P , p are parallel to QQ' , qq' respectively, and the angles at V , v are equal.

Therefore the angles PTA , pta are equal.

Also, by similar segments,

$$QV : PV = qv : pv,$$

while $PL : QV = QV : PV$, and $pl : qv = qv : pv$;

$$\therefore PL : PV = pl : pv.$$

$$\text{But } \left. \begin{array}{l} PL : 2PT = OP : PH \\ pl : 2pt = op : ph \end{array} \right\},$$

and $OP : PH = op : ph$,

by similar triangles.

$$\text{Therefore } PV : PT = pv : pt.$$

II. If the curves be *hyperbolas* or *ellipses*, suppose a similar construction made, and let the ordinates PN , pn be drawn to the major or conjugate axes. We can use the figures of Prop. 112, only remembering that the chords are here QQ' , qq' , and do not pass through A , a .

(1) Since the conics are similar, the ratio of the axis to its parameter is the same for both.

Therefore $PN^2 : CN \cdot NT = pn^2 : cn \cdot nt.$ [Prop. 14]

Also the angles PTN , ptn are equal,

therefore $PN : NT = pn : nt.$

Hence $PN : CN = pn : cn,$

and $\angle PCN = \angle pcn.$

Therefore also $\angle CPT = \angle cpt.$

It follows that the triangles OPH , oph are similar.

Therefore $OP : PH = op : ph.$

But
$$\left. \begin{aligned} OP : PH &= PL : 2PT \\ op : ph &= pl : 2pt \end{aligned} \right\},$$

whence $PL : PT = pl : pt.$

Also, by similar triangles,

$$PT : CP = pt : cp;$$

$$\therefore PL : CP = pl : cp,$$

or $PL : PP' = pl : pp' \dots\dots\dots(A).$

Therefore the "figures" on the diameters PP' , pp' are similar.

Again, we made $PV : PT = pv : pt,$

so that $PL : PV = pl : pv \dots\dots\dots(B).$

We derive, by the method employed in Prop. 112, that

$$QV : PV = qv : pv,$$

and that, if PV , pv be proportionally divided in the points V' , v' , the ordinates through these points are in the same ratios.

Also the angles at V , v are equal.

Therefore the segments are similar.

(2) If the segments are similar, the ordinates are in the ratio of their abscissae, and we have

$$\left. \begin{aligned} QV : PV &= qv : pv \\ PV : PV' &= pv : pv' \\ PV' : Q'V' &= pv' : q'v' \end{aligned} \right\}.$$

Then $QV^2 : Q'V'^2 = qv^2 : q'v'^2;$

$$\therefore PV \cdot VP' : PV' \cdot V'P' = pv \cdot vp' : pv' \cdot v'p',$$

and $PV : PV' = pv : pv',$

so that $P'V : P'V' = p'v : p'v'.$

From these equations it follows that

and
$$\left. \begin{aligned} PV' : VV' &= pv' : vv' \\ P'V' : VV' &= p'v' : vv' \end{aligned} \right\},$$

whence $P'V' : P'V'' = p'v' : p'v';$

$$\therefore P'V' \cdot V'P : P'V'' = p'v' \cdot v'p : p'v'^2.$$

But $PV'^2 : Q'V'^2 = pv'^2 : q'v'^2;$

$$\therefore P'V' \cdot V'P : Q'V'^2 = p'v' \cdot v'p : q'v'^2.$$

But these ratios are those of PP' , pp' to their respective parameters.

Therefore the “figures” on PP' , pp' are similar; and, since the angles at V , v are equal, the conics are similar.

Again, since the conics are similar, the “figures” on the axes are similar.

Therefore $PN^2 : CN \cdot NT = pn^2 : cn \cdot nt,$

and the angles at N , n are right, while the angle CPT is equal to the angle cpt .

Therefore the triangles CPT , cpt are similar, and the angle CTP is equal to the angle ctp .

Now, since $PV \cdot VP' : QV^2 = pv \cdot vp' : qv^2,$

and $QV^2 : PV^2 = qv^2 : pv^2;$

it follows that $PV : P'V = pv : p'v,$

whence $PP' : PV = pp' : pv.$

But, by the similar triangles CPT , cpt ,

$$CP : PT = cp : pt,$$

or $PP' : PT = pp' : pt;$

$$\therefore PV : PT = pv : pt,$$

and the proposition is proved.

Proposition 115.

[VI. 21, 22.]

If two ordinates be drawn to the axes of two parabolas, or the major or conjugate axes of two similar ellipses or two similar hyperbolas, as $PN, P'N'$ and $pn, p'n'$, such that the ratios $AN : an$ and $AN' : an'$ are each equal to the ratio of the respective latera recta, then the segments PP', pp' will be similar; also PP' will not be similar to any segment in the other conic which is cut off by two ordinates other than $pn, p'n'$, and vice versa.

[The method of proof adopted follows the lines of the previous propositions, and accordingly it is unnecessary to reproduce it.]

Proposition 116.

[VI. 26, 27.]

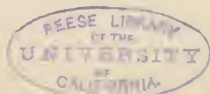
If any cone be cut by two parallel planes making hyperbolic or elliptic sections, the sections will be similar but not equal.

On referring to the figures of Props. 2 and 3, it will be seen at once that, if another plane parallel to the plane of section be drawn, it will cut the plane of the axial triangle in a straight line $p'pm$ parallel to $P'PM$ and the base in a line dme parallel to DME ; also $p'pm$ will be the diameter of the resulting hyperbola or ellipse, and the ordinates to it will be parallel to dme , i.e. to DME .

Therefore the ordinates to the diameters are equally inclined to those diameters in both curves.

Also, if PL, pl are the corresponding parameters,

$$PL : PP' = BF, FC : AF^2 = pl : pp'.$$



Hence the rectangles $PL \cdot PP'$ and $pl \cdot pp'$ are similar.

It follows that the conics are similar. [Prop. 112]

And they cannot be equal, since $PL \cdot PP'$ cannot be equal to $pl \cdot pp'$. [Cf. Prop. 110 (2)]

[A similar proposition holds for the *parabola*, since, by Prop. 1, $PL : PA$ is a constant ratio. Therefore two parallel parabolic sections have different parameters.]

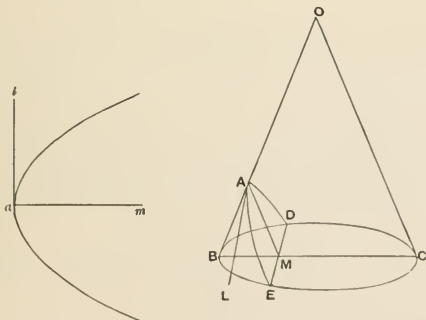
PROBLEMS.

Proposition 117.

[VI. 28.]

In a given right cone to find a parabolic section equal to a given parabola.

Let the given parabola be that of which am is the axis and al the latus rectum. Let the given right cone be OBC , where O is the apex and BC the circular base, and let OBC be a triangle through the axis meeting the base in BC .



Measure OA along OB such that

$$al : OA = BC^2 : BO, OC.$$

Draw AM parallel to OC meeting BC in M , and through AM draw a plane at right angles to the plane OBC and cutting the circular base in DME .

Then DE is perpendicular to AM , and the section DAE is a parabola whose axis is AM .

Also [Prop. 1], if AL is the latus rectum,

$$AL : AO = BC^2 : BO \cdot OC,$$

whence $AL = al$, and the parabola is equal to the given one [Prop. 110].

No other parabola with vertex on OB can be found which is equal to the given parabola except DAE . For, if another such parabola were possible, its plane must be perpendicular to the plane OBC and its axis must be parallel to OC . If A' were the supposed vertex and $A'L'$ the latus rectum, we should have $A'L' : A'O = BC^2 : BO \cdot OC = AL : AO$. Thus, if A' does not coincide with A , $A'L'$ cannot be equal to AL or al , and the parabola cannot be equal to the given one.

Proposition 118.

[VI. 29.]

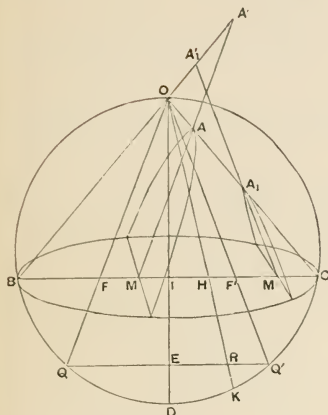
In a given right cone to find a section equal to a given hyperbola. (A necessary condition of possibility is that the ratio of the square on the axis of the cone to the square on the radius of the base must not be greater than the ratio of the transverse axis of the given hyperbola to its parameter.)

Let the given hyperbola be that of which aa' , al are the transverse axis and parameter respectively.

I. Suppose $OI^2 : BI^2 < aa' : al$, where I is the centre of the base of the given cone.

Let a circle be circumscribed about the axial triangle OBC , and produce OI to meet the circle again in D .

Then $OI : ID = OI^2 : BI^2$,
so that $OI : ID < aa' : al$.



Take E on ID such that $OI : IE = aa' : al$, and through E draw the chord QQ' parallel to BC .

Suppose now that AA', A_1A_1' are placed in the angle formed by OC and BO produced, such that $AA' = A_1A_1' = aa'$, and AA', A_1A_1' are respectively parallel to OQ, OQ' , meeting BC in M, M' .

Through $A'AM, A_1'A_1M'$ draw planes perpendicular to the plane of the triangle OBC making hyperbolic sections, of which $A'AM, A_1'A_1M'$ will therefore be the transverse axes.

Suppose OQ, OQ' to meet BC in F, F' .

Then $aa' : al = OI : IE$

$$= OF : FQ \text{ or } OF' : F'Q'$$

$$= OF^2 : OF \cdot FQ \text{ or } OF'^2 : OF' \cdot F'Q'$$

$$= OF^2 : BF \cdot FC \text{ or } OF'^2 : BF' \cdot F'C$$

$$= AA' : AL \text{ or } A_1A_1' : A_1L_1,$$

where AL , A_1L_1 are the parameters of AA' , $A_1A'_1$ in the sections respectively.

It follows, since $AA' = A_1A' = aa'$,
that $AL = A_1L_1 = al$.

Hence the two hyperbolic sections are each equal to the given hyperbola.

There are no other equal sections having their vertices on OC .

For (1), if such a section were possible and OH were parallel to the axis of such a section, OH could not be coincident either with OQ or OQ' . This is proved after the manner of the preceding proposition for the parabola.

If then (2) OH meet BC in H , QQ' in R , and the circle again in K , we should have, if the section were possible,

$$\begin{aligned} aa' : al &= OH^2 : BH \cdot HC \\ &= OH^2 : OH \cdot HK \\ &= OH : HK; \end{aligned}$$

which is impossible, since

$$aa' : al = OI : IE = OH : HR.$$

II. If $OI^2 : BI^2 = aa' : al$, we shall have $OI : ID = aa' : al$, and OQ , OQ' will both coincide with OD .

In this case there will be only one section equal to the given hyperbola whose vertex is on OC , and the axis of this section will be perpendicular to BC .

III. If $OI^2 : BI^2 > aa' : al$, no section can be found in the right cone which is equal to the given hyperbola.

For, if possible, let there be such a section, and let ON be drawn parallel to its axis meeting BC in N .

Then we must have $aa' : al = ON^2 : BN \cdot NC$,
so that $OI^2 : BI \cdot IC > ON^2 : BN \cdot NC$.

But $ON^2 > OI^2$, while $BI \cdot IC > BN \cdot NC$: which is absurd.

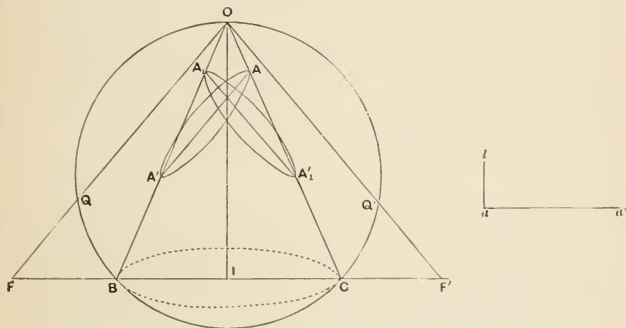
Proposition 119.

[VI. 30.]

In a given right cone to find a section equal to a given ellipse.

In this case we describe the circle about OBC and suppose F, F' taken on BC produced in both directions such that, if OF, OF' meet the circle in Q, Q' ,

$$OF : FQ = OF' : F'Q' = aa' : al.$$



Then we place straight lines AA', A_1A_1' in the angle BOC so that they are each equal to aa' , while AA' is parallel to OQ and A_1A_1' to OQ' .

Next suppose planes drawn through AA', A_1A_1' each perpendicular to the plane of OBC , and these planes determine two sections each of which is equal to the given ellipse.

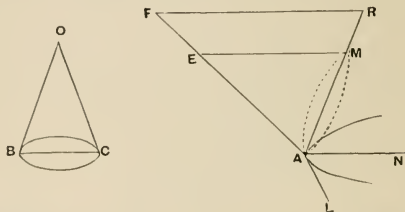
The proof follows the method of the preceding proposition.

Proposition 120.

[VI. 31.]

To find a right cone similar to a given one and containing a given parabola as a section of it.

Let OBC be an axial section of the given right cone, and let the given parabola be that of which AN is the axis and AL the latus rectum. Erect a plane passing through AN and perpendicular to the plane of the parabola, and in this plane make the angle NAM equal to the angle OBC .



Let AM be taken of such a length that $AL : AM = BC : BO$, and on AM as base, in the plane MAN , describe the triangle EAM similar to the triangle OBC . Then suppose a cone described with vertex E and base the circle on AM as diameter in a plane perpendicular to the plane EAM .

The cone EAM will be the cone required.

For $\angle MAN = \angle OBC = \angle EAM = \angle EMA$;

therefore EM is parallel to AN , the axis of the parabola.

Thus the plane of the given parabola cuts the cone in a section which is also a parabola.

$$\begin{aligned} \text{Now} \quad AL : AM &= BC : BO \\ &= AM : AE, \end{aligned}$$

$$\text{or} \quad AM^2 = EA \cdot AL;$$

$$\begin{aligned} \therefore AM^2 : AE \cdot EM &= AL : EM \\ &= AL : EA. \end{aligned}$$

Hence AL is the latus rectum of the parabolic section of the cone made by the plane of the given parabola. It is also the latus rectum of the given parabola.

Therefore the given parabola is itself the parabolic section, and EAM is the cone required.

There can be no other right cone similar to the given one, having its vertex on the same side of the given parabola, and containing that parabola as a section.

For, if another such cone be possible, with vertex F , draw through the axis of this cone a plane cutting the plane of the given parabola at right angles. The planes must then intersect in AN , the axis of the parabola, and therefore F must lie in the plane of EAN .

Again, if AF , FR are the sides of the axial triangle of the cone, FR must be parallel to AN , or to EM , and

$$\angle AFR = \angle BOC = \angle AEM,$$

so that F must lie on AE or AE produced. Let AM meet FR in R .

Then, if AL' be the latus rectum of the parabolic section of the cone FAR made by the plane of the given parabola,

$$\begin{aligned} AL' : AF &= AR^2 : AF \cdot FR \\ &= AM^2 : AE \cdot EM \\ &= AL : AE. \end{aligned}$$

Therefore AL' , AL cannot be equal; or the given parabola is not a section of the cone FAR .

Proposition 121.

[VI. 32.]

To find a right cone similar to a given one and containing a given hyperbola as a section of it. (If OBC be the given cone and D the centre of its base BC , and if AA' , AL be the axis and parameter of the given hyperbola, a necessary condition of possibility is that the ratio $OD^2 : DB^2$ must not be greater than the ratio $AA' : AL$.)

Take a point H on EI such that $FI : IH = AA' : AL$, and through H draw the chord QQ' of the circle parallel to AA' . Join $A'Q$, AQ , and in the plane of the circle draw AR making with AQ an angle equal to the angle OBC . Let AR meet $A'Q$ produced in R , and QQ' produced in N .

Join FQ meeting AA' in K .

Then, since the angle QAR is equal to the angle OBC , and

$$\angle FQA = \frac{1}{2} \angle A'QA = \frac{1}{2} \angle B'OC,$$

AR is parallel to FQ .

Also the triangle QAR is similar to the triangle OBC .

Suppose a cone formed with vertex Q and base the circle described on AR as diameter in a plane perpendicular to that of the circle FQA .

This cone will be such that the given hyperbola is a section of it.

We have, by construction,

$$\begin{aligned} AA' : AL &= FI : IH \\ &= FK : KQ, \text{ by parallels,} \\ &= FK \cdot KQ : KQ^2 \\ &= A'K \cdot KA : KQ^2. \end{aligned}$$

But, by the parallelogram $QKAN$,

$$A'K : KQ = QN : NR,$$

and

$$KA : KQ = QN : NA,$$

whence

$$A'K \cdot KA : KQ^2 = QN^2 : AN \cdot NR.$$

It follows that

$$AA' : AL = QN^2 : AN \cdot NR.$$

Therefore [Prop. 2] AL is the parameter of the hyperbolic section of the cone QAR made by the plane of the given hyperbola. The two hyperbolas accordingly have the same axis and parameter, whence they coincide [Prop. 110 (2)]; and the cone QAR has the required property.

Another such cone is found by taking the point Q' instead of Q and proceeding as before.

No other right cone except these two can be found which is similar to the given one, has its apex on the same side of the plane of the given hyperbola, and contains that hyperbola as a section.

For, if such a cone be possible with apex P , draw through its axis a plane cutting the plane of the given hyperbola at right angles. The plane thus described must then pass through the axis of the given hyperbola, whence P must lie in the plane of the circle FQA . And, since the cone is similar to the given cone, P must lie on the arc $A'QA$.

Then, by the converse of the preceding proof, we must have (if FP meet $A'A$ in T)

$$AA' : AL = FT : TP;$$

$$\therefore FT : TP = FI : IH,$$

which is impossible.

II. Suppose that

$$OD^2 : DB^2 = AA' : AL,$$

so that

$$FI : IE = AA' : AL.$$

In this case Q, Q' coalesce with E , and the cone with apex E and base the circle on AG as diameter perpendicular to the plane of FQA is the cone required.

III. If $OD^2 : DB^2 > AA' : AL$, no right cone having the desired properties can be drawn.

For, if possible, let P be the apex of such a cone, and we shall have, as before,

$$FT : TP = AA' : AL.$$

But $AA' : AL < OD^2 : DB^2$, or $FI : IE$.

Hence $FT : TP < FI : IE$, which is absurd.

Therefore, etc.

Proposition 122.

[VI. 33.]

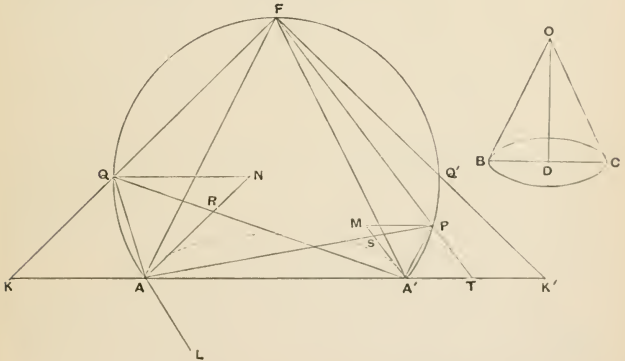
To find a right cone similar to a given one and containing a given ellipse as a section of it.

As before, take a plane through AA' perpendicular to the plane of the given ellipse; and in the plane so drawn describe on AA' as base a segment of a circle containing an angle equal to the angle BOC , the vertical angle of the given cone. Bisect the arc of the segment in F .

Draw two lines FK, FK' to meet AA' produced both ways and such that, if they respectively meet the segment in Q, Q' ,

$$FK : KQ = FK' : K'Q' = AA' : AL.$$

Draw QN parallel to AA' , and AN parallel to QF , meeting in N . Join $AQ, A'Q$, and let $A'Q$ meet AN in R .



Conceive a cone drawn with Q as apex and as base the circle on AR as diameter and in a plane at right angles to that of $AF A'$.

This cone will be such that the given ellipse is one of its sections.

For, since FQ , AR are parallel,

$$\angle FQR = \angle ARQ,$$

$$\therefore \angle ARQ = \angle FAA'$$

$$= \angle OBC.$$

And

$$\angle AQR = \angle AFA'$$

$$= \angle BOC.$$

Therefore the triangles QAR , OBC are similar, and likewise the cones QAR , OBC .

Now

$$AA' : AL = FK : KQ, \text{ by construction,}$$

$$= FK \cdot KQ : KQ^2$$

$$= A'K \cdot KA : KQ^2$$

$$= (A'K : KQ) \cdot (KA : KQ)$$

$$= (QN : NR) \cdot (QN : NA), \text{ by parallels,}$$

$$= QN^2 : AN \cdot NR.$$

Therefore [Prop. 3] AL is the latus rectum of the elliptic section of the cone QAR made by the plane of the given ellipse. And AL is the latus rectum of the given ellipse. Therefore that ellipse is itself the elliptic section.

In like manner another similar right cone can be found with apex Q' such that the given ellipse is a section.

No other right cone besides these two can be found satisfying the given conditions and having its apex on the same side of the plane of the given ellipse. For, as in the preceding proposition, its apex P , if any, must lie on the arc AFA' . Draw PM parallel to $A'A$, and $A'M$ parallel to FP , meeting in M . Join AP , $A'P$, and let AP meet $A'M$ in S .

The triangle $PA'S$ will then be similar to OBC , and we shall have $PM^2 : A'M \cdot MS = AT \cdot TA' : TP^2 = FT \cdot TP : TP^2$, in the same way as before.

We must therefore have

$$AA' : AL = FT : TP;$$

and this is impossible, because

$$AA' : AL = FK : KQ.$$

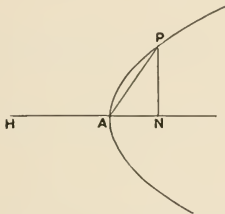
VALUES OF CERTAIN FUNCTIONS OF THE LENGTHS OF CONJUGATE DIAMETERS.

Proposition 123 (Lemma).

[VII. 1.]

In a parabola, if PN be an ordinate and AH be measured along the axis away from N and equal to the latus rectum,*

$$AP^2 = AN \cdot NH. \quad [= AN (AN + p_a)]$$



This is proved at once from the property $PN^2 = p_a \cdot AN$, by adding AN^2 to each side.

Proposition 124 (Lemma).

[VII. 2, 3.]

If AA' be divided at H , internally for the hyperbola, and externally for the ellipse, so that $AH : HA' = p_a : AA'$, then, if PN be any ordinate,

$$AP^2 : AN \cdot NH = AA' : A'H.$$

* Though Book VII. is mainly concerned with conjugate diameters of a central conic, one or two propositions for the parabola are inserted, no doubt in order to show, in connection with particular propositions about a central conic, any obviously corresponding properties of the parabola.

Produce AN to K , so that

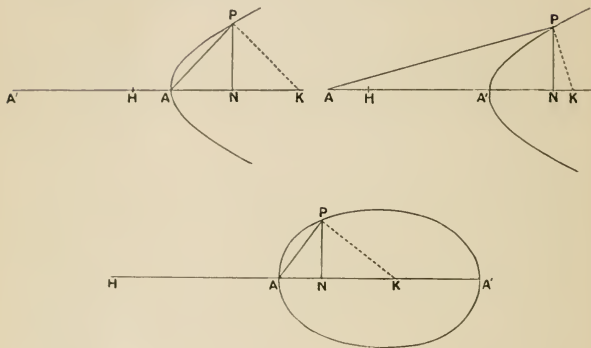
$$AN \cdot NK = PN^2;$$

thus

$$\begin{aligned} AN \cdot NK : AN \cdot A'N &= PN^2 : AN \cdot A'N \\ &= p_a : AA' \quad [\text{Prop. 8}] \\ &= AH : A'H, \text{ by construction,} \end{aligned}$$

or

$$NK : A'N = AH : A'H.$$



It follows that

$$A'N \pm NK : A'N = A'H \pm AH : A'H$$

(where the upper sign applies to the hyperbola).

Hence $A'K : A'N = AA' : A'H;$

$$\therefore A'K \pm AA' : A'N \pm A'H = AA' : A'H,$$

or

$$AK : NH = AA' : A'H.$$

Thus $AN \cdot AK : AN \cdot NH = AA' : A'H.$

But $AN \cdot AK = AP^2$, since $AN \cdot NK = PN^2$.

Therefore $AP^2 : AN \cdot NH = AA' : A'H.$

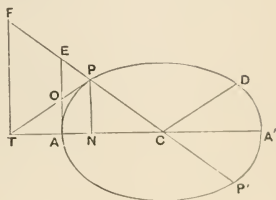
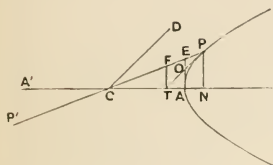
The same proposition is true if AA' is the *minor* axis of an ellipse and p_a the corresponding parameter.

Proposition 125 (Lemma).

[VII. 4.]

If in a hyperbola or an ellipse the tangent at P meet the axis AA' in T , and if CD be the semi-diameter parallel to PT , then

$$PT^2 : CD^2 = NT : CN.$$



Draw AE , TF at right angles to CA to meet CP , and let AE meet PT in O .

Then, if p be the parameter of the ordinates to PP' , we have

$$\frac{p}{2} : PT = OP : PE. \quad [\text{Prop. 23}]$$

Also, since CD is parallel to PT , it is conjugate to CP .

$$\text{Therefore} \quad \frac{p}{2} \cdot CP = CD^2 \dots\dots\dots(1).$$

$$\text{Now} \quad OP : PE = TP : PF;$$

$$\therefore \frac{p}{2} : PT = PT : PF,$$

$$\text{or} \quad \frac{p}{2} \cdot PF = PT^2 \dots\dots\dots(2).$$

From (1) and (2) we have

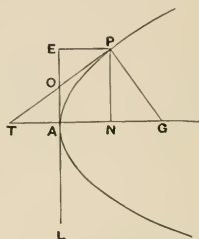
$$\begin{aligned} PT^2 : CD^2 &= PF : CP \\ &= NT : CN. \end{aligned}$$

Proposition 126 (Lemma).

[VII. 5.]

In a parabola, if p be the parameter of the ordinates to the diameter through P , and PN the principal ordinate, and if AL be the latus rectum,

$$p = AL + 4AN.$$



Let the tangent at A meet PT in O and the diameter through P in E , and let PG , at right angles to PT , meet the axis in G .

Then, since the triangles PTG , EPO are similar,

$$GT : TP = OP : PE,$$

$$\therefore GT = \frac{p}{2} \dots\dots\dots(1). \quad [\text{Prop. 22}]$$

Again, since TPG is a right angle,

$$\begin{aligned} TN \cdot NG &= PN^2 \\ &= LA \cdot AN, \end{aligned}$$

by the property of the parabola.

$$\text{But} \quad TN = 2AN. \quad [\text{Prop. 12}]$$

$$\text{Therefore} \quad AL = 2NG \dots\dots\dots(2);$$

$$\begin{aligned} \text{thus} \quad AL + 4AN &= 2(TN + NG) \\ &= 2TG \\ &= p, \text{ from (1) above.} \end{aligned}$$

[*Note.* The property of the normal (NG = half the latus rectum) is incidentally proved here by regarding it as the perpendicular through P to the tangent at that point. Cf. Prop. 85 where the normal is regarded as the minimum straight line from G to the curve.]

DEF. If AA' be divided, internally for the hyperbola, and externally for the ellipse, in each of two points H, H' such that

$$A'H : AH = AH' : A'H' = AA' : p_a,$$

where p_a is the parameter of the ordinates to AA' , then $AH, A'H'$ (corresponding to p_a in the proportion) are called **homologues**.

In this definition AA' may be either the *major* or the *minor* axis of an ellipse.

Proposition 127.

[VII. 6, 7.]

If $AH, A'H'$ be the "homologues" in a hyperbola or an ellipse, and PP', DD' any two conjugate diameters, and if AQ be drawn parallel to DD' meeting the curve in Q , and QM be perpendicular to AA' , then

$$PP'^2 : DD'^2 = MH' : MH.$$

Join $A'Q$, and let the tangent at P meet AA' in T .

Then, since $A'C = CA$, and $QV = VA$ (where CP meets QA in V), $A'Q$ is parallel to CV .

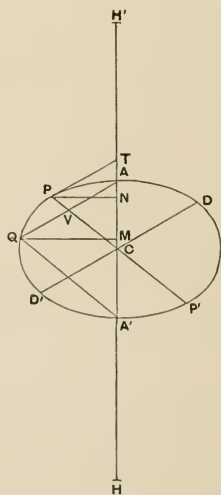
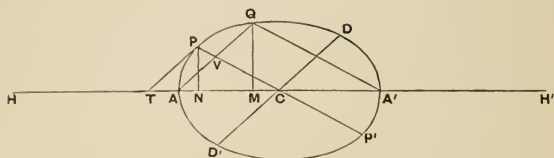
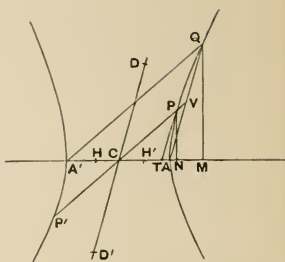
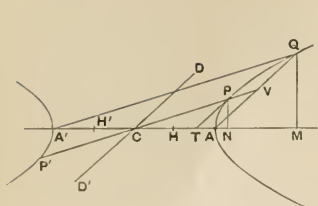
$$\begin{aligned} \text{Now} \quad PT^2 : CD^2 &= NT : CN && [\text{Prop. 125}] \\ &= AM : A'M, \text{ by similar triangles.} \end{aligned}$$

And, also by similar triangles,

$$CP^2 : PT^2 = A'Q^2 : AQ^2,$$

whence, *ex aequali*,

$$\begin{aligned} CP^2 : CD^2 &= (AM : A'M) \cdot (A'Q^2 : AQ^2) \\ &= (AM : A'M) \times (A'Q^2 : A'M \cdot MH') \\ &\quad \times (A'M \cdot MH' : AM \cdot MH) \times (AM \cdot MH : AQ^2). \end{aligned}$$



But, by Prop. 124,

$$A'Q^2 : A'M.MH' = AA' : AH',$$

and $AM.MH : AQ^2 = A'H : AA' = AH' : AA'.$

Also $A'M.MH' : AM.MH = (A'M : AM).(MH' : MH).$

It follows that

$$CP^2 : CD^2 = MH' : MH,$$

or

$$PP'^2 : DD'^2 = MH' : MH.$$

This result may of course be written in the form

$$PP' : p = MH' : MH,$$

where p is the parameter of the ordinates to $PP'.$

Proposition 128.

[VII. 8, 9, 10, 11.]

In the figures of the last proposition the following relations hold for both the hyperbola and the ellipse :

$$(1) \quad AA'^2 : (PP' \pm DD')^2 = A'H.MH' : (MH' \pm \sqrt{MH.MH'})^2,$$

$$(2) \quad AA'^2 : PP'.DD' = A'H : \sqrt{MH.MH'},$$

$$(3) \quad AA'^2 : (PP'^2 \pm DD'^2) = A'H : MH \pm MH'.$$

(1) We have

$$AA'^2 : PP'^2 = CA^2 : CP^2;$$

$$\therefore AA'^2 : PP'^2 = CN.CT : CP^2 \quad [\text{Prop. 14}]$$

$$= A'M.A'A : A'Q^2,$$

by similar triangles.

$$\text{Now} \quad A'Q^2 : A'M.MH' = AA' : AH' \quad [\text{Prop. 124}]$$

$$= AA' : A'H$$

$$= A'M.A'A : A'M.A'H,$$

whence, alternately,

$$A'M.A'A : A'Q^2 = A'M.A'H : A'M.MH'.$$

Therefore, from above,

$$AA'^2 : PP'^2 = A'H : MH' \dots\dots\dots (\alpha),$$

$$= A'H.MH' : MH'^2.$$

Again, $PP'^2 : DD'^2 = MH' : MH \dots (\beta)$, [Prop. 127]
 $= MH'^2 : MH \cdot MH'$;

$$\therefore PP' : DD' = MH' : \sqrt{MH \cdot MH'} \dots (\gamma).$$

Hence $PP' : PP' \pm DD' = MH' : MH' \pm \sqrt{MH \cdot MH'}$,
 and $PP'^2 : (PP' \pm DD')^2 = MH'^2 : (MH' \pm \sqrt{MH \cdot MH'})^2$.

Therefore by (α) above, *ex aequali*,

$$AA'^2 : (PP' \pm DD')^2 = A'H \cdot MH' : (MH' \pm \sqrt{MH \cdot MH'})^2.$$

(2) We derive from (γ) above

$$PP'^2 : PP' \cdot DD' = MH' : \sqrt{MH \cdot MH'}.$$

Therefore by (α) , *ex aequali*,

$$AA'^2 : PP' \cdot DD' = A'H : \sqrt{MH \cdot MH'}.$$

(3) From (β) ,

$$PP'^2 : (PP'^2 \pm DD'^2) = MH' : MH \pm MH'.$$

Therefore by (α) , *ex aequali*,

$$AA'^2 : (PP'^2 \pm DD'^2) = A'H : MH \pm MH'.$$

Proposition 129.

[VII. 12, 13, 29, 30.]

In every ellipse the sum, and in every hyperbola the difference, of the squares on any two conjugate diameters is equal to the sum or difference respectively of the squares on the axes.

Using the figures and construction of the preceding two propositions, we have

$$\begin{aligned} AA'^2 : BB'^2 &= AA' : p_a \\ &= A'H : AH, \text{ by construction,} \\ &= A'H : A'H'. \end{aligned}$$

Therefore

$$AA'^2 : AA'^2 \pm BB'^2 = A'H : A'H \pm A'H'$$

(where the upper sign belongs to the *ellipse*),

$$\text{or } AA'^2 : AA'^2 \pm BB'^2 = A'H : HH' \dots (\alpha).$$

Again, by (α) in Prop. 128 (1),

$$AA'^2 : PP'^2 = A'H : MH',$$

and, by means of (β) in the same proposition,

$$\begin{aligned} PP'^2 : (PP'^2 \pm DD'^2) &= MH' : MH \pm MH' \\ &= MH' : HH'. \end{aligned}$$

From the last two relations we obtain

$$AA'^2 : (PP'^2 \pm DD'^2) = A'H : HH'.$$

Comparing this with (α) above, we have at once

$$(PP'^2 \pm DD'^2) = (AA'^2 \pm BB'^2).$$

Proposition 130.

[VII. 14, 15, 16, 17, 18, 19, 20.]

The following results can be derived from the preceding propositions, viz.

(1) *For the ellipse,*

$$AA'^2 : PP'^2 \sim DD'^2 = A'H : 2CM;$$

and for both the ellipse and hyperbola, if p denote the parameter of the ordinates to PP' ,

$$(2) \quad AA'^2 : p^2 = A'H \cdot MH' : MH^2,$$

$$(3) \quad AA'^2 : (PP' \pm p)^2 = A'H \cdot MH' : (MH \pm MH')^2,$$

$$(4) \quad AA'^2 : PP' \cdot p = A'H : MH, \text{ and}$$

$$(5) \quad AA'^2 : PP'^2 \pm p^2 = A'H \cdot MH' : MH'^2 \pm MH^2.$$

(1) We have

$$AA'^2 : PP'^2 = A'H : MH', \quad [\text{Prop. 128 (1), } (\alpha)]$$

$$\begin{aligned} \text{and} \quad PP'^2 : PP'^2 \sim DD'^2 &= MH' : MH' \sim MH \quad [\text{ibid., } (\beta)] \\ &= MH' : 2CM \text{ in the ellipse.} \end{aligned}$$

Therefore for the *ellipse*

$$AA'^2 : PP'^2 \sim DD'^2 = A'H : 2CM.$$

(2) For either curve

$$AA'^2 : PP'^2 = A'H : MH', \text{ as before,} \\ = A'H . MH' : MH'^2,$$

and, by Prop. 127,

$$PP'^2 : p^2 = MH'^2 : MH^2; \\ \therefore AA'^2 : p^2 = A'H . MH' : MH^2.$$

(3) By Prop. 127,

$$PP' : p = MH' : MH; \\ \therefore PP'^2 : (PP' \pm p)^2 = MH'^2 : (MH \pm MH')^2.$$

And $AA'^2 : PP'^2 = A'H . MH' : MH'^2$, as before ;

$$\therefore AA'^2 : (PP' \pm p)^2 = A'H . MH' : (MH \pm MH')^2.$$

(4) $AA'^2 : PP'^2 = A'H : MH'$, as before,

$$\text{and} \quad PP'^2 : PP' . p = PP' : p \\ = MH' : MH; \quad [\text{Prop. 127}] \\ \therefore AA'^2 : PP' . p = A'H : MH.$$

(5) $AA'^2 : PP'^2 = A'H . MH' : MH'^2$, as before,

$$\text{and} \quad PP'^2 : PP'^2 \pm p^2 = MH'^2 : MH'^2 \pm MH^2,$$

by means of Prop. 127 :

$$\therefore AA'^2 : PP'^2 \pm p^2 = A'H . MH' : MH'^2 \pm MH^2.$$

Proposition 131.

[VII. 21, 22, 23.]

In a hyperbola, if $AA' >_{or<} BB'$, then, if PP', DD' be any other two conjugate diameters, $PP' >_{or<} DD'$ respectively; and the ratio $PP' : DD'$ continually $\left\{ \begin{array}{l} \text{decreases} \\ \text{or increases} \end{array} \right\}$ as P moves further from A on either side.

Also, if $AA' = BB'$, $PP' = DD'$.

(1) Of the figures of Prop. 127, the first corresponds to the case where $AA' > BB'$, and the second to the case where $AA' < BB'$.

Taking then the $\left\{ \begin{smallmatrix} \text{first} \\ \text{second} \end{smallmatrix} \right\}$ figure respectively, it follows from

$$PP'^2 : DD'^2 = MH' : MH \quad [\text{Prop. 127}]$$

that $PP' \begin{smallmatrix} > \\ \text{or} < \end{smallmatrix} DD'$.

$$\begin{aligned} \text{Also } AA'^2 : BB'^2 &= AA' : p_a = A'H : AH, \text{ by construction,} \\ &= AH' : AH, \end{aligned}$$

and $AH' : AH \begin{smallmatrix} > \\ \text{or} < \end{smallmatrix} MH' : MH$,

while $MH' : MH \left\{ \begin{smallmatrix} \text{diminishes} \\ \text{or increases} \end{smallmatrix} \right\}$ continually as M moves further from A , i.e. as Q , or P , moves further from A along the curve.

$$\text{Therefore } AA'^2 : BB'^2 \begin{smallmatrix} > \\ \text{or} < \end{smallmatrix} PP'^2 : DD'^2,$$

and the latter ratio $\left\{ \begin{smallmatrix} \text{diminishes} \\ \text{or increases} \end{smallmatrix} \right\}$ as P moves further from A .

And the same is true of the ratios

$$AA' : BB' \text{ and } PP' : DD'.$$

(2) If $AA' = BB'$, then $AA' = p_a$, and both H and H' coincide with C .

In this case therefore

$$AH = AH' = AC,$$

$$MH = MH' = CM,$$

and $PP' = DD'$ always.

Proposition 132.

[VII. 24.]

In an ellipse, if AA' be the major, and BB' the minor, axis, and if PP' , DD' be any other two conjugate diameters, then

$$AA' : BB' > PP' : DD',$$

and the latter ratio diminishes continually as P moves from A to B .

We have $CA^2 : CB^2 = AN \cdot NA' : PN^2$;

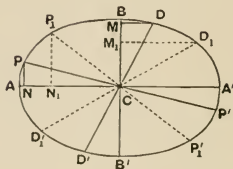
$$\therefore AN \cdot NA' > PN^2,$$

and, adding CN^2 to each,

$$CA^2 > CP^2,$$

or

$$AA' > PP' \dots\dots\dots(1).$$



Also $CB^2 : CA^2 = BM \cdot MB' : DM^2$,

where DM is the ordinate to BB' .

Therefore $BM \cdot MB' < DM^2$,

and, adding CM^2 , $CB^2 < CD^2$;

$$\therefore BB' < DD' \dots\dots\dots(2).$$

Again, if P_1P_1' , D_1D_1' be another pair of conjugates, P_1 being further from A than P , D_1 will be further from B than D .

And $AN \cdot NA' : AN_1 \cdot N_1A' = PN^2 : P_1N_1^2$.

But $AN_1 \cdot N_1A' > AN \cdot NA'$;

$$\therefore P_1N_1^2 > PN^2,$$

and $AN_1 \cdot N_1A' - AN \cdot NA' > P_1N_1^2 - PN^2$.

But, as above, $AN_1 \cdot N_1A' > P_1N_1^2$,

and $AN_1 \cdot N_1A' - AN \cdot NA' = CN^2 - CN_1^2$;

$$\therefore CN^2 - CN_1^2 > P_1N_1^2 - PN^2;$$

thus

$$CP^2 > CP_1^2,$$

or

$$PP' > P_1P_1' \dots\dots\dots(3).$$

In an exactly similar manner we prove that

$$DD' < D_1D_1' \dots\dots\dots(4).$$

We have therefore, by (1) and (2),

$$AA' : BB' > PP' : DD',$$

and, by (3) and (4), $PP' : DD' > P_1P'_1 : D_1D'_1$.

COR. It is at once clear, if p_a, p, p_1 are the parameters corresponding to $AA', PP', P_1P'_1$, that

$$p_a < p, \quad p < p_1, \text{ etc.}$$

Proposition 133.

[VII. 25, 26.]

(1) *In a hyperbola or an ellipse*

$$AA' + BB' < PP' + DD',$$

where PP', DD' are any conjugate diameters other than the axes.

(2) *In the hyperbola $PP' + DD'$ increases continually as P moves further from A , while in the ellipse it increases as P moves from A until PP', DD' take the position of the equal conjugate diameters, when it is a maximum.*

(1) For the hyperbola

$$AA'^2 \sim BB'^2 = PP'^2 \sim DD'^2 \quad [\text{Prop. 129}]$$

or $(AA' + BB') \cdot (AA' \sim BB') = (PP' + DD') \cdot (PP' \sim DD')$,

and, by the aid of Prop. 131,

$$AA' \sim BB' > PP' \sim DD';$$

$$\therefore AA' + BB' < PP' + DD'.$$

Similarly it is proved that $PP' + DD'$ increases as P moves further from A .

In the case where $AA' = BB', PP' = DD'$, and $PP' > AA'$; and the proposition still holds.

(2) For the ellipse

$$AA' : BB' > PP' : DD';$$

$$\therefore (AA'^2 + BB'^2) : (AA' + BB')^2 > (PP'^2 + DD'^2) : (PP' + DD')^2.*$$

But

$$AA'^2 + BB'^2 = PP'^2 + DD'^2; \quad [\text{Prop. 129}]$$

$$\therefore AA' + BB' < PP' + DD'.$$

* Apollonius draws this inference directly, and gives no intermediate steps.

Similarly it may be proved that $PP' + DD'$ increases as P moves from A until PP' , DD' take the position of the equal conjugate diameters, when it begins to diminish again.

Proposition 134.

[VII. 27.]

In every ellipse or hyperbola having unequal axes

$$AA' \sim BB' > PP' \sim DD',$$

where PP' , DD' are any other conjugate diameters. Also, as P moves from A , $PP' \sim DD'$ diminishes, in the hyperbola continually, and in the ellipse until PP' , DD' take up the position of the equal conjugate diameters.

For the *ellipse* the proposition is clear from what was proved in Prop. 132.

For the *hyperbola*

$$AA'^2 \sim BB'^2 = PP'^2 \sim DD'^2,$$

and

$$PP' > AA'.$$

It follows that

$$AA' \sim BB' > PP' \sim DD',$$

and the latter diminishes continually as P moves further from A .

[This proposition should more properly have come before Prop. 133, because it is really used (so far as regards the hyperbola) in the proof of that proposition.]

Proposition 135.

[VII. 28.]

In every hyperbola or ellipse

$$AA' \cdot BB' < PP' \cdot DD',$$

and $PP' \cdot DD'$ increases as P moves away from A , in the hyperbola continually, and in the ellipse until PP' , DD' coincide with the equal conjugate diameters.

We have

$$AA' + BB' < PP' + DD',$$

[Prop. 133]

so that

$$\therefore (AA' + BB')^2 < (PP' + DD')^2.$$

And, for the *ellipse*,

$$AA'^2 + BB'^2 = PP'^2 + DD'^2. \quad [\text{Prop. 129}]$$

Therefore, by subtraction,

$$AA' \cdot BB' < PP' \cdot DD',$$

and in like manner it will be shown that $PP \cdot DD'$ increases until PP' , DD' coincide with the equal conjugate diameters.

For the *hyperbola* [proof omitted in Apollonius] $PP' > AA'$, $DD' > BB'$, and PP' , DD' both increase continually as P moves away from A . Hence the proposition is obvious.

Proposition 136.

[VII. 31.]

If PP' , DD' be two conjugate diameters in an ellipse or in conjugate hyperbolas, and if tangents be drawn at the four extremities forming a parallelogram $LL'MM'$, then

the parallelogram $LL'MM' = \text{rect. } AA' \cdot BB'$.

Let the tangents at P , D meet the axis AA' in T , T' respectively. Let PN be an ordinate to AA' , and take a length PO such that

$$PO^2 = CN \cdot NT.$$

$$\text{Now} \quad CA^2 : CB^2 = CN \cdot NT : PN^2 \quad [\text{Prop. 14}]$$

$$= PO^2 : PN^2,$$

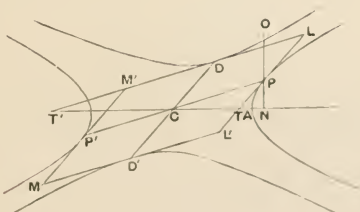
$$\text{or} \quad CA : CB = PO : PN;$$

$$\therefore CA^2 : CA \cdot CB = PO \cdot CT : CT \cdot PN.$$

Hence, alternately,

$$CA^2 : PO \cdot CT = CA \cdot CB : CT \cdot PN,$$

$$\text{or} \quad CT \cdot CN : PO \cdot CT = CA \cdot CB : CT \cdot PN \quad \dots\dots(1).$$



Therefore $AA'^2 : p_a^2 = A'H \cdot AH' : AH^2 \dots\dots\dots(\alpha)$.

We have now $AH > AH'$ but $\nless 2AH'$.

And $MH + HA > 2AH$;

$$\therefore MH + HA : AH > AH : AH',$$

or $(MH + HA) AH' > AH^2 \dots\dots\dots(\beta)$.

It follows that

$$(MH + HA) AM : (MH + HA) AH', \text{ or } AM : AH', \\ < (MH + HA) AM : AH^2.$$

Therefore, *componendo*,

$$MH' : AH' < (MH + HA) AM + AH^2 : AH^2 \\ < MH^2 : AH^2 \dots\dots\dots(\gamma),$$

whence $A'H \cdot MH' : A'H \cdot AH' < MH^2 : AH^2$,

or, alternately,

$$A'H \cdot MH' : MH^2 < A'H \cdot AH' : AH^2.$$

But, by Prop. 130 (2), and by the result (α) above, these ratios are respectively equal to $AA'^2 : p^2$, and $AA'^2 : p_a^2$.

Therefore $AA'^2 : p^2 < AA'^2 : p_a^2$,

or $p_a < p$.

Again, if P_1 be a point further from A than P is, and if $A'Q_1$ is parallel to CP_1 , and M_1 is the foot of the ordinate QM_1 , then, since $AH \nless 2AH'$,

$$MH < 2MH';$$

also $M_1H + HM > 2MH$.

Thus $(M_1H + HM) MH' > MH^2$.

This is a similar relation to that in (β) above except that M is substituted for A , and M_1 for M .

We thus derive, by the same proof, the corresponding result to (γ) above, or

$$M_1H' : MH' < M_1H^2 : MH^2,$$

whence $A'H \cdot M_1H' : M_1H^2 < A'H \cdot MH' : MH^2$,

or $AA'^2 : p_1^2 < AA'^2 : p^2$,

so that $p < p_1$, and the proposition is proved.

(3) Now let AA' be less than $\frac{P_0}{2}$.

Take a point M_0 such that $HH' = H'M_0$, and let Q_0, P_0 be related to M_0 in the same way that Q, P are to M .

Then $P_0P_0' : p_0 = M_0H' : M_0H$. [Prop. 127]

It follows, since $HH' = H'M_0$, that

$$p_0 = 2P_0P_0'.$$

Next, let P be a point on the curve between P_0 and A , and Q, M corresponding points.

Then $M_0H' \cdot H'M < HH'^2$,

since

$$MH' < M_0H'.$$

Add to each side the rectangle $(MH + HH')MH'$, and we have

$$(M_0H + HM)MH' < MH^2.$$

This again corresponds to the relation (β) above, with M substituted for A , M_0 for M , and $<$ instead of $>$.

The result corresponding to (γ) above is

$$M_0H' : MH' > M_0H^2 : MH^2;$$

$$\therefore A'H \cdot M_0H' : M_0H^2 > A'H \cdot MH' : MH^2,$$

or

$$AA'^2 : p_0^2 > AA'^2 : p^2.$$

Therefore

$$p > p_0.$$

And in like manner we prove that p increases as P moves from P_0 to A .

Lastly, let P be more remote from A than P_0 is.

In this case $H'M > H'M_0$,

and we have

$$MH' \cdot H'M_0 > HH'^2,$$

and, by the last preceding proof, interchanging M and M_0 and substituting the opposite sign of relation,

$$AA'^2 : p^2 < AA'^2 : p_0^2,$$

and

$$p > p_0.$$

In the same way we prove that p increases as P moves further away from P and A .

Hence the proposition is established.

Proposition 138.

[VII. 36.]

In a hyperbola with unequal axes, if p_a be the parameter corresponding to AA' and p that corresponding to PP' ,

$$AA' \sim p_a > PP' \sim p,$$

and $PP' \sim p$ diminishes continually as P moves away from A .

With the same notation as in the preceding propositions,

$$A'H : HA = AH' : H'A' = AA' : p_a,$$

whence $AA'^2 : (AA' \sim p_a)^2 = A'H . AH' : HH'^2$.

Also [Prop. 130 (3)]

$$AA'^2 : (PP' \sim p)^2 = A'H . MH' : HH'^2.$$

But $A'H . MH' > A'H . AH'$;

$$\therefore AA'^2 : (PP' \sim p)^2 > AA'^2 : (AA' \sim p_a)^2.$$

Hence $AA' \sim p_a > PP' \sim p$.

Similarly, if P_1, M_1 be further from A than P, M are, we have

$$A'H . M_1H' > A'H . MH',$$

and it follows that

$$PP' \sim p > P_1P_1' \sim p_1,$$

and so on.

Proposition 139.

[VII. 37.]

In an ellipse, if P_0P_0', D_0D_0' be the equal conjugate diameters and PP', DD' any other conjugate diameters, and if p_0, p, p_a, p_b be the parameters corresponding to P_0P_0', PP', AA', BB' respectively, then

(1) $AA' \sim p_a$ is the **maximum** value of $PP' \sim p$ for all points P between A and P_0 , and $PP' \sim p$ diminishes continually as P moves from A to P_0 ,

(2) $BB' \sim p_b$ is the **maximum** value of $PP' \sim p$ for all points P between B and P_0 , and $PP' \sim p$ diminishes continually as P passes from B to P_0 ,

(3) $BB' \sim p_b > AA' \sim p_a$.

The results (1) and (2) follow at once from Prop. 132.

(3) Since $p_b : BB' = AA' : p_a$, and $p_b > AA'$, it follows at once that $BB' \sim p_b > AA' \sim p_a$.

Proposition 140.

[VII. 38, 39, 40.]

(1) In a hyperbola, if AA' be not less than $\frac{1}{3} p_a$,

$$PP' + p > AA' + p_a,$$

where PP' is any other diameter and p the corresponding parameter; and $PP' + p$ will be the smaller the nearer P approaches to A .

(2) If $AA' < \frac{1}{3} p_a$, there is on each side of the axis a diameter, as P_0P_0' , such that $P_0P_0' = \frac{1}{3} p_0$; and $P_0P_0' + p_0$ is less than $PP' + p$, where PP' is any other diameter on the same side of the axis. Also $PP' + p$ increases as P moves away from P_0 .

(1) The construction being the same as before, we suppose

$$(a) \quad AA' \nless p_a.$$

In this case [Prop. 137 (1)] PP' increases as P moves from A , and p along with it.

Therefore $PP' + p$ also increases continually.

$$(b) \quad \text{Suppose } AA' < p_a \text{ but } \nless \frac{1}{3} p_a;$$

$$\therefore AH' \nless \frac{1}{3} AH;$$

thus

$$AH' \nless \frac{1}{4} (AH + AH'),$$

and

$$(AH + AH'). 4AH' \nless (AH + AH')^2.$$

Hence $4(AH + AH')AM : 4(AH + AH')AH'$, or $AM : AH'$,

$$\nless 4(AH + AH')AM : (AH + AH')^2;$$

and, *componendo*,

$$MH' : AH' \nless 4(AH + AH')AM + (AH + AH')^2 : (AH + AH')^2.$$

Now

$$(MH + MH')^2 - (AH + AH')^2 = 2AM(MH + MH' + AH + AH') \\ > 4AM(AH + AH');$$

$$\therefore 4AM(AH + AH') + (AH + AH')^2 < (MH + MH')^2.$$

It follows that

$$MH' : AH' < (MH + MH')^2 : (AH + AH')^2,$$

or $A'H . MH' : (MH + MH')^2 < A'H . AH' : (AH + AH')^2;$

$$\therefore AA'^2 : (PP' + p)^2 < AA'^2 : (AA' + p_a)^2 \text{ [by Prop. 130 (3)].}$$

Hence $AA' + p_a < PP' + p.$

Again, since $AH' < \frac{1}{4}(AH + AH'),$

$$MH' > \frac{1}{4}(MH + MH');$$

$$\therefore 4(MH + MH')MH' > (MH + MH')^2.$$

And, if P_1 be another point further from A than P is, and Q_1, M_1 points corresponding to Q, M , we have, by the same proof as before (substituting M for A , and M_1 for M),

$$A'H . M_1H' : (M_1H + M_1H')^2 < A'H . MH' : (MH + MH')^2.$$

We derive $PP' + p < P_1P_1' + p_1;$

and the proposition is established.

(2) We have $AH' < \frac{1}{3}AH$, so that $AH' < \frac{1}{2}HH'.$

Make $H'M_0$ equal to $\frac{1}{2}HH'$, so that $M_0H' = \frac{1}{3}M_0H.$

Then $P_0P_0' : p_0 = M_0H' : M_0H = 1 : 3,$

and $P_0P_0' = \frac{p_0}{3}.$

Next, since $M_0H' = \frac{1}{3}M_0H,$

$$M_0H' = \frac{1}{4}(M_0H + M_0H').$$

Now suppose P to be a point between A and P_0 , so that

$$M_0H' > MH';$$

$$\therefore (M_0H + M_0H')^2 > (M_0H + MH') . 4M_0H'.$$

Subtracting from each side the rectangle $(M_0H + MH') \cdot 4MM_0$,

$$(MH + MH')^2 > (M_0H + MH') \cdot 4MH';$$

$$\therefore (M_0H + MH') \cdot 4MM_0 : (M_0H + MH') \cdot 4MH', \text{ or } MM_0 : MH', \\ > (M_0H + MH') \cdot 4MM_0 : (MH + MH')^2.$$

Therefore, *componendo*,

$$M_0H' : MH' > (M_0H + MH') \cdot 4MM_0 + (MH + MH')^2 : (MH + MH')^2 \\ > (M_0H + M_0H')^2 : (MH + MH')^2.$$

Hence

$$A'H \cdot M_0H' : (M_0H + M_0H')^2 > A'H \cdot MH' : (MH + MH')^2.$$

Therefore [Prop. 130 (3)]

$$AA'^2 : (P_0P_0' + p_0)^2 > AA'^2 : (PP' + p)^2,$$

and

$$PP' + p > P_0P_0' + p_0.$$

Again, if P_1 be a point between P and A , we have

$$(MH + MH')^2 > (MH + M_1H') \cdot 4MH',$$

and we prove exactly as before that

$$P_1P_1' + p_1 > PP' + p,$$

and so on.

Lastly, if $MH > M_0H$, we shall have

$$(MH + M_0H') \cdot 4M_0H' > (M_0H + M_0H')^2.$$

If to both sides of this inequality there be added the rectangle $(MH + M_0H') \cdot 4MM_0$, they become respectively

$$(MH + M_0H') \cdot 4MH' \text{ and } (MH + MH')^2,$$

and the method of proof used above gives

$$P_0P_0' + p_0 < PP' + p,$$

and so on.

Hence the proposition is established.

Proposition 142.

[VII. 42.]

In a hyperbola, if PP' be any diameter with parameter p ,

$$AA'.p_a < PP'.p,$$

and $PP'.p$ increases as P moves away from A .

We have $A'H : HA = AA'^2 : AA'.p_a$,

and $A'H : MH = AA'^2 : PP'.p$, [Prop. 130 (4)]

while $AH < MH$;

$$\therefore AA'.p_a < PP'.p,$$

and, since MH increases as P moves from A , so does $PP'.p$.

Proposition 143.

[VII. 43.]

*In an ellipse $AA'.p_a < PP'.p$, where PP' is any diameter, and $PP'.p$ increases as P moves away from A , reaching a **maximum** when P coincides with B or B' .*

The result is derived at once, like the last proposition, from Prop. 130 (4).

[Both propositions are also at once obvious since

$$PP'.p = DD'^2.]$$

Proposition 144.

[VII. 44, 45, 46.]

In a hyperbola,

(1) *if $AA' \nless p_a$, or*

(2) *if $AA' < p_a$, but $AA'^2 \nless \frac{1}{2}(AA' \sim p_a)^2$, then*

$$AA'^2 + p_a^2 < PP'^2 + p^2,$$

where PP' is any diameter, and $PP'^2 + p^2$ increases as P moves away from A ;

(3) if $AA'^2 < \frac{1}{2}(AA' \sim p_a)^2$, then there will be found on either side of the axis a diameter P_0P_0' such that $P_0P_0'^2 = \frac{1}{2}(P_0P_0' \sim p_0)^2$, and $P_0P_0'^2 + p_0^2$ will be less than $PP'^2 + p^2$, where PP' is any other diameter. Also $PP'^2 + p^2$ will be the smaller the nearer PP' is to P_0P_0' .

(1) Let AA' be not less than p_a .

Then, if PP' be any other diameter, $p > p_a$, and p increases as P moves further from A [Prop. 137 (1)]; also $AA' < PP'$, which increases as P moves further from A ;

$$\therefore AA'^2 + p_a^2 < PP'^2 + p^2,$$

and $PP'^2 + p^2$ increases continually as P moves further from A .

(2) Let AA' be less than p_a , but $AA'^2 \nless \frac{1}{2}(AA' \sim p_a)^2$.

Then, since $AA' : p_a = A'H : AH = AH' : A'H'$,

$$2AH'^2 \nless HH'^2,$$

and

$$2MH' \cdot AH' > HH'^2.$$

Adding $2AH \cdot AH'$ to each side of the last inequality,

$$\begin{aligned} 2(MH + AH')AH' &> 2AH \cdot AH' + HH'^2 \\ &> AH^2 + AH'^2; \end{aligned}$$

$$\begin{aligned} \therefore 2(MH + AH')AM : 2(MH + AH')AH', \text{ or } AM : AH', \\ < 2(MH + AH')AM : AH^2 + AH'^2. \end{aligned}$$

Therefore, *componendo*,

$$MH' : AH' < 2(MH + AH')AM + AH^2 + AH'^2 : AH^2 + AH'^2,$$

$$\text{and } MH^2 + MH'^2 = AH^2 + AH'^2 + 2AM(MH + AH'),$$

$$\text{so that } MH' : AH' < MH^2 + MH'^2 : AH^2 + AH'^2,$$

$$\text{or } A'H \cdot MH' : MH^2 + MH'^2 < A'H \cdot AH' : AH^2 + AH'^2;$$

$$\therefore AA'^2 : PP'^2 + p^2 < AA'^2 : AA'^2 + p_a^2. \quad [\text{Prop. 130 (5)}]$$

Thus

$$AA'^2 + p_a^2 < PP'^2 + p^2.$$

Again, since

$$2MH'^2 > HH'^2,$$

$$\text{and (if } AM_1 > AM) \quad 2M_1H' \cdot MH' > HH'^2,$$

we prove in a similar manner, by substituting M for A and M_1 for M , that

$$PP'^2 + p^2 < P_1P_1'^2 + p_1^2.$$

(3) Let AA' be less than $\frac{1}{2}(AA' \sim p_a)^2$,

so that $2AH'^2 < HH'^2$.

Make $2M_0H'^2$ equal to HH'^2 .

Now $M_0H' : M_0H = P_0P_0' : p_0$, [Prop. 127]

so that $P_0P_0'^2 = \frac{1}{2}(P_0P_0' \sim p_0)^2$.

Next, if P be between A and P_0 ,

$$2M_0H'^2 = HH'^2,$$

and $2M_0H' \cdot MH' < HH'^2$.

Adding $2MH \cdot MH'$ to each side,

$$2(M_0H + MH') MH' < MH^2 + MH'^2,$$

and, exactly in the same way as before, we prove that

$$P_0P_0'^2 + p_0^2 < PP'^2 + p^2.$$

Again, if P_1 be between A and P ,

$$2MH' \cdot M_1H' < HH'^2,$$

whence (adding $2M_1H \cdot M_1H'$)

$$2(MH + M_1H') M_1H' < M_1H^2 + M_1H'^2,$$

and, in the same way,

$$PP'^2 + p^2 < P_1P_1'^2 + p_1^2.$$

Similarly $P_1P_1'^2 + p_1^2 < AA'^2 + p_a^2$.

Lastly, if $AM > AM_0$,

$$2MH' \cdot M_0H' > HH'^2,$$

and, if $AM_1 > AM$,

$$2M_1H' \cdot MH' > HH'^2;$$

whence we derive in like manner that

$$PP'^2 + p^2 > P_0P_0'^2 + p_0^2,$$

$$P_1P_1'^2 + p_1^2 > PP'^2 + p^2,$$

and so on.

Proposition 145.

[VII. 47, 48.]

In an ellipse,

(1) if $AA'^2 \nless \frac{1}{2}(AA' + p_a)^2$, then $AA'^2 + p_a^2 < PP'^2 + p^2$, and the latter increases as P moves away from A , reaching a **maximum** when P coincides with B ;

(2) if $AA'^2 > \frac{1}{2}(AA' + p_a)^2$, then there will be on each side of the axis a diameter P_0P_0' such that $P_0P_0'^2 = \frac{1}{2}(P_0P_0' + p_0)^2$, and $P_0P_0'^2 + p_0^2$ will then be less than $PP'^2 + p^2$ in the same quadrant, while this latter increases as P moves from P_0 on either side.

(1) Suppose $AA'^2 \nless \frac{1}{2}(AA' + p_a)^2$.

Now $A'H \cdot AH' : A'H^2 + A'H'^2 = AA'^2 : AA'^2 + p_a^2$.

Also $AA'^2 : BB'^2 = p_b : BB' = AA' : p_a = A'H : A'H'$
 $= A'H \cdot A'H' : A'H'^2$,

and $BB'^2 : (BB'^2 + p_b^2) = A'H'^2 : A'H^2 + A'H'^2$;

hence, *ex aequali*,

$$AA'^2 : (BB'^2 + p_b^2) = A'H \cdot A'H' : A'H^2 + A'H'^2,$$

and, as above,

$$AA'^2 : (AA'^2 + p_a^2) = A'H \cdot A'H' : A'H^2 + A'H'^2.$$

Again, $AA'^2 \nless \frac{1}{2}(AA' + p_a)^2$,

$$\therefore 2A'H \cdot AH' \nless HH'^2,$$

whence $2A'H \cdot MH' < HH'^2$.

Subtracting $2MH \cdot MH'$, we have

$$2A'M \cdot MH' < MH^2 + MH'^2 \dots \dots \dots (1),$$

$$\therefore 2A'M \cdot AM : 2A'M \cdot MH', \text{ or } AM : MH',$$

$$> 2A'M \cdot AM : MH^2 + MH'^2,$$

and, since $2A'M \cdot AM + MH^2 + MH'^2 = A'H^2 + A'H'^2$,

we have, *componendo*,

$$AH' : MH' > A'H^2 + A'H'^2 : MH^2 + MH'^2,$$

$$\therefore A'H \cdot AH' : A'H^2 + A'H'^2 > A'H \cdot MH' : MH^2 + MH'^2,$$

$$\text{whence } AA'^2 : (AA'^2 + p_a^2) > AA'^2 : (PP'^2 + p^2),$$

[Prop. 130 (5)]

$$\text{or } AA'^2 + p_a^2 < PP'^2 + p^2.$$

Again, either $MH < M_1H'$, or $MH \nless M_1H'$.

$$(a) \text{ Let } MH < M_1H'.$$

$$\text{Then } MH^2 + MH'^2 > M_1H^2 + M_1H'^2,$$

$$\text{and } M_1H^2 + M_1H'^2 > M_1H' \cdot 2(M_1H' - MH)^* ;$$

$$\therefore MM_1 \cdot 2(M_1H' - MH) : M_1H' \cdot 2(M_1H' - MH), \text{ or } MM_1 : M_1H', \\ > MM_1 \cdot 2(M_1H' - MH) : M_1H^2 + M_1H'^2.$$

$$\text{But } MH^2 + MH'^2 - (M_1H^2 + M_1H'^2) = 2(CM^2 - CM_1^2);$$

$$\therefore MM_1 \cdot 2(M_1H' - MH) + M_1H^2 + M_1H'^2 = MH^2 + MH'^2;$$

thus, *componendo*, we have

$$MH' : M_1H' > MH^2 + MH'^2 : M_1H^2 + M_1H'^2;$$

therefore, alternately,

$$A'H \cdot MH' : MH^2 + MH'^2 > A'H \cdot M_1H' : M_1H^2 + M_1H'^2,$$

$$\text{and } AA'^2 : PP'^2 + p^2 > AA'^2 : P_1P_1'^2 + p_1^2, [\text{Prop. 130 (5)}]$$

$$\text{so that } PP'^2 + p^2 < P_1P_1'^2 + p_1^2.$$

$$(b) \text{ If } MH \nless M_1H',$$

$$MH^2 + MH'^2 \nless M_1H^2 + M_1H'^2,$$

and it results, in the same way as before, that

$$A'H \cdot MH' : MH^2 + MH'^2 > A'H \cdot M_1H' : M_1H^2 + M_1H'^2,$$

$$\text{and } PP'^2 + p^2 < P_1P_1'^2 + p_1^2.$$

Lastly, since

$$A'H \cdot A'H' : A'H^2 + A'H'^2 = AA'^2 : BB'^2 + p_b^2,$$

$$\text{and } A'H \cdot M_1H' : M_1H^2 + M_1H'^2 = AA'^2 : P_1P_1'^2 + p_1^2,$$

* As in (1) above,

$$M_1H^2 + M_1H'^2 > 2A'M_1 \cdot M_1H'$$

$$> M_1H' \cdot 2(M_1H' - A'H')$$

$$> M_1H' \cdot 2(M_1H' - MH), \text{ a fortiori.}$$

it is shown in the same manner that

$$P_1 P_1'^2 + p_1^2 < BB'^2 + p_b^2.$$

$$(2) \quad \text{Suppose} \quad AA'^2 > \frac{1}{2}(AA' + p_a)^2,$$

$$\text{so that} \quad 2AH'^2 > HH'^2.$$

Make $2M_0H'^2$ equal to HH'^2 , so that

$$M_0H'^2 = \frac{1}{2}HH'^2 = HH' \cdot CH';$$

$$\therefore HH' : M_0H' = M_0H' : CH'$$

$$= HH' \sim M_0H' : M_0H' \sim CH',$$

$$\text{whence} \quad M_0H : CM_0 = HH' : M_0H',$$

$$\text{and} \quad HH' \cdot CM_0 = M_0H \cdot M_0H'.$$

$$\text{If then (a)} \quad AM < AM_0,$$

$$4CM_0 \cdot CH' > 2MH \cdot M_0H'.$$

Adding $2MM_0 \cdot M_0H'$ to each side,

$$4CM_0 \cdot CH' + 2MM_0 \cdot M_0H' > 2M_0H \cdot M_0H',$$

and again, adding $4CM_0^2$,

$$2(CM + CM_0) M_0H' > (M_0H^2 + M_0H'^2).$$

It follows that

$$2(CM + CM_0) MM_0 : 2(CM + CM_0) M_0H', \text{ or } MM_0 : M_0H', \\ < 2(CM + CM_0) MM_0 : (M_0H^2 + M_0H'^2).$$

$$\text{Now} \quad 2(CM + CM_0) MM_0 + M_0H^2 + M_0H'^2 \\ = MH^2 + MH'^2,$$

so that, *componendo*,

$$MH' : M_0H' < MH^2 + MH'^2 : M_0H^2 + M_0H'^2,$$

and

$$A'H \cdot MH' : MH^2 + MH'^2 < A'H \cdot M_0H' : M_0H^2 + M_0H'^2,$$

$$\text{whence} \quad P_0P_0'^2 + p_0^2 < PP'^2 + p^2.$$

$$\text{Similarly, if} \quad AM_1 < AM,$$

$$2HH' \cdot CM > 2M_1H \cdot MH',$$

and we prove, in the same manner as above,

$$PP'^2 + p^2 < P_1P_1'^2 + p_1^2.$$

And, since $2HH' \cdot CM_1 > 2AH \cdot M_1H'$,
in like manner

$$P_1P_1'^2 + p_1^2 < AA'^2 + p_a^2.$$

Lastly (b), if $AM > AM_0$, the same method of proof gives

$$P_0P_0'^2 + p_0^2 < PP'^2 + p^2,$$

etc.

Proposition 146.

[VII. 49, 50.]

In a hyperbola,

(1) *if $AA' > p_a$, then*

$AA'^2 \sim p_a^2 < PP'^2 \sim p^2$, where PP' is any diameter, and $PP'^2 \sim p^2$ increases as P moves further from A ;

also $PP'^2 \sim p^2 > AA'^2 \sim p_a \cdot AA'$ but $< 2(AA'^2 \sim p_a \cdot AA')$:

(2) *if $AA' < p_a$, then*

$AA'^2 \sim p_a^2 > PP'^2 \sim p^2$, which diminishes as P moves away from A ;

also $PP'^2 \sim p^2 > 2(AA'^2 \sim p_a \cdot AA')$.

(1) As usual, $A'H : AH = AH' : A'H' = AA' : p_a$;

$$\therefore A'H \cdot AH' : AH'^2 \sim AH^2 = AA'^2 : AA'^2 \sim p_a^2.$$

Now $MH' : AH' < MH : AH$;

$$\therefore MH' : AH' < MH' + MH : AH' + AH$$

$$< (MH' + MH) HH' : (AH' + AH) HH',$$

i.e.

$$< MH'^2 \sim MH^2 : AH'^2 \sim AH^2.$$

Hence

$$A'H \cdot MH' : MH'^2 \sim MH^2 < A'H \cdot AH' : AH'^2 \sim AH^2;$$

$$\therefore AA'^2 : PP'^2 \sim p^2 < AA'^2 : AA'^2 \sim p_a^2, [\text{Prop. 130 (5)}]$$

or

$$AA'^2 \sim p_a^2 < PP'^2 \sim p^2.$$

Again, if

$$AM_1 > AM,$$

$$M_1H' : MH' < M_1H : MH;$$

$$\therefore M_1H' : MH' < M_1H' + M_1H : MH' + MH,$$

and, proceeding as before, we find

$$PP'^2 \sim p^2 < P_1 P_1'^2 \sim p_1^2,$$

and so on.

Now, if PO be measured along PP' equal to p ,

$$PP'^2 \sim p^2 = 2PO \cdot OP' + OP'^2;$$

$$\therefore PP'^2 \sim p^2 > PP' \cdot OP' \text{ but } < 2PP' \cdot OP'.$$

But

$$PP' \cdot OP' = PP'^2 - PP' \cdot PO$$

$$= PP'^2 - p \cdot PP'$$

$$= AA'^2 - p_a \cdot AA'; \quad [\text{Prop. 129}]$$

$$\therefore PP'^2 \sim p^2 > AA'^2 \sim p_a \cdot AA' \text{ but } < 2(AA'^2 \sim p_a \cdot AA').$$

(2) If $AA' < p_a$,

$$MH' : AH' > MH : AH;$$

$$\therefore MH' : AH' > MH' + MH : AH' + AH,$$

and

$$A'H \cdot MH' : A'H \cdot AH' > (MH' + MH)HH' : (AH' + AH)HH',$$

i.e.

$$> MH'^2 \sim MH^2 : AH'^2 \sim AH^2.$$

Therefore, proceeding as above, we find in this case

$$PP'^2 \sim p^2 < AA'^2 \sim p_a^2.$$

Similarly

$$P_1 P_1'^2 \sim p_1^2 < PP'^2 \sim p^2,$$

and so on.

Lastly, if PP' be produced to O so that $PO = p$,

$$AA'^2 \sim p_a \cdot AA' = PP'^2 \sim p \cdot PP' \quad [\text{Prop. 129}]$$

$$= PP' \cdot OP'.$$

And

$$PP'^2 \sim p^2 = PP'^2 \sim PO^2$$

$$= 2PP' \cdot P'O + P'O^2$$

$$> 2PP' \cdot OP'$$

or

$$> 2(AA'^2 \sim p_a \cdot AA').$$

Proposition 147.

[VII. 51.]

In an ellipse,(1) *if* PP' *be any diameter such that* $PP' > p$,

$$AA'^2 \sim p_a^2 > PP'^2 \sim p^2,$$

and $PP'^2 \sim p^2$ *diminishes as* P *moves further from* A ;(2) *if* PP' *be any diameter such that* $PP' < p$,

$$BB'^2 \sim p_b^2 > PP'^2 \sim p^2,$$

and $PP'^2 \sim p^2$ *diminishes as* P *moves further from* B .

(1) In this case (using the figure of Prop. 141)

$$AH' : MH' < AC : CM$$

$$\therefore A'H \cdot AH' : A'H \cdot MH' < 2HH' \cdot AC : 2HH' \cdot CM$$

$$\text{i.e.} \quad < AH'^2 \sim AH^2 : MH'^2 \sim MH^2.$$

Therefore, alternately,

$$A'H \cdot AH' : AH'^2 \sim AH^2 < A'H \cdot MH' : MH'^2 \sim MH^2.$$

Hence

$$AA'^2 : AA'^2 \sim p_a^2 < AA'^2 : PP'^2 \sim p^2, \text{ [Prop. 130 (5)]}$$

$$\text{and} \quad AA'^2 \sim p_a^2 > PP'^2 \sim p^2.$$

Also, if $AM_1 > AM$, we shall have in the same way

$$A'H \cdot MH' : A'H \cdot M_1H' < MH'^2 \sim MH^2 : M_1H'^2 \sim M_1H^2,$$

and therefore $PP'^2 \sim p^2 > P_1P_1'^2 \sim p_1^2$, and so on.

(2) P must in this case lie between B and the extremity of either of the equal conjugate diameters, and M will lie between C and A' if P is on the quadrant AB .

Then, if M_1 corresponds to another point P_1 , and $AM_1 > AM$, we have

$$MH' > M_1H', \text{ and } CM < CM_1;$$

$$\therefore A'H.MH' : A'H.M_1H' > CM : CM_1$$

$$> 2CM.HH' : 2CM_1.HH',$$

$$\text{i.e.} \quad > MH^2 \sim MH'^2 : M_1H^2 \sim M_1H'^2,$$

whence, in the same manner, we prove

$$PP'^2 \sim p^2 < P_1P_1'^2 \sim p_1'^2;$$

and $PP'^2 \sim p^2$ increases as P moves nearer to B , being a *maximum* when P coincides with B .



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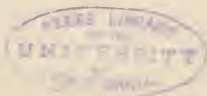
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